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PCB Design

From Start to Finish

Step-by-step guide on printed circuit board design with information suitable for beginners to graduate-level users.

by John Burkhert

Series Overview

This series, by John Burkhert, is a step-by-step guide on printed circuit board design with information suitable for beginners to graduate-level users.

Foreword.

This series is for anyone who wants to design their own printed circuit boards or perform designs for others. While an electronics degree is not required, beginners would do well to consider basic electronic theory as a prerequisite. Each part ends with a Graduate School Zone where more advanced techniques are discussed. Beginners may want to skip those sections and those who have the rudiments down might skip ahead to those sections.

PCB Design is at the crossroads of mechanical and electrical domains. One day, you're working with geometric dimensioning and tolerancing, and the next day you're calculating temperature rise on a power plane. In between these two camps, there is a raft of technology that is particular to printed circuit boards.

Combining an assortment of components that perform a function, and integrating that function with one or more others is at the heart of PCB layout. A component can have two pins or two thousand pins. It can cost a penny or hundreds of dollars. It can be bought off the shelf or custom ordered with a lead time that runs into months. Every PCB is a custom carrier for a selection of parts that, hopefully, do something useful. Let's get started with a summary of each section to come.

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1. PCB Stack-Ups

Section one talks about the importance of having a balanced stackup and discusses single, multi-layer, and rigid-flex design.

1.1 PCB Stackups

1.2 Single Layer Boards

1.3 What about PC Boards With Multi-Layer Stack-ups?

1.4 Graduate School Zone

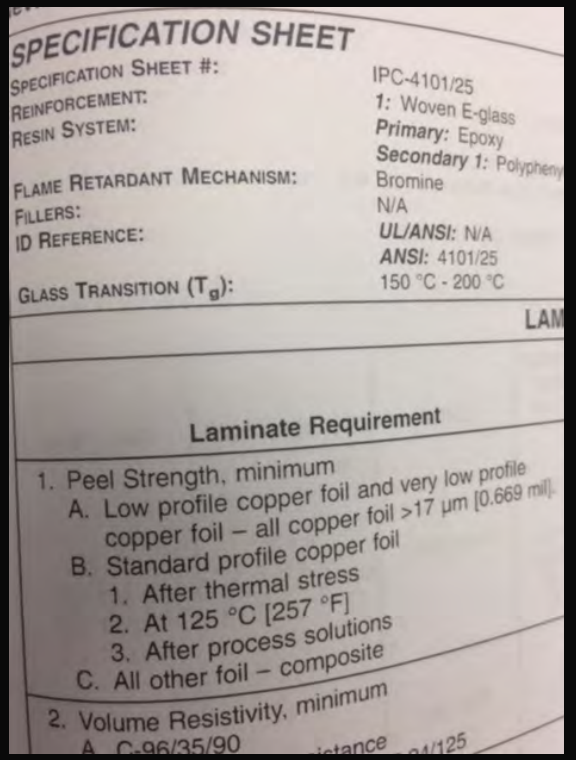
1.1 PCB Stackups

The first thing to understand about printed circuit boards is the stack-up. A stack-up is made of alternating layers of conductors and dielectric materials. Most PCBs rely on dielectric material consisting of strands of glass woven into a resin material. Glass resin has insulating properties much like the fiberglass that is used in home insulation. While the home insulation uses air between the fibers, the stack-up employs certain types of resin and varying compositions of glass and other materials such as aramid paper.

Just as the clothing industry weaves cotton or synthetic materials in different ways, so too does the PCB industry. The nature of the resin and method of the weave are broken out into what is called slash sheets by the IPC. (1) They have two main documents that are relevant to rigid stack-up materials; IPC-4101 for typical materials and IPC-4103 for high speed/high-frequency materials. Flexible materials are catalogued in IPC-4102 and high density interconnect materials are covered by IPC-4104.

Calling out these materials is part of the fabrication drawing notes. The most famous PCB material is called FR4. If you want to have a clear note, avoid commercial terms like that and stick to specific terms like this. LAMINATE AND PREPREG MATERIAL SHALL BE WOVEN E-GLASS / EPOXY IN ACCORDANCE WITH IPC-4101/126. SEE STACK-UP DIAGRAM FOR DETAILS.

Establishing a set of dielectric materials in this way includes factors such as the maximum operating temperature and the coefficient of thermal expansion.


SPECIFICATION SHEET

SPECIFICATION SHEET #:	IPC-4101/25
REINFORCEMENT:	*1:* Woven E-glass
RESIN SYSTEM:	*Primary:* Epoxy
	Secondary 1: Polypheny
FLAME RETARDANT MECHANISM:	Bromine
FILLERS:	N/A
ID REFERENCE:	*UL/ANSI:* N/A
	ANSI: 4101/25
GLASS TRANSITION (T_g):	150 °C - 200 °C

LAM

Laminate Requirement

1. Peel Strength, minimum
   A. Low profile copper foil and very low profile copper foil – all copper foil >17 µm [0.669 mil].
   B. Standard profile copper foil
      1. After thermal stress
      2. At 125 °C [257 °F]
      3. After process solutions
   C. All other foil – composite
2. Volume Resistivity, minimum
   A. C-96/35/90

Section 1. Figure 1. Image 1.1. Credit: EPEC - The basic details of PCB materials are catalogued in the various IPC slash sheets.

Calling these attributes out individually is more of a feel-good thing. Material will meet the Reduction of Hazardous Substances (RoHS) rules or it won't. The slash sheets have that covered.

That statement comes from doing defense projects. In an alternate universe of reference designs or globally sourced boards, the words "OR EQUIVALENT" would follow the slash sheet designation. At that point, it becomes useful to spell out the attributes, especially the T_g minimum and the RoHS requirement.

Fab shops in other countries will use locally sourced materials that are similar but may not meet the high-temperature requirements for lead-free processing or may contain the bromides that are prohibited for mass production in Europe.

The stack-up will be further described in a diagram that details the layer thickness of all dielectrics, copper, solder mask, and even ink if necessary.

Laminate materials come in two basic forms. The core material is a fully formed sheet with a cured layer of dielectric sandwiched between two layers of copper. Prepreg is basically a sheet of copper with an uncured layer of resin/fiber that is kind of like glue. A PCB stack-up is built around a core to which prepreg layers are added.

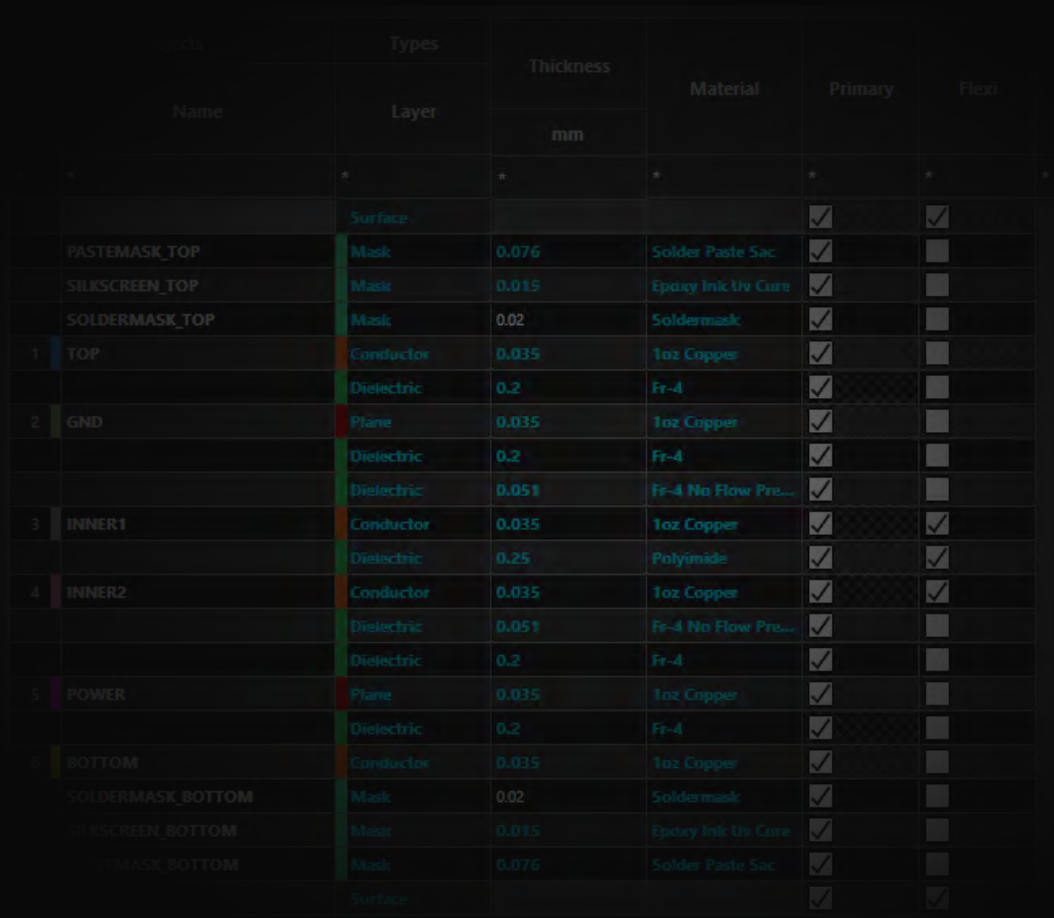
Higher layer-count stack-ups may have multiple cores with prepreg layers used to combine them. The variations are practically unlimited. Guessing the exact material layup that your vendor is going to use for a 24-layer board would be like winning the lottery.

Even an eight-layer board can be stacked up a thousand different ways. Anything more than two layers means that I will defer to the vendor for the actual materials.

When it comes to flexible printed circuits, a whole list of possibilities are in play. In addition to the polyimide base material and copper, there is additional copper plating that is specified. From there, adhesive and coverlay are added. Where components and edge connectors reside, there are additional call-outs for more adhesive and stiffeners under the components as well as the solder mask and ink on the component side.

Flexes and rigid-flexes are distinguished by the fact that the stack-up normally varies from one location to another.

All of these segments should be specified on the fab drawing and a separate stack-up diagram would be used for each area. It can also be handled in tabular form. There will be at least two areas and often more depending on the nature of the flexible printed circuit. We'll get into these attributes along with bend regions and other design rules for flexes in another chapter.



	Name	Types Layer	Thickness mm	Material	Primary	Flexi
		Surface			✓	✓
	PASTEMASK_TOP	Mask	0.076	Solder Paste Sac	✓	
	SILKSCREEN_TOP	Mask	0.015	Epoxy Ink Uv Cure	✓	
	SOLDERMASK_TOP	Mask	0.02	Soldermask	✓	
1	TOP	Conductor	0.035	1oz Copper	✓	
		Dielectric	0.2	Fr-4	✓	
2	GND	Plane	0.035	1oz Copper	✓	
		Dielectric	0.2	Fr-4	✓	
		Dielectric	0.051	Fr-4 No Flow Pre...	✓	
3	INNER1	Conductor	0.035	1oz Copper	✓	✓
		Dielectric	0.25	Polyimide	✓	✓
4	INNER2	Conductor	0.035	1oz Copper	✓	✓
		Dielectric	0.051	Fr-4 No Flow Pre...	✓	
		Dielectric	0.2	Fr-4	✓	
5	POWER	Plane	0.035	1oz Copper	✓	
		Dielectric	0.2	Fr-4	✓	
6	BOTTOM	Conductor	0.035	1oz Copper	✓	
	SOLDERMASK_BOTTOM	Mask	0.02	Soldermask	✓	
	SILKSCREEN_BOTTOM	Mask	0.015	Epoxy Ink Uv Cure	✓	
	PASTEMASK_BOTTOM	Mask	0.076	Solder Paste Sac	✓	
		Surface			✓	✓

1.2 Single Layer Boards

The main thing that distinguishes a stack-up is the number of layers of copper. Although every PCB has a top and a bottom side, there are many PCBs that are considered to be one layer. All of the circuitry is located on the primary side. (2)

On a single-layer board traces that cross over are handled by routing one trace through a resistor while the other trace crosses between the pads of the resistor. Another option is to go around the long way or being creative with placement.

Single layer boards are bound to be simple circuits. A lot of makers fabricate this type of PCB in their garage. A single-layer flex is often used as a custom flat cable.

Just as the clothing industry weaves cotton or synthetic materials in different ways, so too does the PCB industry.

The stack-up diagram will show the overall thickness along with the "weight" of the copper. This is a peculiarity of printed circuits where we refer to the thickness of the copper as a weight.

If we took one ounce of copper and rolled it flat to the point where it covered one square foot, then the copper would end up being 0.0014 inches thick. That is one-ounce copper.

Copper weights can be less or more so we can work in fractions of an ounce or multiple ounces for higher current capacity. Note that thinner lines will require thinner copper to be producible. Half-ounce copper is traditional for inner layers but would be less for fine-line etching. One ounce is normal for outer layers but they start with a half-ounce and plate up.

As the fab-shop adds the plating to the outer layers, they're also plating the surface of the holes. They do this plate-up prior to etching away the circuit pattern for the outer layers. This is why the outer layers will have a greater minimum trace width and air-gap than the inner layers.

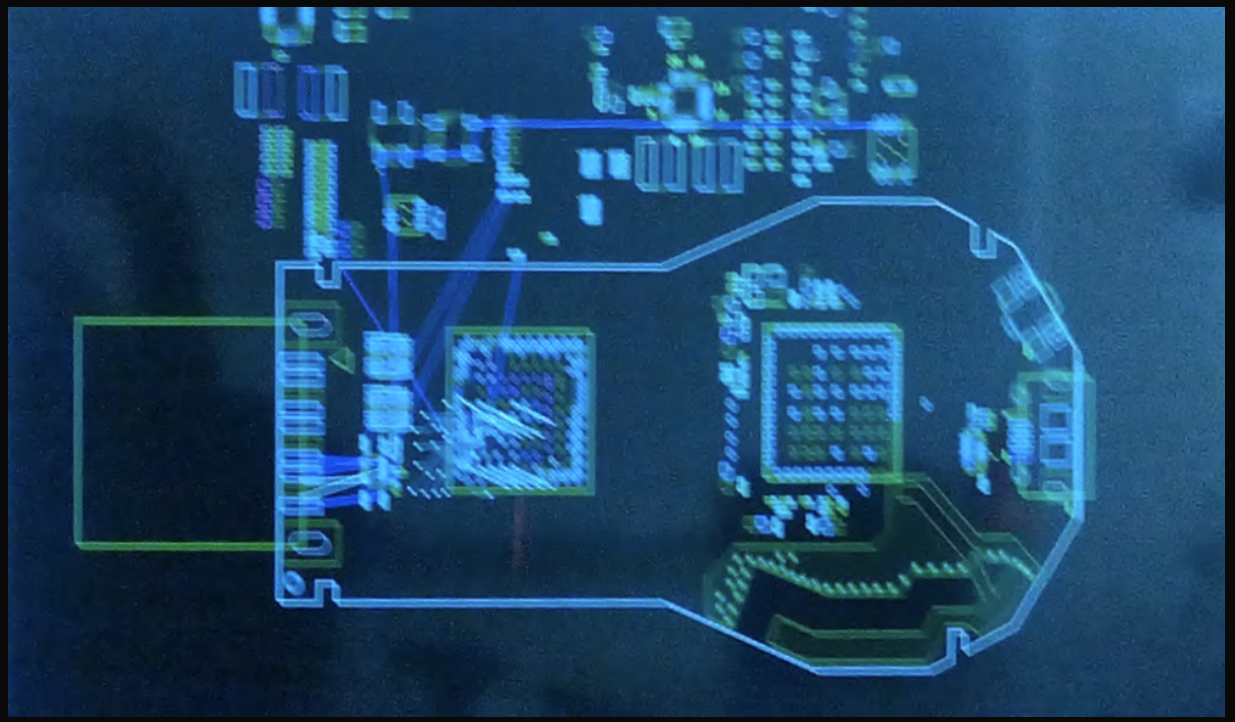
Vias are a vital part of PCB construction and will be covered in another chapter. They are introduced here so we can move on to multi-layer boards.

1.3 What about PC Boards With Multi-Layer Stack-ups?

A two-layer board will have a circuit pattern etched into both sides. This makes it easier to deal with crossed signals. It is also the starting point of what we normally consider for fabrication using vias and larger plated through holes to mount components.

With careful placement and routing, it is possible to support areas of controlled impedance. In this case, the secondary side of the PCB is mostly a ground plane with a few short traces or localized power planes mixed in. The idea is to keep a ground plane under any high-speed or high frequency lines.

When density increases, the venerable four-layer board comes into play. A typical four-layer board will be 0.062 inches in thickness with a relatively thick core between layers 2 and 3.



Section 1. Figure 2. Image 1.2. Credit: Author - An early version of Chromecast where my job was to reduce the layer count from eight to six. It was a complete do-over but the product sales warranted the investment.

In many cases, layer-two is a full ground plane and layer-three is a power plane or a mixture of planes and signals. The reason to allocate some ground plane to layer-three is if there is any controlled impedance used on layer-four.

1.4 Graduate School Zone

The main reason designers wind up using more than four layers has to do with circuit density. Some component packages will require more layers to successfully fan-out the chip. Some devices will require more layers for additional power domains, even redundant layers for the same voltage. Optimizing the layer-count starts with the device with the highest level of integration. Isolating noisy traces and protecting victim traces requires more layers to avoid cross-talk.

As I mentioned, the number of potential stack-ups is endless. The copper can be varied with additional thickness used on layers that carry high current. The hallmark of a good stack-up is balance. Starting from the central core dielectric layer, the layers should be a mirror image working from the center up and the center down.

All multi-layer rigid boards will thus have an even number of layers. There can be multiple cores in a PCB stack-up and each will have metal on both sides. When prepreg layers are added, they always go on both sides of the core in equal distribution with one layer of copper added per layer of prepreg. If the vendor tries to add one layer without the other side getting one, the stack-up will be unbalanced and prone to warpage.

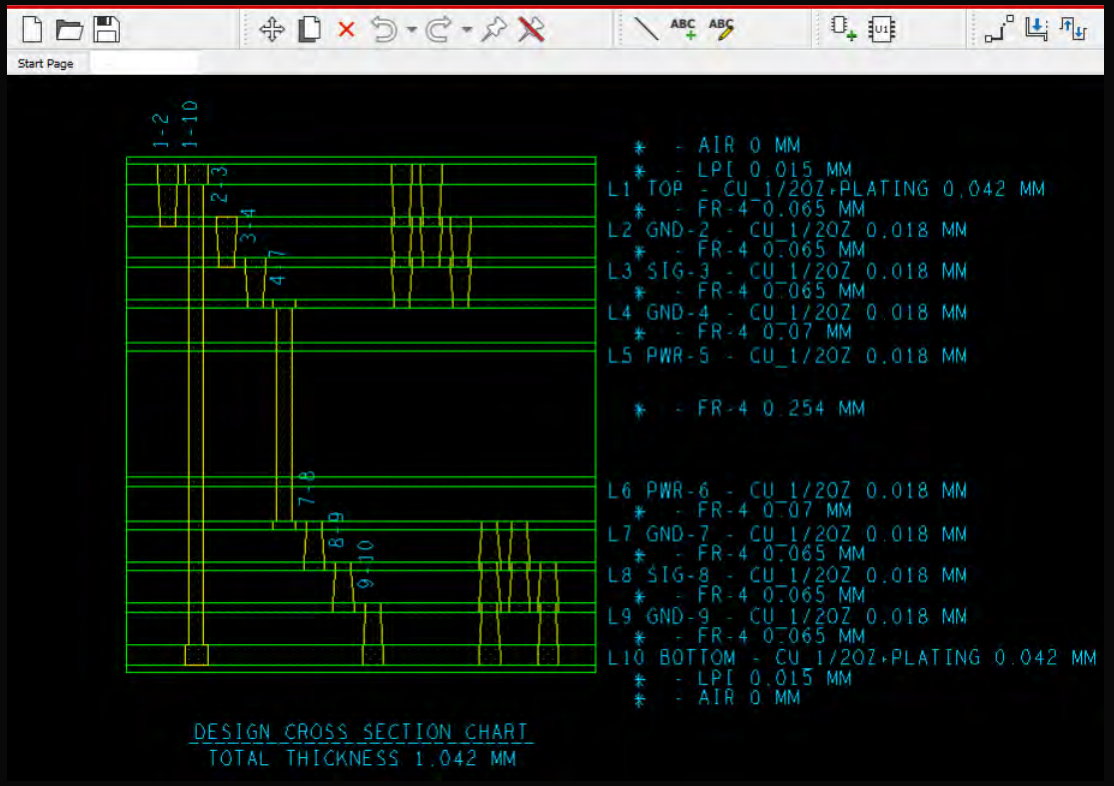
Capturing all of that data can be tedious, especially if you wind up translating the data from one unit of measure to another. Beyond the stack-up, there is also the matter of line widths and air-gaps for controlled impedance requirements. One way to reduce potential errors is to ask your vendor to share the stack-up as IPC-2581 data.

I don't know about all ECAD tools but the ones I do know will import the data seamlessly. In any case, work with the vendor. Don't try this at home!

Meanwhile, it is common for rigid-flex stack-up geometry to use from one to three layers of flexible material as the core. The rigid layers are built above and below the flex layers in equal measure. Three layers of flex material work out well as the flex circuit can have traces on the inner layer with a mesh of ground on the outer two layers. These three layers would protrude out from the rigid section with another rigid section added to the end of the protrusion. When flexibility is more important than EMI/EMC, a single layer of flex material is another possibility. We'll explore flexible printed circuits in more detail later on.

(1) What does IPC mean? It used to stand for Institute of Printed Circuits but they've broadened their scope while retaining the initials. IPC is a trade group that sets and documents the industry standards. The relevant information on a given topic can be hard to find among the many publications. I will be referring to some of those documents while distilling the knowledge into a manageable form.

(2) We used to call the side with components the top side. Primary and secondary have replaced top and bottom. Primary refers to the side with the majority or all of the components. Secondary indicates fewer or no components. The Primary side isn't always facing up in the product. The orientation is typically defined by the physical design team along with the board outline and interconnects.



Section 1. Figure 3. Image 1.3. Credit: Author - This would be known as a 3-N-3 stack-up to denote three sequential lamination cycles and N number of layers for the initial lamination. In this case, N = 4. Lamination takes time and uses a press which is an expensive piece of equipment. A through-hole board uses one lamination cycle while an HDI board will use an additional cycle for each layer of micro-vias. It's a layout advantage that comes at a cost.

2. Components

Section two covers the different types of components and some of the data included in footprints to facilitate DFF, DFA, and DFT.

2.1 Components

2.2 Passive Components - The Basic Building Blocks of a Circuit

2.3 Active Components - Where the Action Starts

2.4 Surface Mount vs. Plated Through Hole Components

2.5 Library Footprints - How to Create The Virtual Parts

2.6 Graduate School Zone

2.1 Components

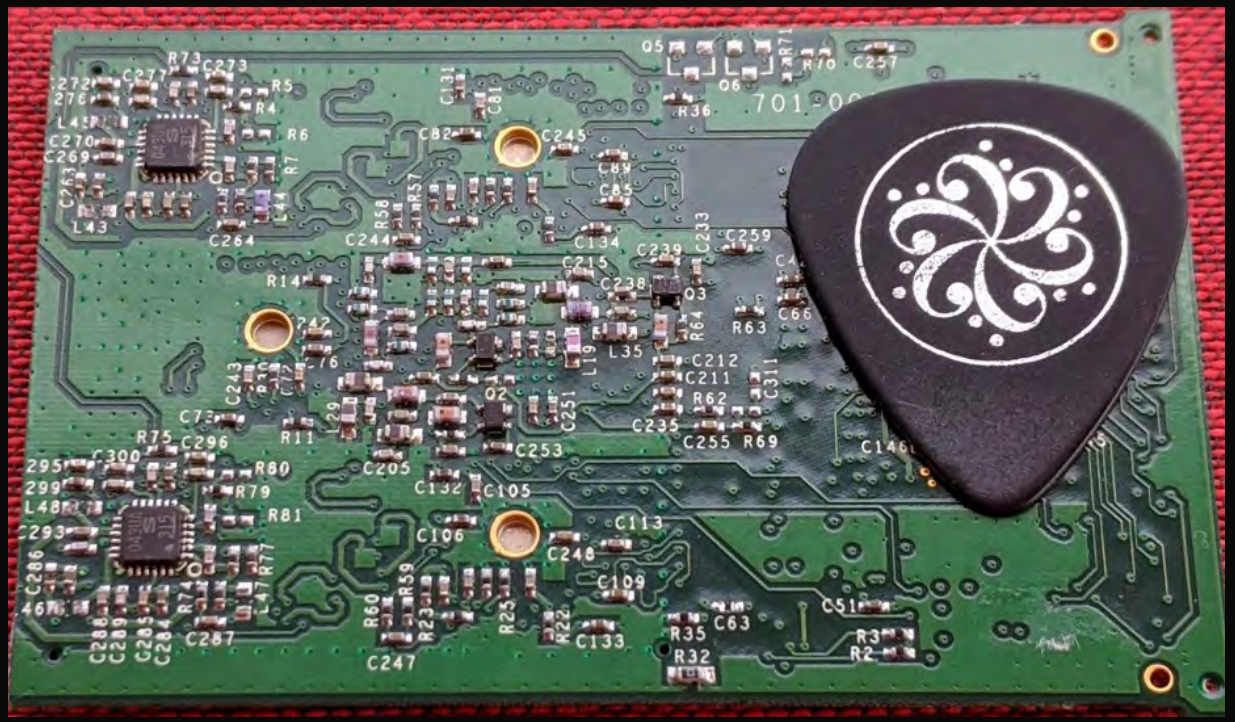
A PCB is only as good as the sum of its parts. Components are categorized in various ways. One way is to identify parts as either passive or active. Passive parts are the basic building blocks, resistors, capacitors, and inductors are examples. Quite often, they have two pins but they can also be packaged with several individual circuits in a single package. Resistor packs are the most common passive components to be gathered into groups.

2.2 Passive Components - The Basic Building Blocks of a Circuit

Resistors are widely used as voltage dividers and terminations. They are also used as pull-ups and pul-downs where the idea is to tie an integrated circuit pin to either a high or low logical state. They are measured in ohms and rated by wattage and tolerance.

Capacitors are formed using two metal plates with a dielectric material in between them. When one lead is tied to the ground net, they can be used to store small amounts of energy. They act like a little bucket of electrons ready to go to work whenever a logic state changes. Capacitors may also be used in-line to block Direct Current from propagating down a transmission line.

Capacitors can be polarized or non-polarized. They are measured in farads, but most caps are way less than one farad.



Section 2. Figure 1. Image 2.1. Credit: Author - The secondary side is a popular location for passive components and lightweight ICs.

Microfarads and picofarads are common measures on a printed circuit board. They come with a voltage rating, a tolerance and an equivalent series resistance or ESR. Super caps are kind of like batteries in that they can store a great deal of energy.

Paired up with resistors, capacitors can act as filters to screen out higher frequency signals. This is known as an RC filter. When combined with an inductor, a cap can be used to reduce the ripple on a voltage. They are commonly paired up this way on the output pin of a switch-mode power supply. That would be an LC filter.

Inductors resist changes in current. Also known as coils, they are notorious for generating magnetic fields. Putting an inductor on either side of a ferrite core will create a transformer.

Depending on the number of coils on either side, a transformer can increase or decrease whatever voltage is supplied. A PCB has all of the necessary ingredients to have its own resistance, capacitance, and inductance. Often, it is by design and just as often not.

Diodes are another type of passive device. They act as a one-way sign along the path. Diodes have various uses including electrostatic discharge suppression. ESD diodes are common when there is a connector that faces the outside world. They shunt the static electricity to the ground before it can zap the active electronics.

2.3 Active Components - Where the Action Starts

Active components are the integrated circuits - or ICs for short. They require a voltage-in along with the signal pins in order to do their thing. A transistor is the most basic of active components. You can still buy a single transistor in a three-pin package. The original transistors were vacuum tubes.

Modern transistors are created in silicon at scales that are two orders of magnitude smaller than the wavelength of visible light. What that means is that there will never be a lens strong enough to view the actual circuits. We infer their existence using electron scanning microscopes.

It's debatable whether there are more transistors in the world than there are stars in the sky. You could easily have a billion transistors in your pocket. If that's all you've got then you better catch up!

Some devices surpass 10,000,000,000 gates. The parts that go onto a PCB are utterly astonishing.

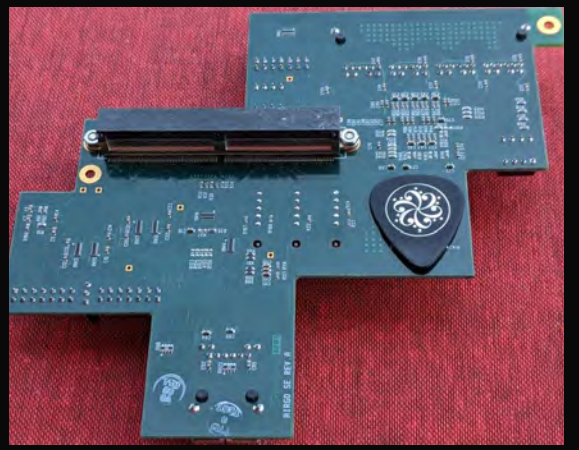
2.4 Surface Mount vs. Plated Through Hole Components

Aside from passive and active, components are also broken out as surface mount and plated-through-hole technology. Surface mount technology (SMT) is the vast majority of available parts. Plated through-hole (PTH) technology is still relevant to connectors for a stronger mechanical bond with the PCB.

Old-school dual-inline-package (DIP) types remain available. There are plenty of applications where they are still sufficient. The idea that you can take a DIP package and turn it upside down and glue it to the board is a "dead bug" solution. Wires are soldered to the legs and you're off and running. Those axial leaded through-hole resistors with their color code bands are a great tool for uncrossing the connections in a simple layout.

Makers and prototype designs can find PTH components beneficial. They are scaled to make rework and tinkering easier than with SMT components. Mass produced goods, on the other hand, rely on surface mount technology. The assembly machines are geared for the tight registration required to accurately place them on the PCB. The reflow ovens favor the SMT process as well. For mass production, it's advisable not to mix the two types.

In any case, PCB design requires a specific geometry for each different package type. While there is some standardization, there is also a proliferation of different packages. Crystal oscillators seem to have a slightly different size for every frequency. The package size is part of what determines the operating characteristics.



Section 2. Figure 2. Image 2.2. Credit: Author - Note the pitch of the through-hole component pins relative to the SMD resistor networks and the stacking connector.

Analog chips are also in their own world. They may be based on a JEDEC (1) standard but there are so many variables to encounter. Even “standard” ball grid array (BGA) packages will have a few random pins removed just because. What this means to the designer is that the component footprint library is never complete.

Managing all of the underlying data from the schematic and simulation to the layout and solid models takes someone (like you) who is on the ball.

2.5 Library Footprints - How to Create The Virtual Parts

A typical library part will establish the size and location for the solderable pads. PTH parts would also include a drill diameter that is sufficient to clear the pins. Both will require a solder mask opening, usually a little larger than the metal pad size. The exact expansion will be 50 to 100 microns beyond the overall pad size. There are cases where it is necessary to flip the geometry to use a mask size that is smaller than the metal. Solder mask defined pads are typical of fine pitch devices where there is insufficient space to lay down a workable solder dam between the pins.

The library footprint will also represent the actual size of the component for the Assembly Drawing. The outline may be drawn again a little further away from the part's actual outline for the silkscreen.

A placeholder for the reference designator would also be part of the assembly and silkscreen. An important part of any polarized component is to show the orientation of the part. It could be the pin-one, a positive terminal, a cathode or whatever feature determines the rotation.

2.6 Graduate School Zone

More data can be appended to the footprint including the package height, any keep-outs for connecting parts and paste stencil openings, step models, or anything else required by manufacturing. The origin of the individual components is usually the body center. This data is fed to the pick and place machine so it's a good idea to have a consistent method of defining components.

The PCB can be no better than the underlying footprint geometry. It pays dividends to faithfully recreate the information contained in the component vendor's data sheet and application notes. Most integrated circuits will have some specific information regarding the placement of passive devices that are critical to the function. Do not take existing footprints for granted if there is any chance that it isn't designed as it should be.

Component footprints can be generated (or bought) in three separate sizes; minimum, nominal, and maximum. Minimum size is useful for high density or where the capacitance of a larger pad is undesirable. Nominal is the general use case and maximum gets you higher reliability and enhances rework. In most cases, the default should be nominal.

The designer is coddled these days. Not sorry. The component vendors create footprints of their devices in an array of CAD formats. If they don't, then there are many services out there that will generate footprints for you. They'll even throw in 3D models so that the renderings look more realistic. You feed them your Bill of Materials and they give you models based on a few preset parameters. Change a parameter and generate a whole new library.

If you're lucky enough to have an in-house librarian, then they're the one(s) being coddled. The librarian always has a backstop for overflow demand. They (or you) still must check the geometry prior to use. Managing all of the underlying data from the schematic and simulation to the layout and solid models takes someone (like you) who is on the ball.

It can be tempting to use a mixture of minimum for RF chains and nominal elsewhere. That will confuse the assembly line. They can adjust the process to one size but not two or three sizes. The levers they can pull have to do with the size of the paste apertures and the profile of the soldering process. A certain amount of time is spent for ramping up the temperature followed by another period of dwell time at the reflow temperature.

A baker can make cookies or cakes but not both at the same time with good results. Soldering parts down is similar in that regard.

(1) Just like the IPC, the meaning of JEDEC has morphed. What used to be the Joint Electron Device Engineering Council is now known as the JEDEC Solid State Technology Association. They are the ones that produce mechanical outlines for common package types like SOIC (small outline integrated circuit) which is then appended with the number of pins and possibly other details.

Section 2. Figure 3. Image 2.3. Credit: Author - Passive surface-mount components make up the bulk of a typical PCB Assembly. That makes it important to take care of the little things. The three tapped holes are for a bracket that holds a heat spreader.

3. PCB Outlines

Section three discusses various methods of depanelization, V-score, and routing out the board outline with a mill or a laser.

3.1 PCB Outlines: Boards and Panels

3.2 Getting Ahead of the Tolerance Problem

3.3 Assembly Sub-Panels - Making Room for Manufacturing

3.4 Graduate School Zone

3.1 PCB Outlines: Boards and Panels

Before you can place the virtual components on the board, you need a virtual board. In addition to describing the outer limits of a PCB, the outline drawing will cover several things. Expect connector placement, mounting holes, keep-outs for mechanical hardware like heat pipes and RF shields.

When you get a new outline drawing, it pays to do a thorough evaluation. There may be an area where the PCB is overly constrained. The longer you wait to express your concern the less likely it is that there will be wiggle room. Once the interconnections that feed the board are drawn up, it may be too late to rotate a connector or move a mounting hole.

3.2 Getting Ahead of the Tolerance Problem

If the outline drawing includes any high-precision positional tolerances, thickness, flatness, or unilateral tolerances, it would be necessary to run all of these past the fabrication vendor(s). A lot of component vendors put unrealistic tolerances on their datasheets. The mechanical engineer or librarian would be inclined to pass these tolerances right along.

As the expert, it's up to you to confirm that the tight tolerances can be met in volume production. From my experience, the most common request that can't be met is regarding PCB thickness. A two layer PCB made of ceramic material has a chance of holding a tight thickness tolerance. Multilayer boards made of glass, resin copper along with various plating and coating layers will generally be held to 10% of the nominal thickness.

Wishful thinking in product design would like to have a better handle on this variable. Control of the Z-stack would be nice but a PCB is not like a block of aluminum. The dielectric breathes in moisture like a sponge. It expands and contracts with temperature. Even if the vendor "inspected in" the thickness by discarding all of the panels that didn't meet the tolerance, there's no guarantee that it wouldn't go into an altered state.

To some extent, this applies to warpage as well. While I'm confident in this statement, the mechanical engineer may not want to believe it. You may have gone down this road yourself. The thing to do is bring in the final arbitrator. The fab shop is just a phone call or an e-mail away. You still might get a "yes-man" answer from the sales team so ask the planner or the CAM engineer.

If you don't know them, ask for a tour of the factory and get to know them. If they're too remote, you may wind up doing a little bit of networking to get the unvarnished truth.

3.3 Assembly Sub-Panels - Making Room for Manufacturing

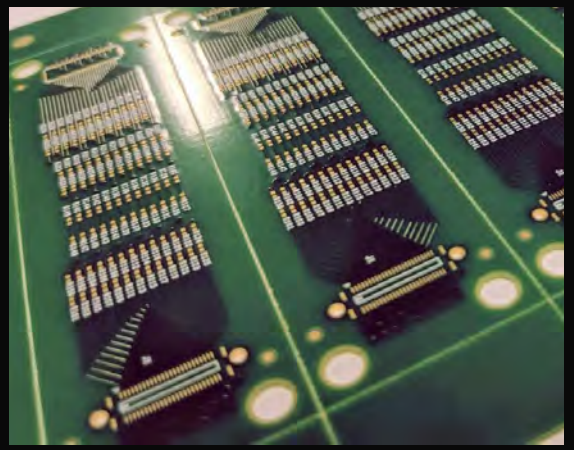
While you're contacting the vendors about the new outline criteria, it would be a good time to broach the subject of assembly sub-panels. Edge connectors have to be factored into the sub-panel so that they can be placed without depanelizing the individual boards. Break-off tabs that allow removal of the assembly may intrude into the board a little. Knowing where they will go allows you to accommodate them. The end-game is probably the worst time to negotiate details that could have been solved earlier. The fewer surprises, the better.

For the simplest board outlines, it may be possible to design the assembly sub-panel yourself. One such situation may be when the outline is a rectangle or a square. Those are often the ones where there was no outline drawing or other mechanical engineering support.

Using V-score grooves eliminates the need to calculate how many break-off tabs would be sufficient.

To design the sub-panel in-house, you would need to know the upper and lower size limits of the assembly machines. There are a few popular sizes for tooling holes that you'd want to use. The rails that support the board have a minimum width. I hesitate to put actual numbers to these features because no two assembly houses are alike and they usually get better over time.

Depending on the format of the outline drawing, you may have to interpret some things. On day one, it's rare to see a completed drawing. More likely the input is a file that you can import and maybe a screenshot of the PCB in its native habitat.

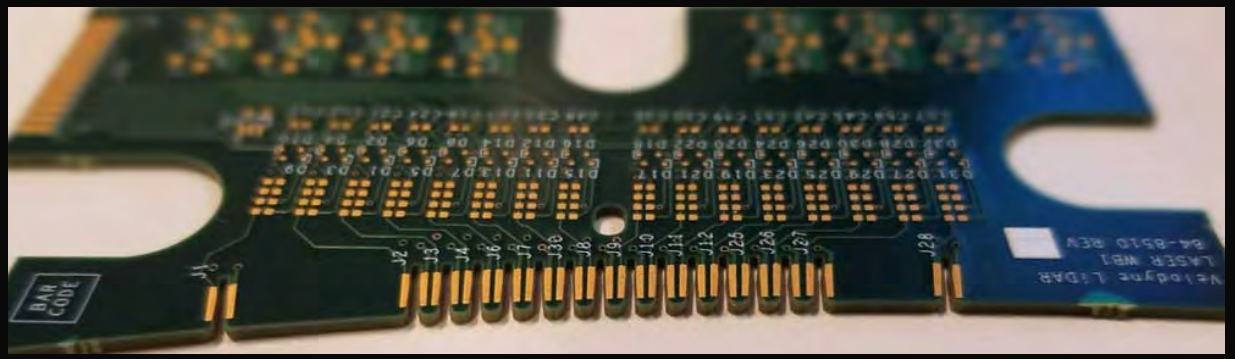


Section 3. Figure 1. Image 3.1. Credit: Author - A rectangular PCB that is panelized with V-score grooves so it can be depanized by hand as the unneeded sections can be snapped right off.

The end-game is probably the worst time to negotiate details that could have been solved earlier. The fewer surprises, the better.

3.4 Graduate School Zone

The layers used for data transfer don't always make sense. For instance, the component keep-out area is often used to indicate the location of a feature like a connector. No part goes there except for the part that goes there! That's a loop you'll want to close with the physical designer.



Section 3. Figure 2. Image 3.2. Credit: Credit: Author - The outline can be the most challenging aspect for the fabricator. Without the little grooves, this six-layer board would have been half the price.

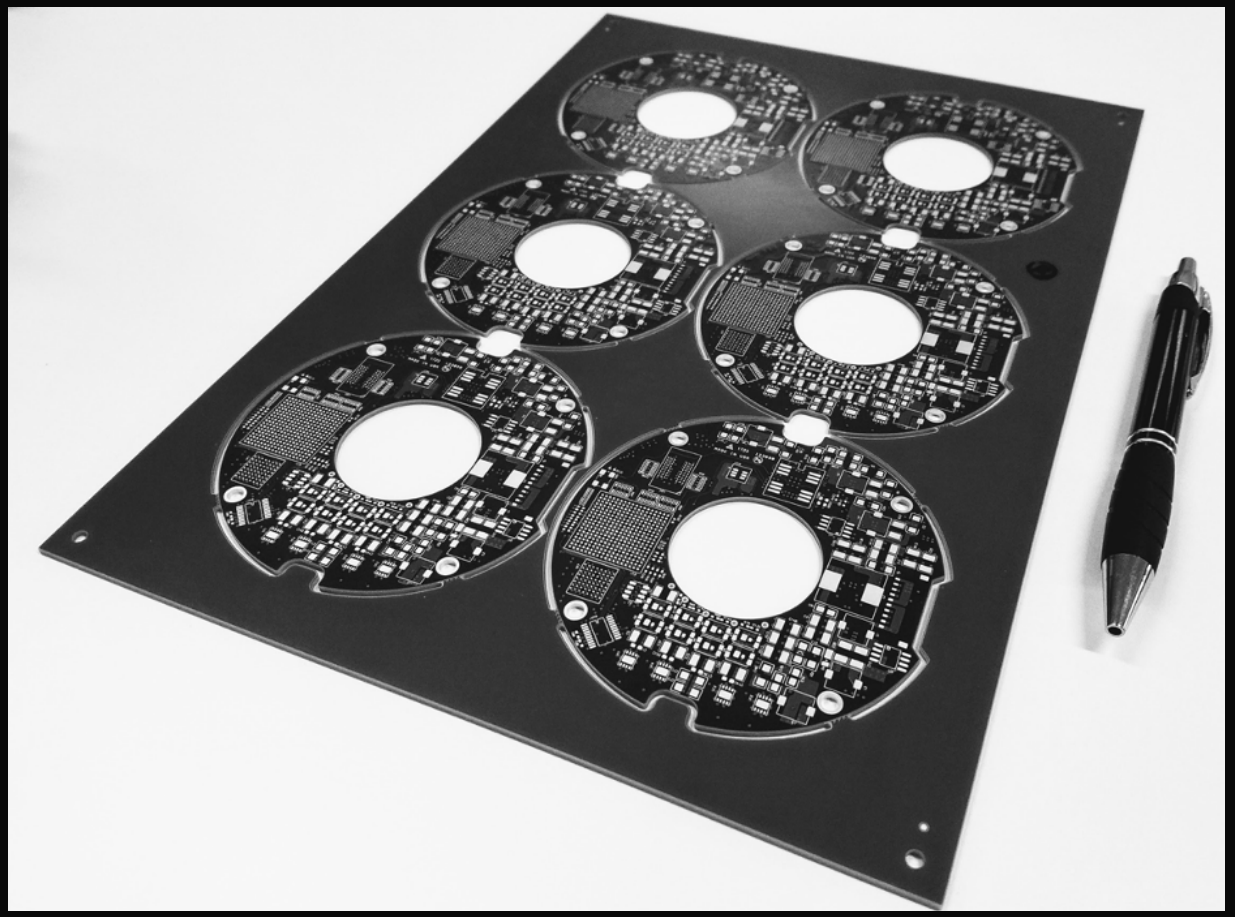
Their software supports a particular set of layers that may not be optimal, particularly for flex circuits where there are additional layers to consider.

The distance from the edge of the PCB to the copper features is an important factor. A pull-back of 200 to 250 microns is typical. I took a walk around a PCB Design conference and asked every fab shop what the most aggressive board edge-to-copper clearance was possible. The common answer among the advanced vendors was 127 microns.

I needed something more extreme. There weren't many fabricators who were willing to use lasers to form the edges of the PCBs. The ones who could insist on a maximum PCB thickness of 400 microns. There was a common issue with carbon deposits on the edges. Those were some hairy boards and a significant cost driver.

Part of the reason for the $70,000 price tag on the flagship product was the number of boards that had to be cut out with lasers.

The gap between the metal and the board edge can be reduced to zero in two ways. The most obvious is wrapping the plating around the edge. Another is so-called castellated vias. The castellation is going to be larger than a typical via and will be drilled right on the edge of the board before the board outline is routed. What is left is a half-via that can be used to connect inner layers as well as the outer layers.



Section 3. Figure 3. Image 3.3. Credit: Credit: Author - Subpanels are necessary for oddly shaped PCBs and any time the components are close to the board edge.

4. Component Placement

Section four goes over some rules of thumb and steps to consider for robust component placement on your board.

4.1 Component Placement

4.2 Driving the Component Placement With the Schematic Diagram

4.3 Verifying the Logic

4.4 Floorplanning - Like Organizing a City Into Neighborhoods

4.5 Analog Component Placement - How to RF

4.6 Graduate School Zone

4.1 Component Placement

If your aim is to become a full-stack developer, you're going to be capturing the logic in a schematic diagram. Excuse me while I back-track a little bit. Most tutorials would make this chapter one but here we go. When I first started doing the layout, the electrical engineers handed over a hand-drawn sketch and the CAD people generated all of the symbols, married them to the footprints, and created the whole schematic from there. It was a chance to decompress from the previous board while getting familiar with the next one.

Then, schematic capture on CAD tools became a part of the EE curriculum and the task became their job. Time marches on and now, a lot of PCB Designer opportunities are asking for this skill once again. Some are also asking for us to do 3D mechanical engineering as well.

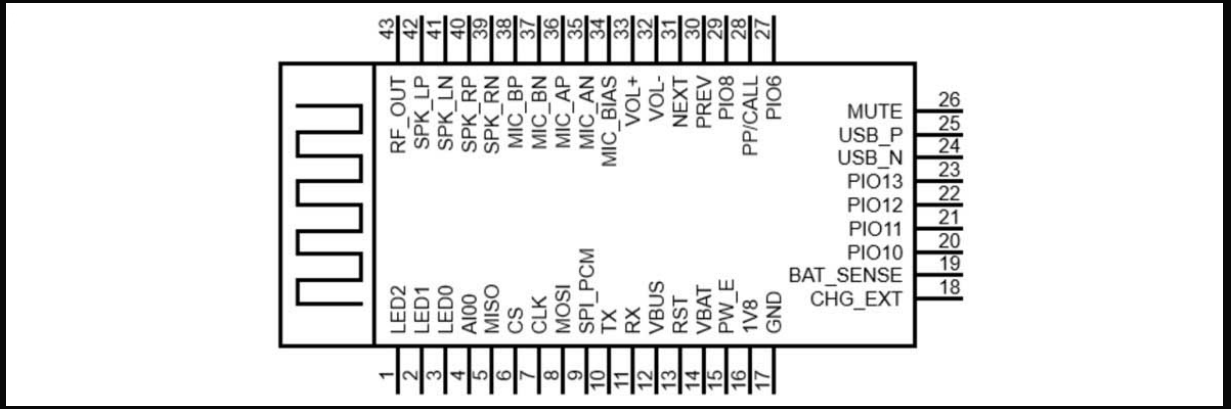
Welcome to the renaissance! A degree of some kind is more than likely to be desired as we go forward.

4.2 Driving the Component Placement With the Schematic Diagram

Whatever the case, the schematic is where we develop the connections between devices. It is also a potential place to set up some of the design guidance. For instance, components may be grouped into functional sections that are kept in close proximity and also kept separated from other circuits.

The placement guidance may be based on what goes inside of an EMI shield or clustered around the primary devices. This can be in the form of attributes assigned to individual components or simply rectangles around sections or entire pages of the schematic with a label describing the properties and physical locations around the board.

In addition to driving placement, general routing rules may also be captured in one way or another.



Section 4. Figure 1. Image 4.1. Credit: Bois83 - A typical schematic symbol for an analog chip.

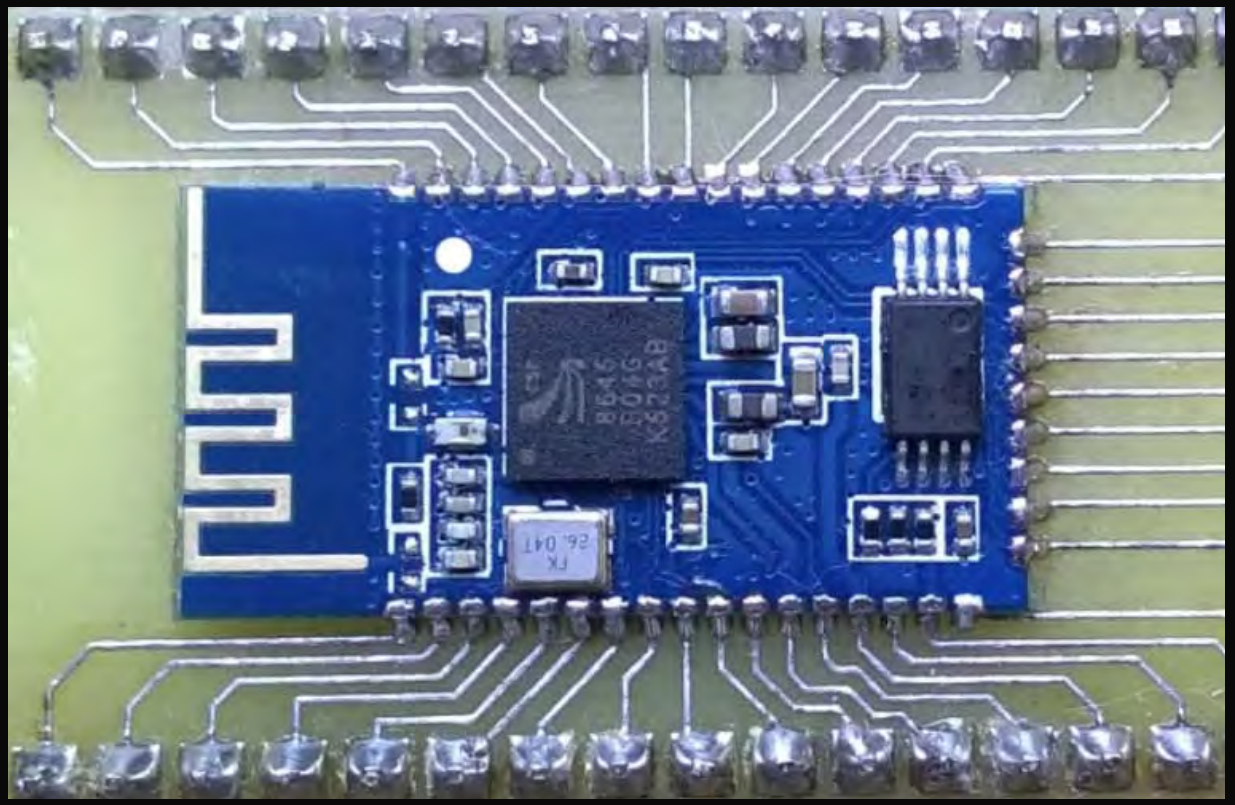
While we may not know the actual geometry of a transmission line at this stage, we can at least assign the impedance requirements to the nets so that they may be considered during placement. When you're putting this additional information into the schematic in the form of a local note, it's good to have a consistent approach that makes all of the information of this type searchable.

Begin each piece of data with the words "Layout Note" or something similar.

Busses of all types can be established by the use of common net names. The actual length matching requirements may also be added but this is often left to the layout.

The main idea is to give the general framework of the board and allow the designer to proceed from there. That's not to say that it can't be taken further to the point of establishing the location of each bypass capacitor by calling out the associated reference designator and pin number of the device pins next to each cap.

A block diagram is useful for establishing the floorplanning and placement parameters from a high level. If the board is complex enough to warrant a block diagram, chances are good that a power tree will also come in handy. A power tree begins with the root voltage at the top and spreads out through the various power supplies and voltage dividers down to each domain. The purpose of each voltage node is included along with the expected current consumption.



Section 4. Figure 2. Image 4.2. Credit: Bois83 - Layout of the analog chip, see figure 1.

If you're going to add only one extra piece of intelligence, the power tree would usually be the most useful to capture for posterity.

In addition to providing the high-level layout information, the Bill of Materials and the net-list are both extracted from the schematic. This is, after all, the intrinsic purpose of the schematic diagram. It's easy to make little mistakes that can affect the output.

Most mistakes are related to the misuse of a grid when generating or placing symbols and wires. Even if you're not capturing the logic, here is where you can save the day, so to speak.

Wise engineers will want to capture their instructions for posterity.

4.3 Verifying the Logic

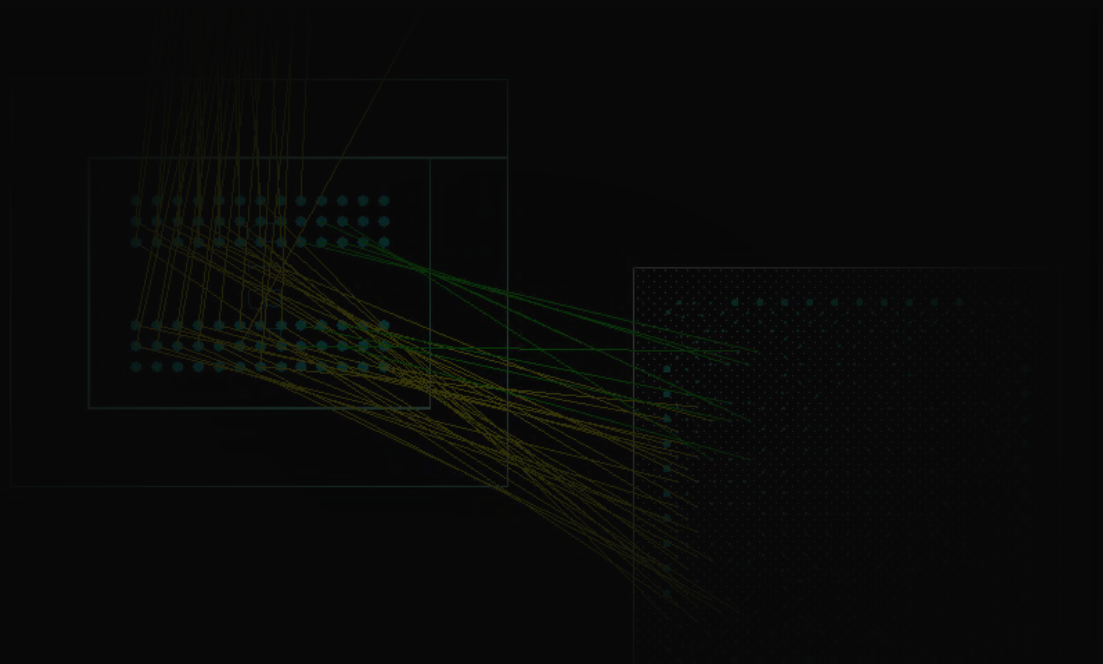
The most common problem I see is a wire just missing the pin. It looks connected but is not. Any two-pin component where only one pin is connected is a suspect. The second most common issue is a typo in a net name. Two nodes with identical net names except one has CLK and the other has TCK in the name should be reported back to the person who captured the schematic, even if you're the one.

Always go through the pages of the schematic prior to digging into the layout. Wise engineers will want to capture their instructions for posterity. You may find out that two resistors are optional and that their common pins should overlap to create a "poor man's switch" with three pins made from four. Select-at-test component values or the use of zero ohm resistors are earmarks of a well-thought-out schematic.

Take as much context as you can from the presentation, especially with analog circuits. Did I mention that I'm skipping the part about schematics? Let's get on with placement.

4.4 Floorplanning - Like Organizing a City Into Neighborhoods

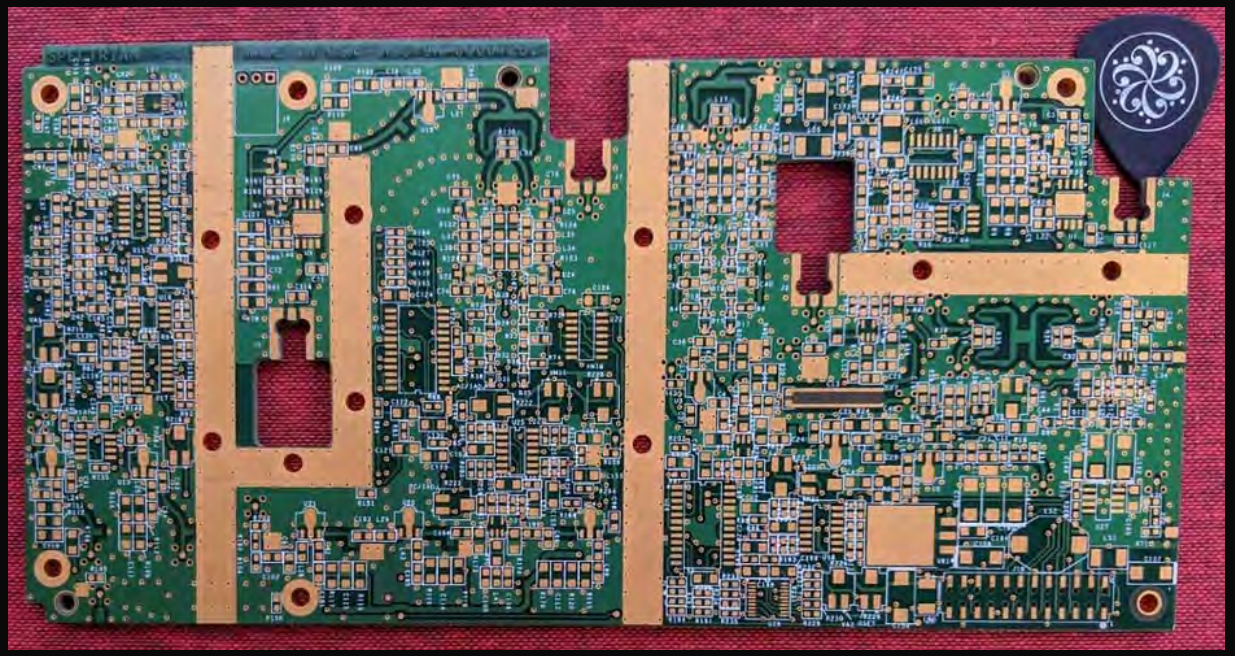
Floorplanning starts with one part. Which part will depend on the nature of the board. A digital board is likely to have a chip or a chip-set that dominates the landscape. The designer would be well served to work from the center outward. Make good use of the link-budget to prioritize which peripheral components get to be near the main device.



Memory devices usually get the first priority. DDR comes before flash. Crystals and bypass capacitors want to be kept close to their respective pins. The power to the bias pins that makes all of the switching happen comes directly from the bypass caps. The power supply's job is to recharge those caps. Even so, power supplies work best when they are placed near their loads.

4.5 Analog Component Placement - How to RF

It's really only the odd sensor or radio chip that doesn't want to be close to the processor. Sensors and antennas thrive in the boondocks. The PCB edge isn't always the quietest location but a good place to start. Context matters. The optimum location for the analog circuits depends on the rest of the chip-set.



Section 4. Figure 3. Image 4.3. Credit: Author - Analog boards like good fences. This was a rapid development system where the wide strips were for walls that could be screwed into the housing right through the PCB.

It's not unusual for an RF device to be housed inside a faraday cage. Designing each analog circuit as a compact rectangle with room around each circuit for shield walls gives you that option. RF connectors are a special breed with a single signal pin surrounded by ground pins. The next part in the chain is usually a blocking capacitor.

One of the most common features in an analog circuit is a pi pad.

It takes three elements to make one; a shunt resistor then a series resistor and then another shunt resistor.

When you see that combination on a schematic, keep the group packed together.

Op amps are another common circuit. The key to a good op amp circuit is the feedback resistor which straddles the negative side of the input and output pin. That loop must be kept short.

The other short loop will be the bypass or decoupling capacitor; sometimes called a decoupling cap. That one goes from the voltage pin to the ground. Assuming surface mount components, routing the bypass cap starts at the power via with a wide trace that passes through the bypass cap and into the power pin. Via-in-pad is also acceptable but the via goes on the cap's pin rather than the voltage pin of the device.

I went into specifics about the routing from the cap to the power pin because stubs can be a problem.

They can also be a solution to a problem. Adding a little bit of extra metal along an RF transmission line can be used for impedance matching. There can be more than one of these wide spots carefully placed along the trace.

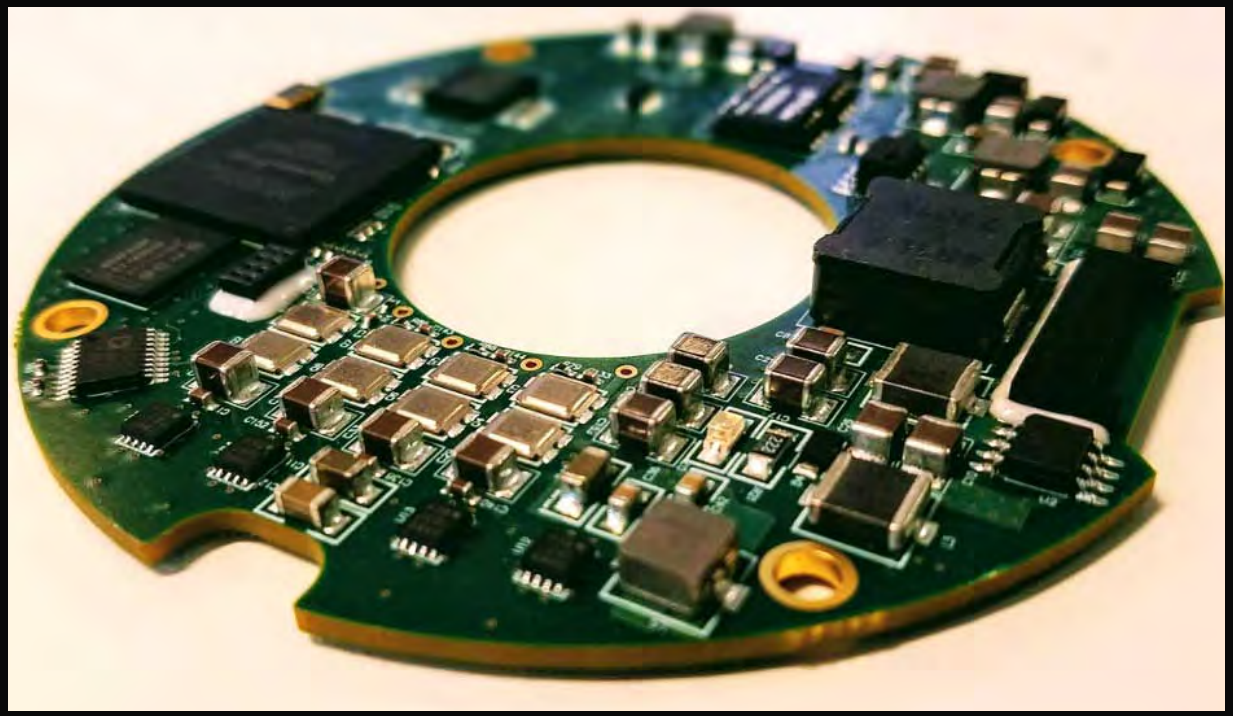
Components are not attached to the stub but you might want to consider it a component. The footprint would be a single pin with no associated part sort of like a test point except with a shape instead of a circle. Fine-tune the radio may require lengthening or shortening the stub.

Little slivers of extra metal are left floating at the end of the stub so the tech can create a solder bridge or use a metal tab to solder it from the stub to the extension.

4.6 Graduate School Zone

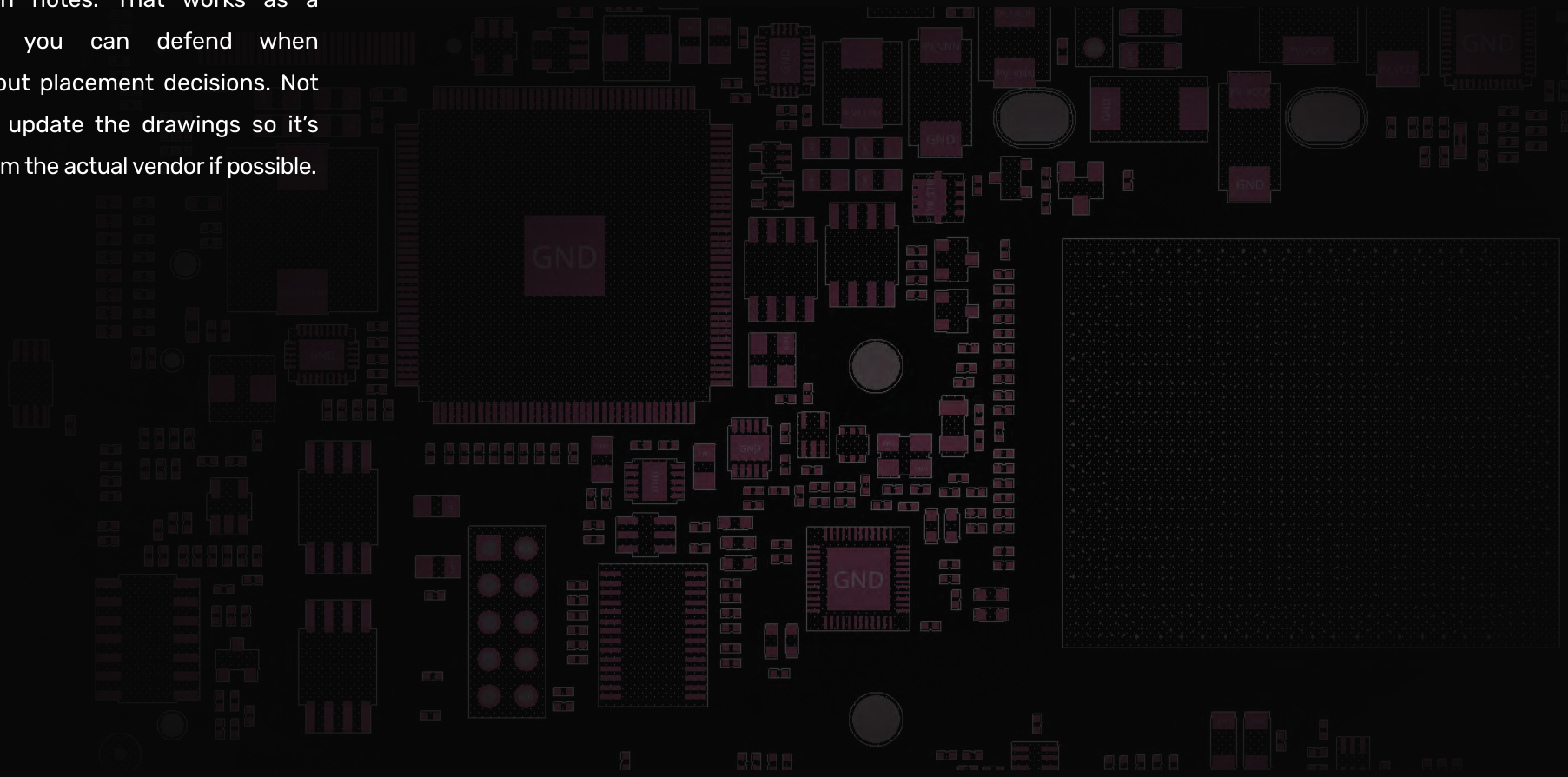
Hardworking components have high junction temperatures. Spreading them out helps dissipate the thermal energy that they generate. Heat is detrimental to reliability and concentrated hot spots will throttle the system until it no longer works as intended. Space that is plated with ground shapes that are perforated with vias are a hot component's best friend.

Harsh environments and Chip-On-Board applications will benefit from underfill. It is best to have two or more sides accessible for the underfill nozzle. Reworking BGA footprints requires clearance on all four sides for the tool. Either way, this area can be set aside as part of the footprint's courtyard.



Section 4. Figure 4. Image 4.4. Credit: Author - Oversized SMT pads and underfilled devices are a characteristic of high-reliability PCBs. A bead of adhesive around the connectors is another insurance policy.

Every PCB is going to be different. Getting off to a good start involves going online and browsing for the devices to find data sheets and application notes. That works as a baseline that you can defend when questioned about placement decisions. Not all distributors update the drawings so it's best to get it from the actual vendor if possible.



5. Fan Out Studies

Section five covers fanout in the context of high-speed differential pairs and miniaturization of components.

5.1 Fan Out Studies

5.2 Miniaturization Effects on PCB Design

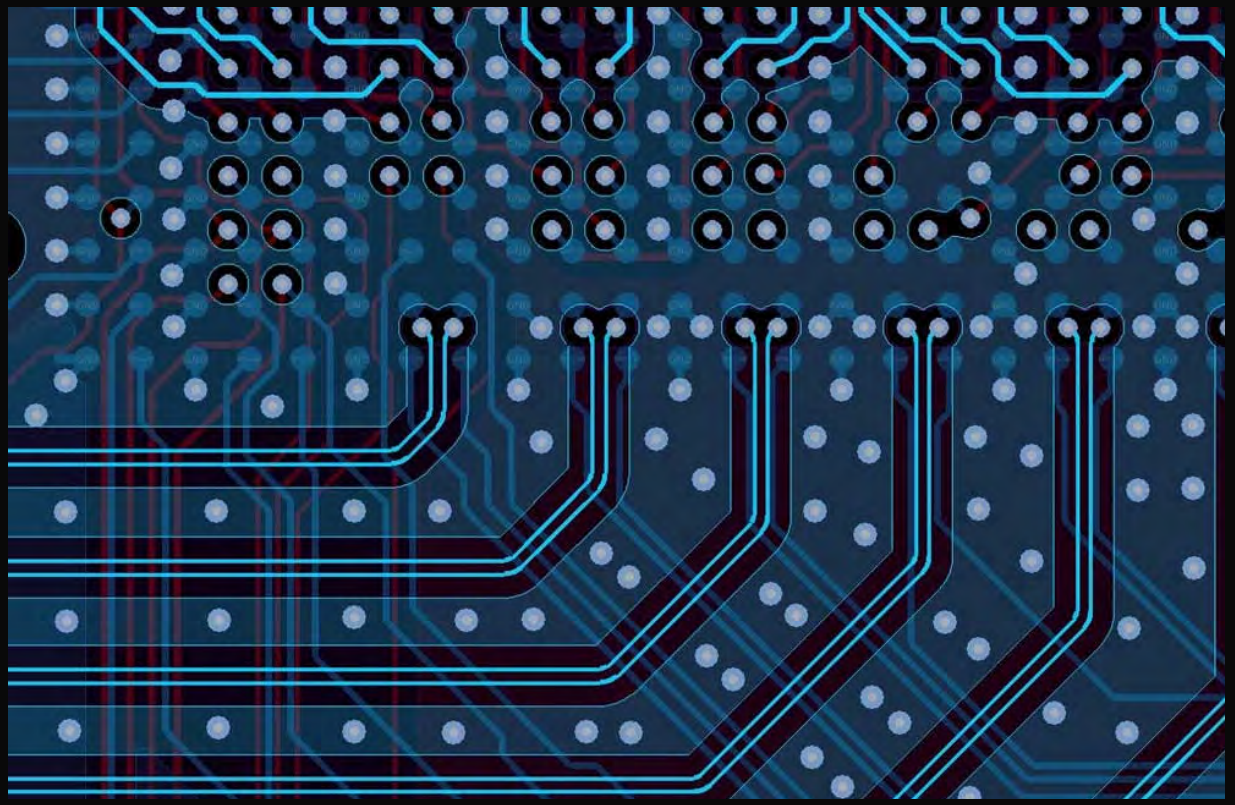
5.3 Shaping Up the Power Planes

5.4 Graduate School Zone

5.1 Fan Out Studies

Determining the layer-count and the via construction are the main goals of a fan out a study. The same device that was at the root of the floorplanning study is likely to be the one that drives the layer count and overall board construction. Those worst-case geometry components are sometimes the only device of that type that drive the rest of the criteria including PCB price and lead-time. After solving for that one pain point, the rest of the layout is relatively easy.

The easy ones are traps! Remember the words, “Fan OUT” and add vias accordingly. A well-planned fanout will get every signal to pass through the ground plane in its own void in the copper. When two vias share the same hole through the copper plane, there may be magnetic coupling that affects the integrity of the circuit.



Section 5. Figure 1. Image 5.1. Credit: Author - Isolated differential pairs fanning out from a connector.

Fine-pitch connectors and quad-packs need some breathing room for the fact that the vias will be larger than the pins. Plan for it during placement.

High-speed differential pairs like to be routed on an inner layer to reduce spectral emissions. A copper flood of ground further isolates each pair. Note that the signal vias are gathered towards one another because that's what the pin pitch of the connector allows. Other signals and power/ground connections follow a more familiar BGA fan-out.

The top layer, shown in blue is used to reduce the via-count by routing on the primary component layer.

This opens up the channels for the second row up to where all of the high-speed differential pairs are launched on Layer-3 shown in cyan.

Furthermore, the third row and all subsequent rows fan-out in an upward direction to give more space to the diff-pairs.

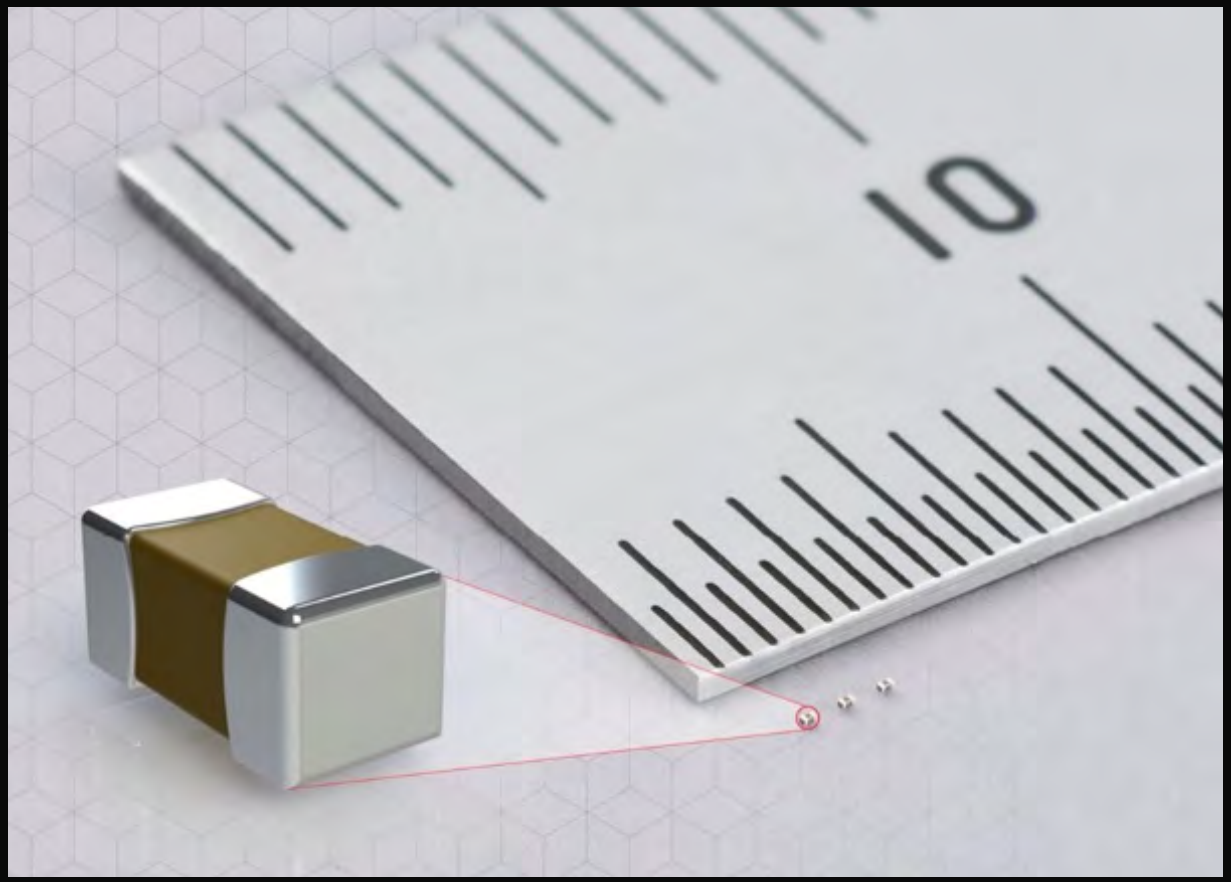
The exception to the "fan-up" is the ground vias. Not only do they fan-down, but some of the ground pins on the second row also have two ground-vias so that there is one ground-via near each high-speed signal pin. The ground pins on the outer row fan-out straight down rather than at an angle. This allows access to pins in the interior of the connector without using a via.

5.2 Miniaturization Effects on PCB Design

The never-ending trend is for smaller components. The main goal of the component package design is to spread out the chip's connections to a more reasonable pitch. Generation after generation, the pitch decreases while the pin-count goes up.

This has continued to the point where the Chip Scale Package (CSP) has become a normal technology.

CSP is defined by the ratio of the silicon to the package. To begin with, the ratio was set at 1.2:1 where the whole package was only nominally larger than the die itself. Those package sizes persist while the dice have gone through further shrinkage nodes down into the single-digit nanometer scale for individual features.



Section 5. Figure 2. Image 5.2. Credit: Murata - Capacitors are shrinking too. 0201 packages are common, 01005 packages have been around for almost 10 years. Introducing the 008005 as the next tiny thing. Translated, that's 10 mils long and 5 mils.

Chip Scale Packaging is, by definition, High Density Interconnect where micro-vias are the primary inter-layer conduit. The two-layer substrate that houses the chip is quick to design and fabricate. The resulting component package is small. That's good for Marketing but not so great when you're trying to design a robust PCB.

The PCB Designer eventually becomes responsible for the job traditionally handled by the substrate. The silicon die type known as a flip chip will be attached to a substrate and the connections will be passed through a via to the SMD pads.

The alternative to a flip chip is die-attach done having the copper slug attached to the bottom of the die. The solid metal base is reflowed inside a cavity in the substrate.

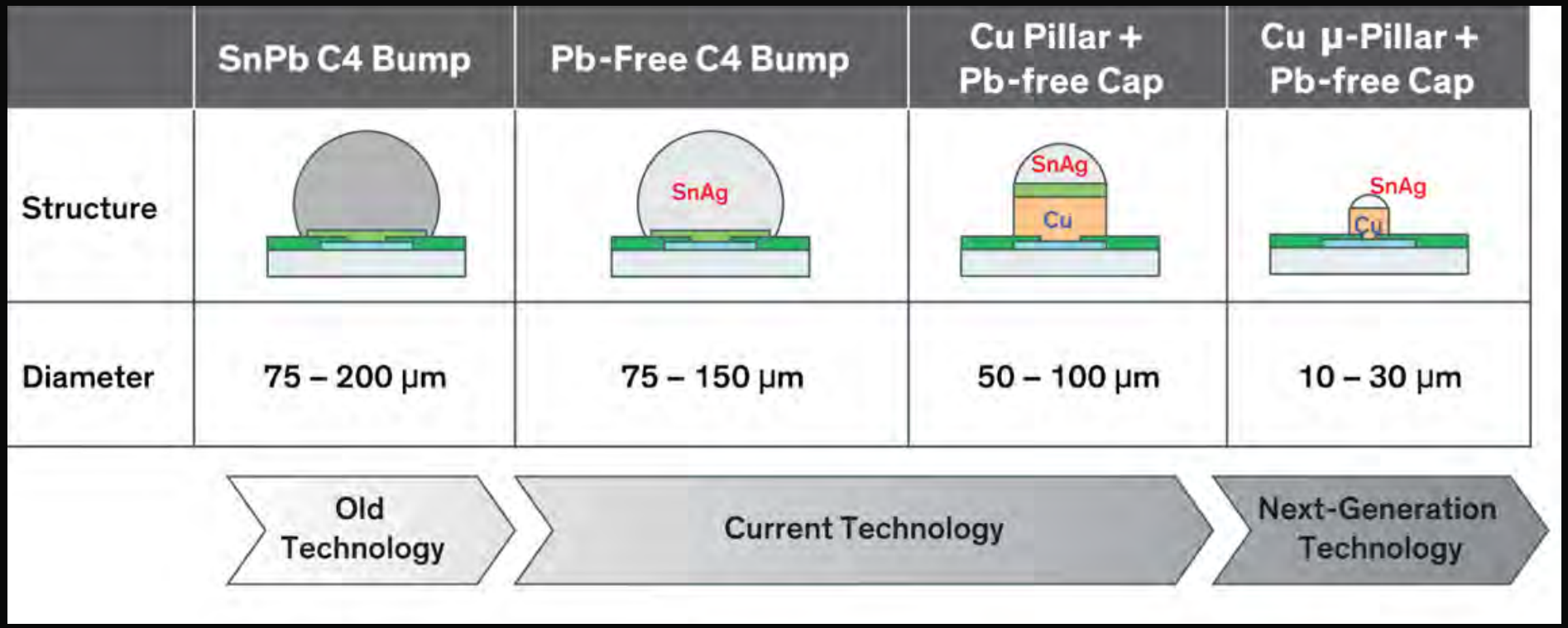
Then, wire bonds are used to connect the upward-facing pins of the chip to the substrate. The only expansion of the pin-pitch would come from the wirebond cage in that situation.

Even when the chip vendor uses an interposer to spread out the pins of a flip-chip, the results may require High Density Interconnect to fan-out. HDI is an expensive and time consuming process. (1) A board could have twenty devices with only one of them being too fine-pitched to get done with plated though-hole vias.

"...the hot new technology is copper pillars in place of the balls of solder".

Chip makers continue to push the limits of HDI technology. Leading edge ball grid array (BGA) packages have a pitch of 350 microns; not quite 14 mils center to center. At that rate the pads will be solder mask defined. The metal pad size defined by the smallest via the fab shop can produce.

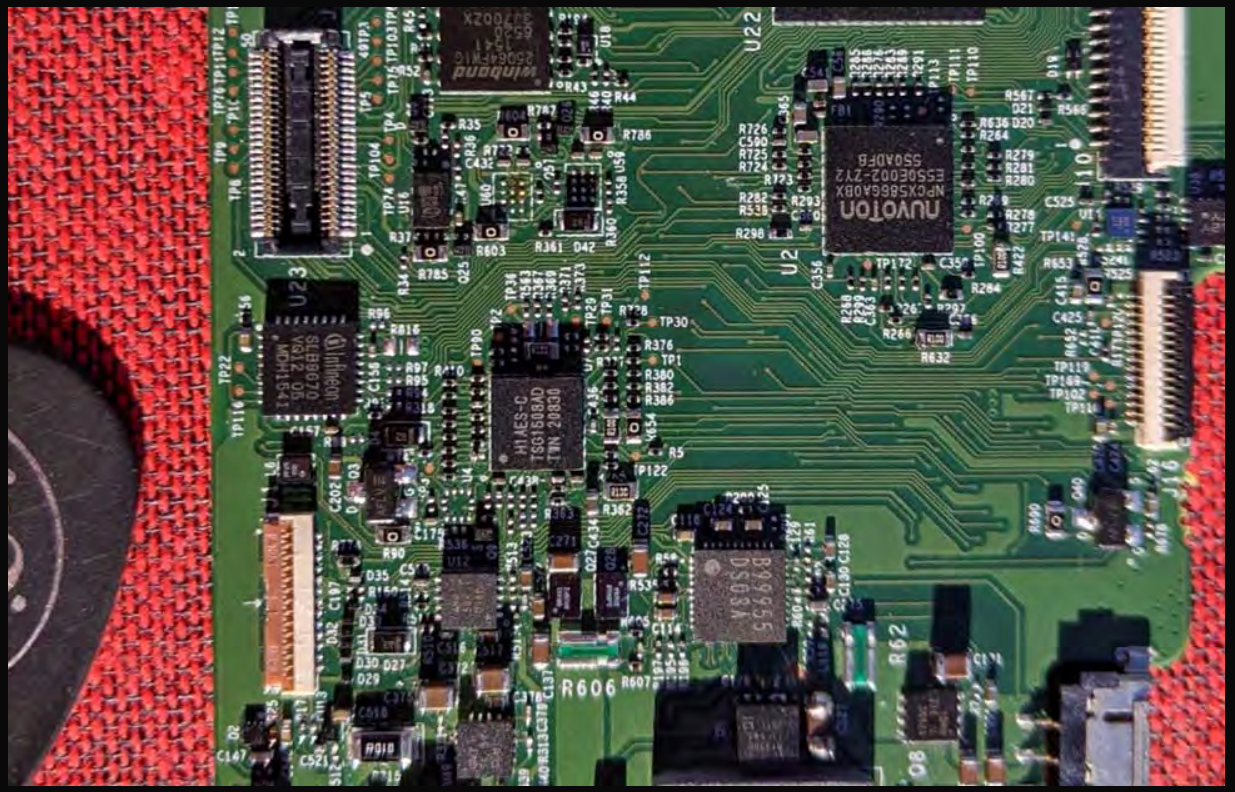
The fab shops have to get better at it. Shrinking the package further still, the hot new technology is copper pillars in place of the balls of solder on a standard flip-chip. The via-in-pad is the foundation for all of these technologies. It's the light blue part of figure 2 that remains mostly the same in spite of the shrinking pins.



Section 5. Figure 3. Image 5.3. Credit: Dupont - It seems like every time the industry moves to the next smaller node, the burden is on the PCB design and fabrication people to solve the technical challenges.

According to Dupont, this maturing technology allows the substrate packaging to be reduced by as much as 300%. What are you going to do when the pin pitch gets down to 135 microns? That's pretty ridiculous. Meanwhile, Murata is making 01005 size chip caps look huge with their 008004 packages.

More than likely that one device is going to drive extra lamination cycles to build up the number of micro-vias while the rest can be done at a lower cost. Each additional layer of micro-vias adds a measurable percentage to the cost. I'm driving this home because we keep score with money and the PCB can be a substantial part of the Bill Of Materials cost.



Section 5. Figure 4. Image 5.4. Credit: Author - A few examples of QFN and QFP packages. In this six-layer HDI stack-up, the primary component layer is also the horizontal routing layer.

These cost-driving components are typically ball grid array (BGA) packages. A quad-flat-pack-no-lead (QFN) package can have the same 0.5 mm or 0.4 mm pin-pitch and not require an HDI stack-up. A little communication with the stakeholders including management and procurement might be enough to change the value proposition. If nothing else, it lets them know that you have your head in the right place. If you have time to do a little searching to see if there is a viable substitute, so much the better.

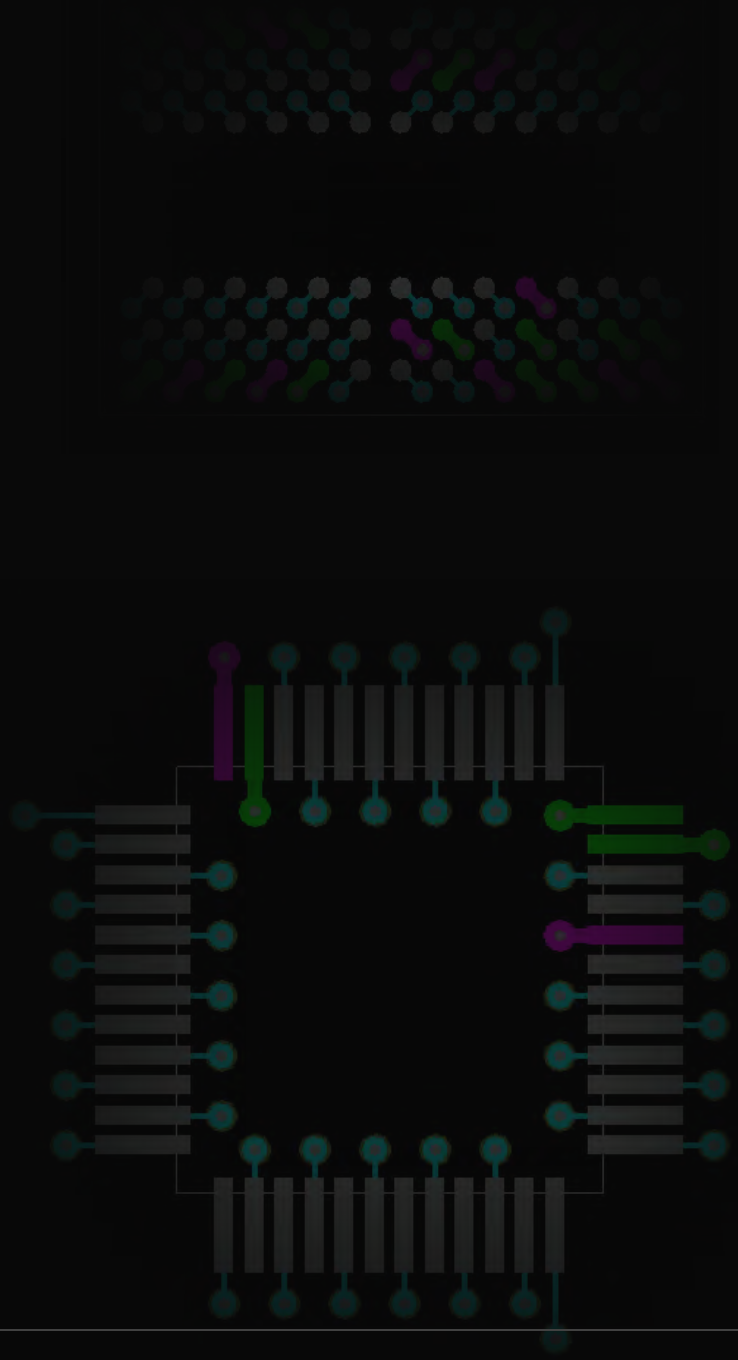
Remember the words, “Fan OUT” and add vias accordingly.

Fan-out studies do not require every component to be placed on the board right away. The challenging device(s) can be done in isolation. Since the traces are not routing away from the cluster of components that complete the chip, the whole thing can be moved without much effort.

One method that I’ve found successful is to place each page in its own off-board area. As each page is complete, draw a rectangle around the parts and label them with the schematic page number and the circuit description that is usually provided on each page.

It’s not that much work to present the components as they are organized in the schematic. Floor planning is easier for those who are also involved when the circuits are identified.

Use a text size that is still visible even when the whole board space is on the screen; just a few words for a brief description. Getting all of those chips organized into their optimum placement allows you to explore the connections between each function with higher confidence. Priority will go to the length of matched groups and any controlled impedance nets.



5.3 Shaping Up the Power Planes

Power distribution starts with the floor planning and gets proven out during the fan-out study. It may take some iterating to find the most advantageous power plan. Making things better along the way is a central part of PCB design. That's why we want to do a fan-out study before knitting everything together.

Colorizing various voltage domains is important since you want to keep power planes tidy. If you have a power tree, then assigning colors that are meaningful is a great idea. Using shades of deep purple all the way to pink to show relative power requirements is my favorite. If you also happen to have various ground domains, try shades of green for the cleaner ground and brown when it gets to the dirty digital stuff.

Sorting the component clusters by their return path is every bit as important as keeping like voltages together.

Generally speaking, high voltage requires extra attention when it comes to spacing. It's the amps and milliamps that matter in terms of the width of the trace or shape. Current capacity in tight areas can be augmented by going vertical. Use another shape on an additional layer or more layers until the total copper weight is sufficient to manage temperature rise. Make sure the shapes are tied together with multiple vias at the source and the loads.

A power-management integrated circuit (PMIC) will have a concentration of current with all of the output pins around the edges.

The area between the output pins and the inductors are particularly problematic when it comes to noise. The size of the inductors compared to the pitch of the pins creates a situation where the shapes in between will look like a fan as they spread out from the pins.

5.4 Graduate School Zone

The pain-points come into a clearer view as the power shapes are fleshed out. One main goal of the fan-out study is to reduce churn during the routing phase. If you are fortunate enough to have a power integrity engineer in-house, getting their opinion, or better yet, simulation results is a valuable step. The larger the power-tree, the more this type of interaction becomes necessary.

Power integrity is the main source of drama in mobile hardware. Correcting the power distribution network will solve issues with battery life, thermal degradation, and overall performance. Most every design will have some compromises. Giving the devices clean and plentiful power will help cover for other shortcomings.

Given the trend towards miniaturization, we're going to have to work harder to manage the junction temperatures from the silicon outward. There will be a little less current to pump out but in a lot smaller space. Adding the higher speeds of the next generation of devices to the high current density situation only compounds the problem. We want "more" so there's no going back. Look forward to lots of simulation and iteration. Here it comes.

(1) HDI is a construction method that allows for smaller vias that traverse one layer-pair at a time. It makes it possible to use via-in-pad without too much extra work. It is a game-changer when you don't have to allow extra room for fan-out vias during placement.

The traces start on an inner layer that is best suited for the type of signal. Note the stack-up diagram at the end of Section one.

6. Digital Routing

Section six dives into digital routing constraints with concrete examples and anecdotes on low versus high frequency sound waves.

6.1 Digital Routing

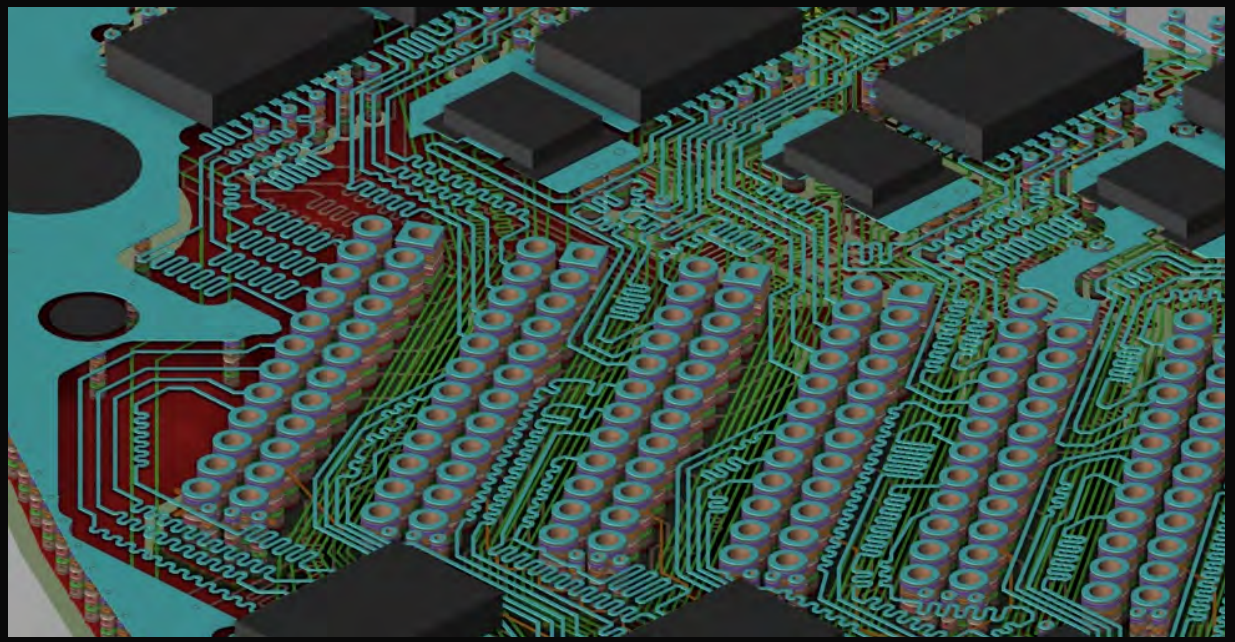
6.2 Differential pairs routed on a 12-Layer PCB

6.3 Graduate School Zone

6.1 Digital Routing

There's routing and then there's "routing." Before any of that begins, it is necessary to identify any connections that require special handling. As I outlined in the chapter on fan-out, there's a progression of interfaces to consider. A group of nets may have a maximum length and would also be tightly matched within the group.

From there, other groups may also be matched to that group but with a little more slop allowed between groups. You'll see that a lot in Double Data Rate (DDR) memory. Address busses and data busses will be broken into byte lanes that have eight individual traces with a clock or a strobe pair that sets the target length.



Section 6. Figure 1. Image 6.1. Credit: Author - Single ended transmission lines using tight serpentine to sync up a LiDAR sensor.

6.2 Differential Pairs - The Basis For Lowering Noise

Let's unpack those groups. The most common scenario is a group of two nets we call a differential pair. A bit of electronic magic goes on when two lines are routed together vs. a single line. A single line carrying a stream of ones and zeros, actually just jumping between having a voltage present and not having a voltage will encounter other traces and shapes along its path. Those traces all want to share their noise conditions.

More exposure over longer lengths of trace lead to more crosstalk. Higher voltages extend the reach of the coupling effect. By the time the trace reaches its destination the journey may have added or subtracted enough stray voltage that the receiving end may not be able to distinguish a high state from a low state.

The noise outweighed the signal and the receiver had to say, "What?" Then the sender sends the message again.

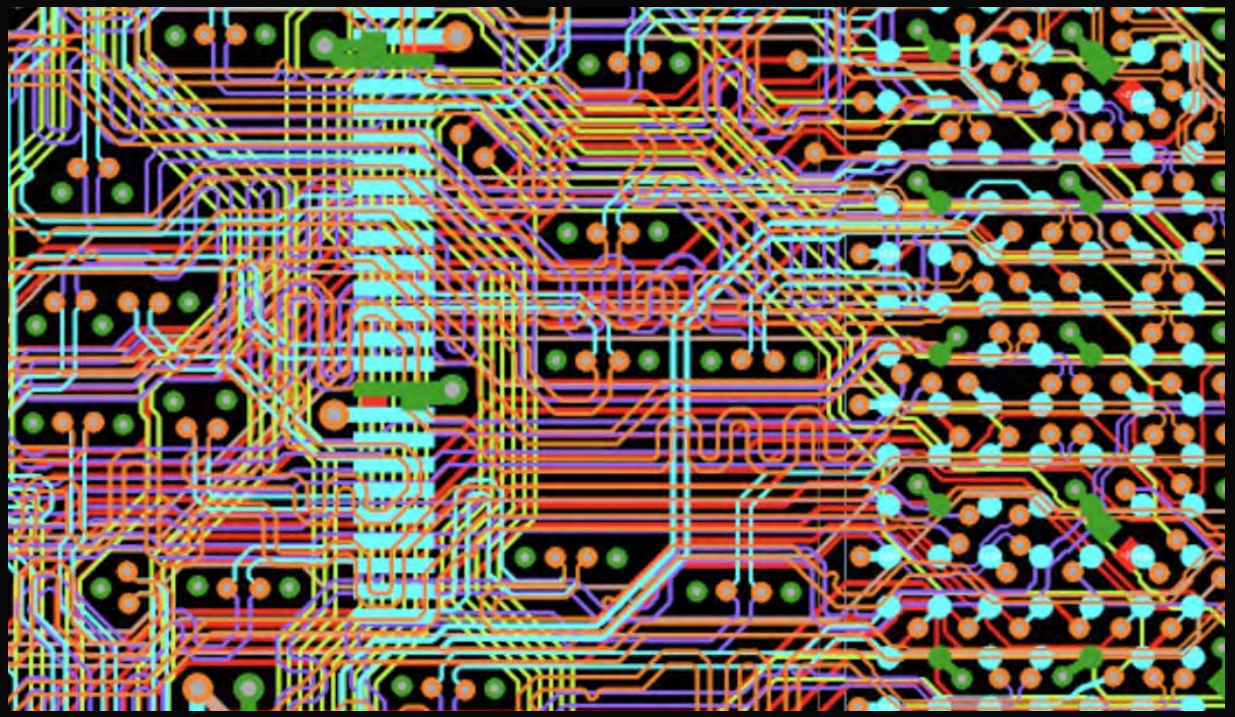
It's acceptable for that to happen occasionally. There are usually a few clock cycles to spare. If it happens too often, the whole system breaks down. Every dropped packet of data increases the bit-error-rate (BER) and the system can only tolerate so many faults. Video can stutter, a call is dropped, the program crashes, the "Blue Screen of Death" ensues. Long high speed connections are the most susceptible to failure due to line losses and noise.

Enter differential pairs: They have much more immunity to the world around them because of the redundancy.

Equal but opposite signals travel along the two paths and the receiver can compare the two signals and filter out the common noise. What's left is a better representation of the original bit stream. Two is better than one when it comes to faster and longer communication across the board.

Note in Figure 2 that the primary component layer is colored in cyan with magenta and orange on the inner routing layers. The red traces are single-ended and routed on the secondary component side. The gaps between the differential traces are equal to three times the line width. The FPGA (1) on the right side is fanned out with the diff-pairs gathered towards each other. Some pin-pairs are horizontal, some vertical, and others are diagonal in their orientation between the positive and negative terminals. The ground vias (green) tend to lean in towards the pairs.

Over on the QFP side, the signal vias are closely spaced and the ground vias are placed near the transitions. Ideally, all of the foursomes of signal and ground vias would be identical.



Section 6. Figure 2. Image 6.2. Credit: Author - Differential pairs routed on a 12-Layer PCB.

The threesome on the lower-left-hand corner is a compromise. Halfway up that side, there are two pairs sharing three ground vias; another compromise.

When there is symmetry, the ground vias should be close to the signal pins. When they are asymmetric, it's better to space the ground vias out a little.

The sheer amount of horizontal routing drove the decision to place the ground vias in parallel with the signal vias. It was during the actual routing and tuning when the compromises became necessary.

Differential pairs perform at their best if their lengths are closely matched. We use what I call "speed-bumps" where one of the two traces makes a little jog to compensate for any length mismatch.

In most systems, all that matters is the overall length and the phase-matching speed bumps can go anywhere along the lines.

In some cases, a turn in the two lines will get the waveforms out of sync enough to matter. At this point, we introduce a speed-bump to phase tune the pair at every corner. This is known as dynamic phase matching. A tight S-turn might be a wash but we're aiming to keep the two signals running neck-and-neck for the entire flight. In this way, any transient voltage spike that washes over the layout will hit the two traces at the exact same time allowing the receiver to do its cancellation trick.

Imagine the interface of a camera when, all of a sudden, there's a flash going off.

The camera modules are usually located at a distance from the centrally located processor while they're pretty close to the flash unit. The power for the flash unit does not have to be right next to the signal to add an instantaneous jolt of noise.

Those signals are a prime candidate for differential routing. The standard MIPI interface takes this into account.

If a high-speed interface does not use differential signaling, then it is likely to have a maximum length that is on the shorter side. As an example, a type of flash memory that uses eMMC architecture comes to mind. The impedance for single-ended connections in a DDR memory bus is usually between 40 ohms and 50 ohms.

6.3 Graduate School Zone

A printed circuit board acts as a low-pass filter in the same way that a long coax would also attenuate high-frequency components of a signal. I use an LPF circuit on my bass rig to cut down the scratchy noise from the strings. That's a good thing. When we're talking about high-speed digital switching, the scratchy stuff is the part that matters. Turning down the treble is like slowing down the rise times which increases the bit error rate. The openings in the eye diagram get smaller. Low-loss materials give the signal more breathing room.

A phenomenon called "skin effect" pulls the highest frequency part of the signal out to the perimeter of the traces. That makes the trace look narrower, more resistive as far as the pulses of data are concerned.

Wider traces help but then you're using more space in the X and Z dimensions in order to maintain impedance requirements.

An old trick from the analog world is to pre-distort the signal. Have you ever been tailgating and tried to listen to your car stereo from behind the car with the trunk open? It will be muffled and not beautiful. You can go in and set the tone control to where it's like an ice-pick to your eardrums. Close the door and go back to the trunk and it may not sound perfect but it will be clearer to the point of being almost beautiful again.

That's what predistortion can do at the driver pin side of the transmission line. There are feedback loops that are part of the firmware that kick in when too many bits fail to cross the divide between the send and receive ends of the signal.

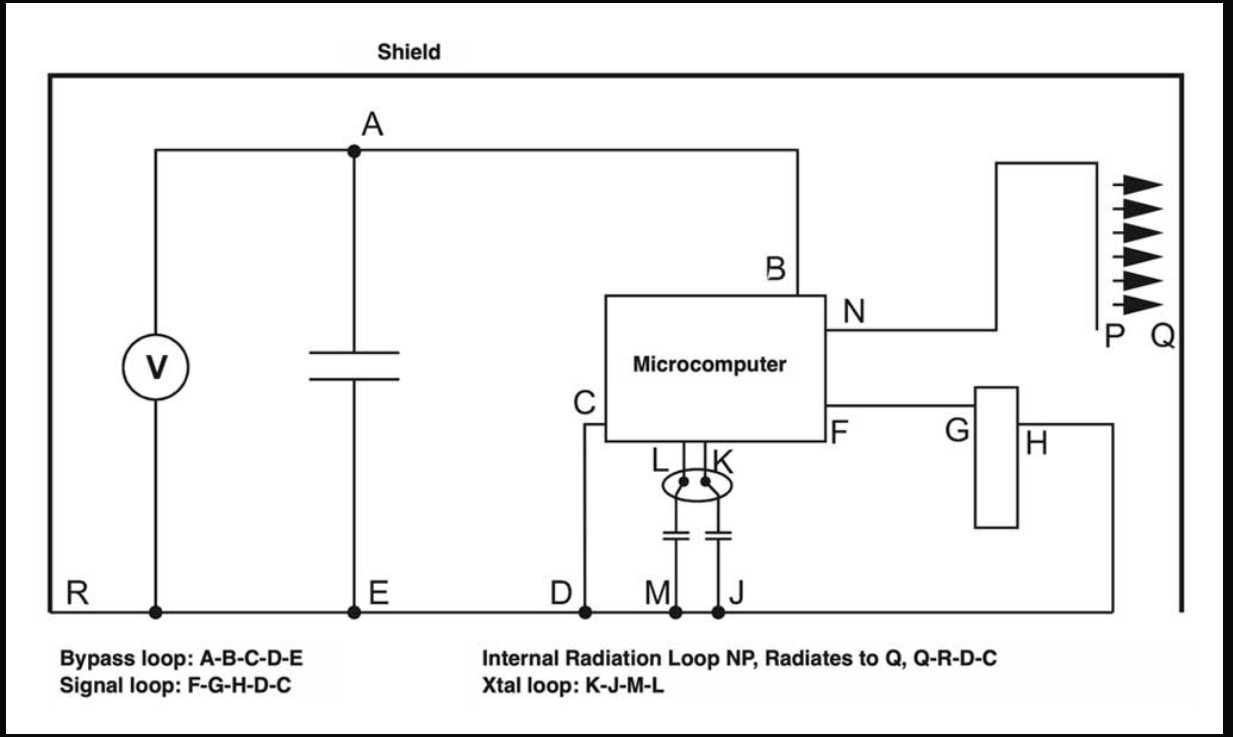
This kind of intelligent communication has to be part of any high-speed bit-stream. The additional pre-emphasis will consume more power so the better we can route those traces, the more efficient the system will be. Keep those eye diagrams open!

Becoming an expert at high speed and high-frequency signaling takes more than this chapter can offer. Therefore, I'd like to recommend the data sheets and application notes of long-established electronics firms such as Texas Instruments. More than mere data sheets, consider them as papers on topics of interest to the PCB designer.

They may be application notes for a specific processor but the guidelines are universal. They are written in plain enough language to understand the concepts.

Their explanation of ground loops maybe twenty years old but the behavior is the same as it ever was. I know some folks discredit ap-notes and datasheets. While nothing is perfect, they're far from useless.

"Every edge transition that is sent from the microcomputer to another chip is a current pulse. The current pulse goes to the receiving device, exits through that device's ground pin, then returns via the ground traces, to the ground pin of the microcomputer (see Figure 3). The pulse does not exit the ground lead of the receiving device and return to the battery but travels in a loop to where it originates. Loops exist everywhere. Any noise voltage and its associated current travel the path(s) of lowest impedance back to the place where it was generated."



Section 6. Figure 3. Image 6.3. Credit: TI - A diagram of the return paths on a typical PCB.

The crystal loop is one to watch out for. I like to sever the direct connection of M-J from the rest of the ground plane. There is usually a ground pin or two near the XO pins that can be used to tie those caps back to ground. If not, then a moat around the crystal serves the same purpose.

There's a symmetry in reducing noise emissions and increasing noise immunity. It's a double benefit, or a double-edged sword if not handled correctly. Digital links become more like analog as the data rates climb. Serialized data lines will benefit from the same noise suppression techniques used in the analog world.

(1) An FPGA or Field Programmable Gate Array is a type of chip that can have logic assigned to the pins after the device leaves the foundry. Compare this to an ASIC or Application Specific Integrated Circuit where everything is hardwired. The FPGA is usually larger than an equivalent ASIC and is used in development. An ASIC is often a follow-on chip for production but FPGAs can be used at scale when development cost and time are at a premium.

> **More exposure over longer lengths of trace lead to more crosstalk.**

7. Understanding/Prioritizing Busses

Section seven breaks down a number of different bus architectures such as PCIe, MIPI, USB, Ethernet, SATA, HDMI and others.

7.1 Understanding and Prioritizing Busses

7.2 Common Bus Types Found on PCBs

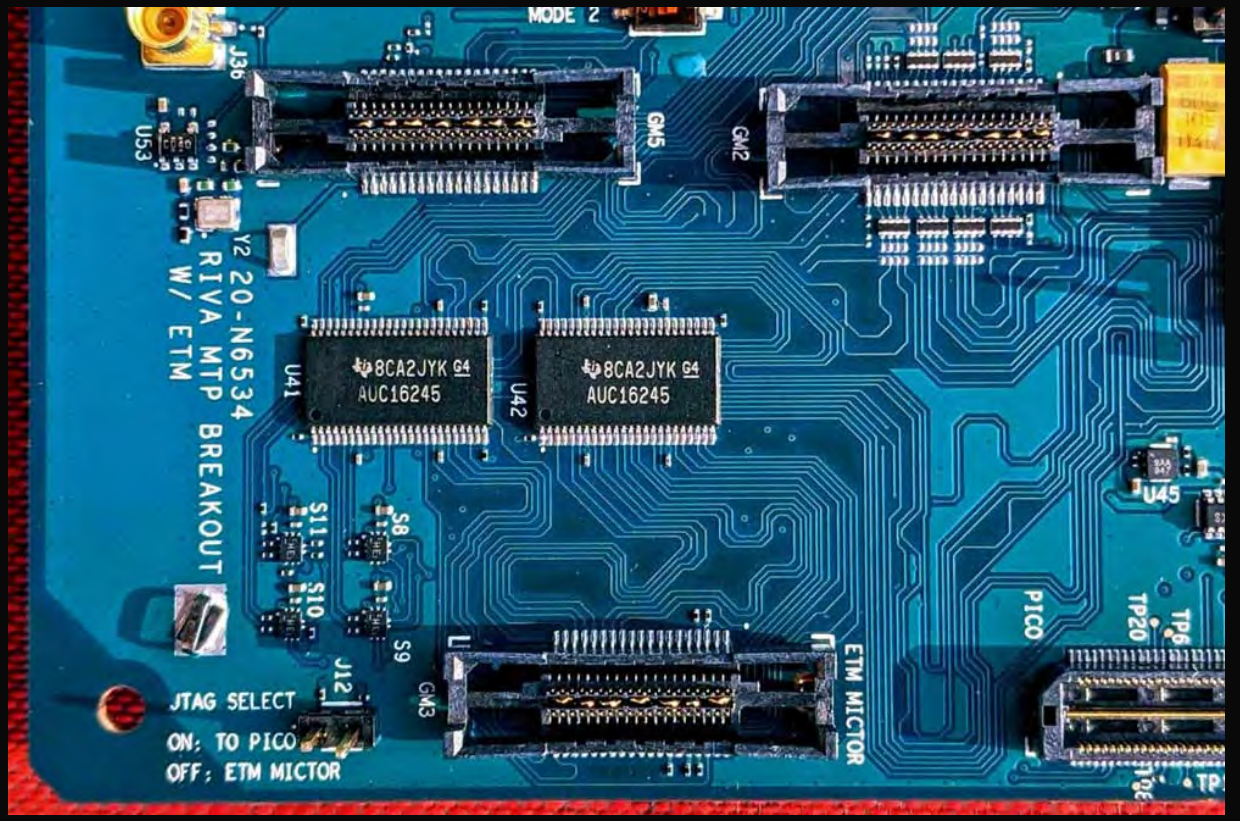
7.3 Graduate School Zone

7.1 Understanding and Prioritizing Busses

Memory, (see Section 8) PCIe, SATA, USB.3, and Ethernet interfaces are likely the most sensitive to the layout. (1) Organize the fan-out vias so that the traces flow without crossing one another. If there must be twisted connections, try to minimize them during placement and even going back so far as schematic capture.

It may be possible to use pin swaps on an FPGA device to clean up the routing lanes. Busses should be given extra care starting with ideal component placement. Always consider the fan-out stage and any fine tuning that comes after routing during the pin re-assignment.

A rule-of-thumb on spacing is to use three times the thickness of the dielectric as a trace-to-trace air-gap.



Section 7. Figure 1. Image 7.1. Credit: Author - Nested serpentine will use fewer corners than folding each trace into an accordion.

I've also seen it expressed in terms of line width times three but it's the z-stack number that matters in most cases. The relationship between coupling and spacing is nonlinear. Cutting the space by half will square the amount of coupling so a little goes a long way whether adding or subtracting space.

More extreme isolation comes from putting a guard band around the bus or even between pairs within the bus. A guard band is a trace in the ground net that is staked to any relevant reference planes with a picket fence of vias.

The idea is to create a "full metal jacket" of ground around the signals. The ideal spacing of the vias is a function of the rise time of the signals and the isolation requirements.

What we have in Figure 2 (see below) is an inner layer used to route four MIPI busses that are used as interfaces for camera modules. Each bus consists of four data pairs and one clock pair. The clocks are highlighted in yellow. Each pair in the bus is tightly matched within the pair so that the complementary P/N lines are the same length. That is the basis of differential routing.

Once that is achieved, the meanders are added so that the four data pairs are in sync with the clock pair. Then the four camera modules are grouped into left and right groups for the final length alignment. Each trace is a member of three cascading match-groups.

Start with the lowest tolerance and work your way out to where there is the most latitude in terms of length matching.

In terms of the full metal jacket, it is lost towards the bottom-right corner of Figure 2 because I had to avoid routing under a big inductor. When it comes to magnetics, the sheets of copper planes do not contain the EMI field.

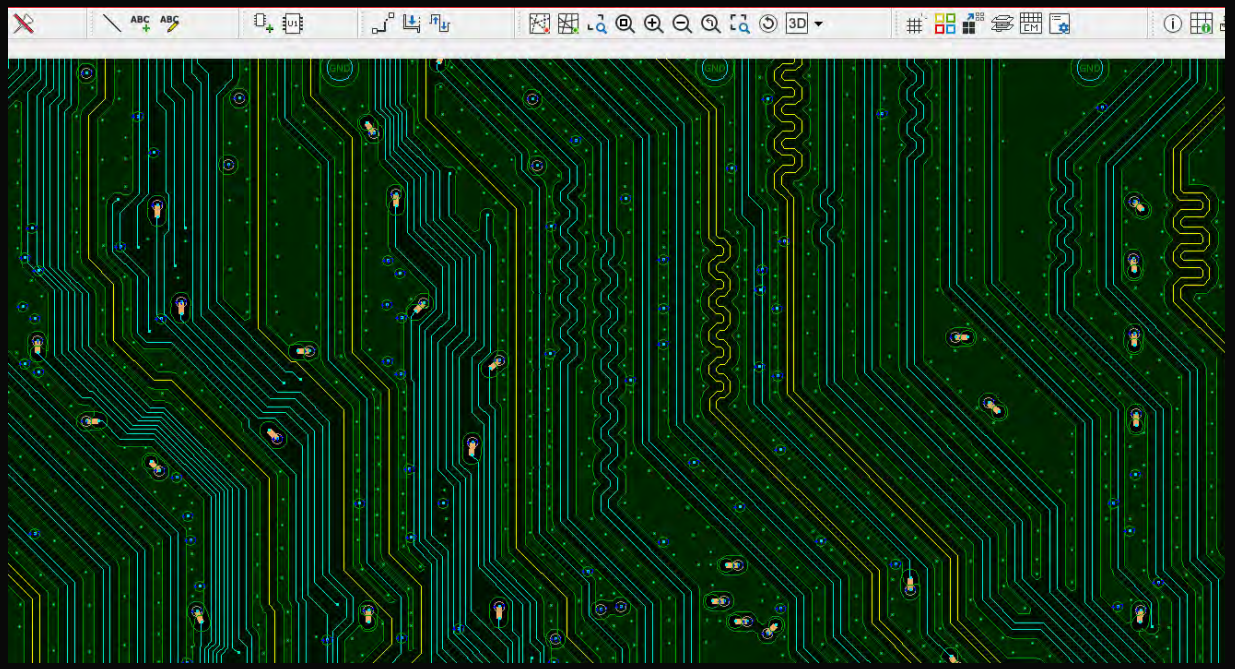
Doing without the guard band between members of the same bus is preferable to routing below an inductor or transformer.

You might notice four single-ended lines that are also tinted to yellow. They are also clocks that are used for different purposes. Again, isolating them was done on a best-effort basis. Compromises had to be considered as they routed down past the power section where a stray via can cause noise problems. Doing real-world PCB design often involves managing risks in the least harmful way.

As always, this job was a four-dimensional puzzle involving space and time.

Lower speed interfaces like I2C, I2S and JTAG can be given the less desirable real estate. That means that they can tolerate having a ground plane above and a power plane below the routing layer. It is best if the power plane is the same one that provides the voltage that drives the bus. It's better not to hop over different power planes even with the low-speed connections.

In cases where the board does not have any high speed interfaces, the lower tier becomes the top tier. A motherboard may deprioritize the I2C where the daughtercard treats it like the main event. In that context, it is the most important set of connections.



Section 7. Figure 2. Image 7.2. Credit: Author - Differential pairs routed on an inner layer.

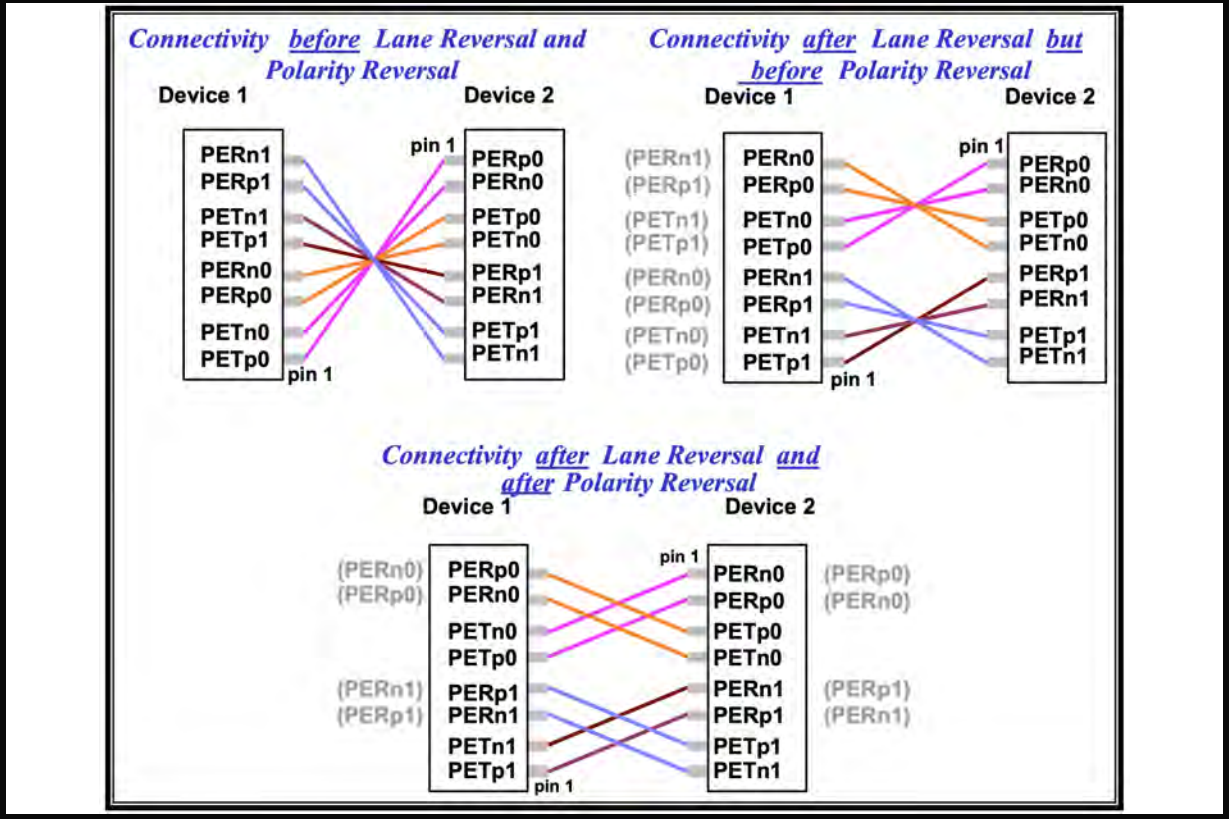
There will always be a hierarchy and there will usually be a layer of the board that is best suited for the routing. Adjust your effort accordingly.

7.2 Common Bus Types Found on PCBs

PCIe - Peripheral Component Interconnect Express: This protocol comes in numerous types with anything from one to sixteen groups of TX/RX pairs. Data rates will depend on what is being connected. The phase matching between the P and N traces are insane.

MIPI - Mobile Industry Processor Interface: Honestly, the only place I've seen this interface is between the SoC and the camera modules. Hololens AR and Pixel phones are the main culprits. Again, length-matching puts you on a grid of microns to clear the design rule violations.

USB 3.0 - Universal Serial Bus gen 3 introduced "super-speed" to the design vocabulary. It's a popular connection to the outside world but may also be implemented within the confines of a board.



Section 7. Figure 3. Image 7.3. Credit: Intel - PCIe is one of the few interfaces that have some latitude in the connectivity. If mirroring or rotating a component doesn't solve the problem then polarity inversion and lane reversal are valid options.

Ethernet - It's been around for a long time and has gone through several improvements. MII stands for Media Independent Interface. Gigabit Ethernet uses GMII and more recently, Reduced Gigabit Media Independent Interface.

They didn't reduce the speed, just the number of wires for the same or greater bandwidth. Once the data becomes serialized, it also becomes differential. Ethernet goes from "not so bad" to "pretty bad" when it comes to dialing in the trace length.

SATA - Serial Advanced Technology Attachment: That already spells trouble, doesn't it? In the glory-days or parallel attachment, the processor talked to the hard drive with a 40-pin connector. Condensing that stream to a 15 pin connector and continually raising the speed limit makes this an interesting connection. Raising the bar further we have SATAe to put the data on the express train. See number 1 for what that can mean. SATAe goes into PCIe territory.

HDMI - High Definition Multimedia Interface: The jump from VHS tapes to DVD disks was a quantum leap in data streaming. We cut the cord at Google when they put me on the Chromecast version 1.

The idea is to create a "full metal jacket" of ground around the signals.

The drama was whether we could keep the routing on the top through a slot in the RF shield or bury the trace and suffer the consequences of using vias to tunnel under. EMI concerns won out and we buried the traces.

SDIO - Secure Digital Input Output: This was the way Qualcomm talked to the outside world from the Snapdragon family of devices. It's only four wires divided into TX and RX pairs but we were really picky about the routing with 25 micron maximum deviation between the positive and negative sides of the pairs.

LVDS - Low Voltage Differential Signalling: A generic term with multiple uses, the low voltage indicates that the dielectric material for the PCB should be a low-loss/high-speed variant.

If not, then it's best to keep the lines relatively short. It's a safe statement to make about any of the interfaces listed above.

There are a number of less critical interfaces where the penalty for each via is less severe. When you see functions like low-resolution video, high definition audio, the above-mentioned I2C or SPI, CANBUS, UART, or GPIO, the signal integrity is less of a concern but if any of those crop up as the board's main features, go ahead and treat them with care.

7.3 Graduate School Zone

Just as there is a best layer, there is likely to be a sketchy layer where routing is challenged by numerous split planes on one of the reference layers. Perhaps, the via-stubs on that layer present a problem for high speed routing. That layer can still be useful for escaping difficult sections. Routing short segments, just enough to find daylight on a better layer, will leave clusters of local traces with space between the clusters.

The layer may also be designated for power planes or as a last resort for emergencies. Define emergencies as the 11th-hour addition of a net that runs coast to coast. Putting the first revision to bed with one layer where the routing is sparse sets you up for a successful rev two. Room for improvement is always a good thing.

(1) Generally speaking, each bus will have its own set of constraints. Some are tighter than others. When the length matching tolerances are small, it's a good sign that the bus is also susceptible to interference.

8. Memory Routing

Section eight talks about volatile and nonvolatile memory along with ten steps to keep in mind when routing.

8.1 Memory Routing

8.2 What is Random Access Memory?

8.3 Learning About Double Data Rate Dynamic Random Access Memory

8.4 High Bandwidth Memory

8.5 How to Approach Routing of DDR DRAM

8.1 Memory Routing

Memory routing can easily fill a book but I'll try to boil it down to a chapter so you don't fall asleep. There are two basic types that we'll call volatile and nonvolatile. Volatility has nothing to do with the personality of the chip although some types can be more finicky when it comes to routing rules. Volatile memory is a bit more humanoid in that it forgets everything when it goes to sleep. Non-volatile memory chips remember everything when you wake them up in the morning. Go get a cup of coffee if your eyelids start getting heavy. Here we go!

Tape drives, hard disk drives, and thumb drives are examples of nonvolatile memory. Hard disk drives are still relevant but a lot of their sockets are getting filled with solid state memory chips that are smaller and more efficient.

In any case, these devices are marketed based on the number of Megabytes, Gigabytes, or even Terabytes of storage.

In addition to SATA, the devices may converse with the SoC or microcontroller using a "scuzzy" (SCSI) interface. Like any interface that has been around for a while, the Small Computer System Interface has been given a few makeovers. Serial Attached Scuzzy (SAS) is a common protocol.

Just as with Ethernet and everything else, serializing the data cuts down on the number of wires. Fewer wires make it easier to keep the entire data stream in sync with the clock. We like it easy! You will be routing to a connector for a disk drive and to a chip for solid state drive.

Routing to a connector means that there are more links in the chain so managing the timing budget can be tricky.

8.2 What is Random Access Memory?

When you shop for a new device, they talk up the storage but then they always have to mention the RAM. Random-Access Memory is that other kind and it comes in smaller increments. Bill Gates famously misstated that “64 Kilobytes ought to be enough for everybody.” That was back when a good hard drive was 40 Megs.

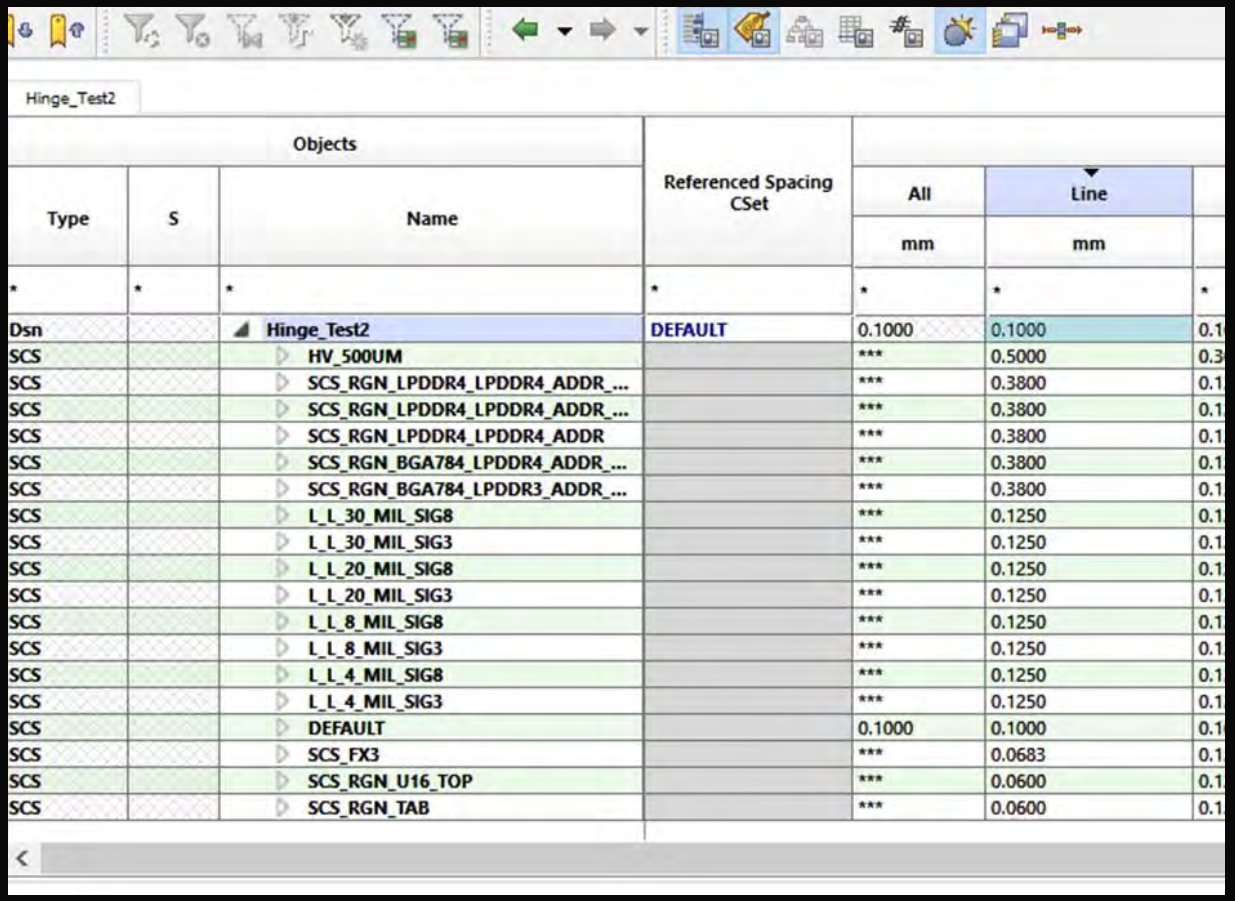
These days, the first Gigabyte of RAM is free and you start paying extra for multiple Gigs. Of course, RAM comes in many flavors. Dynamic RAM and Static RAM are two variants. DRAM is the marketing tool mentioned above and SRAM uses a different architecture that doesn't require periodic refresh cycles. SRAM is also smaller, faster, and more expensive. You’ll find SRAM used on the same silicon as the CPU.

DRAM cells are based on a capacitor and a transistor. So we know DRAM and SRAM, what about SDRAM? Trick question; Synchronous Dynamic Random-Access Memory has better efficiency because of the multitasking during read and write cycles. You can consider the rest of this chapter to lean towards the Graduate School Zone but read on if you’re curious.

8.3 Learning About Double Data Rate Dynamic Random Access Memory

Want more? What about Double Data Rate DRAM? Yeah, DDR DRAM is the stuff. Again with the efficiency but this time it's using both of the clock edges. The Data group is the one that switches on the rising and falling parts of the clock cycle. Numerous improvements come in each generation. There are a lot of pins on a DDR chip relative to most memory. Access to more registers gives DDR serious bandwidth. The dimensions of the registers are set in the firmware.

The address section includes separate connections that describe the bank, (BA) the row (RAS), and the column (CAS) of each bit of data. Various other hooks are used to monitor and control the memory banks; write enable, clock enable, reset, chip select and the synchronization pin come to mind.



Section 8. Figure 1. Image 8.1. Credit: Author - Spacing rules for memory busses are just as complex as length and width constraints. They are often the most difficult aspect to meet. Localized regional spacing constraints are common in dense pin fields.

Fewer wires make it easier to keep the entire data stream in sync with the clock. We like it easy!

Each bit of the data bus must arrive and settle into its state ahead of the clock cycle. This is the setup time. The entire bus must also remain steady while being read in or out after the clock. They call this hold time.

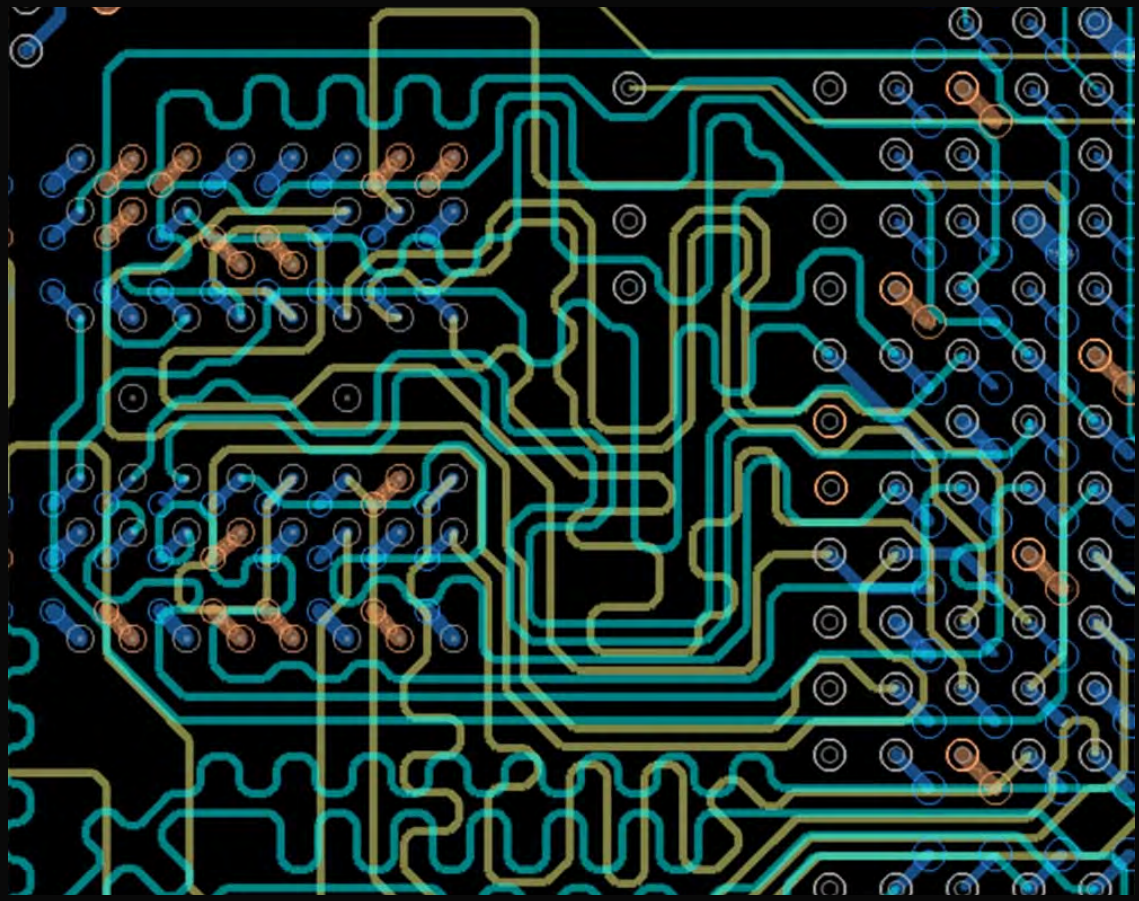
The little cap in each memory cell only holds its charge for so long. Even if the memory isn't in read or write mode, the information decays as the cap discharges.

To solve this little problem, there are refresh cycles going on quite often. That's another pin.

8.4 High Bandwidth Memory

Another wrinkle and the new hotness is High Bandwidth Memory. (HBM) They've learned to stack up the dice and join them up with through-silicon-vias. All of that happens right next to the CPU which takes a load off of the board. If you're doing those substates, look forward to 1000+ pins between the processor and the memory stack. You'll probably have fond memories of the good old DDR5 days.

If those DDR5-days are also your potential future days, then listen up. DDR 3, 4, and 5 come from similar cloth at the PHY level. That's our domain so let's cut it up into chunks of PCB Design. The beauty is that while no two memory implementations are the same, the design guidelines crossover pretty well. Here's a general step-through for a typical DDR bus routing.



Section 8. Figure 2. Image 8.2. Credit: Author - A typical DDR memory bus routing using three layers for signals. Other layers not shown for clarity.

8.5 How to Approach Routing of DDR DRAM

Color code byte lanes > brown, red, orange, yellow, green, blue, violet, grey, etc. These colors correspond with the color code for resistors. Clock pairs (CLK) set the length of the Address (AD) and Command (CMD) single ended lines. Clocks also regulate the strobes (DQ) but normally to a lesser extent. The strobes, in turn, regulate the data. You'll wind up using different shades of the same colors to keep up with all of these interrelationships.

There are a lot of data lines but they are broken up into sub-groups of eight. Each octet of data lines (D0-D7, D8-16, D17-D24, etc.) form a byte lane and each byte lane has its own strobe. There is tight matching within each byte lane and relatively loose matching between lanes.

So, D0-D7 is a tightly knit unit and D8-D16 is another tightly knit unit but there is more latitude from lane to lane. Data is the bulk of the routing and tuning and would be the first priority.

The actual length-matching numbers within and among all of these groups are outside the scope of this guide. It will depend on the memory device and how it is being used. As you get into newer versions of DDR, the margins shrink. They shrink so much that you have to start accounting for delays inside the packages.

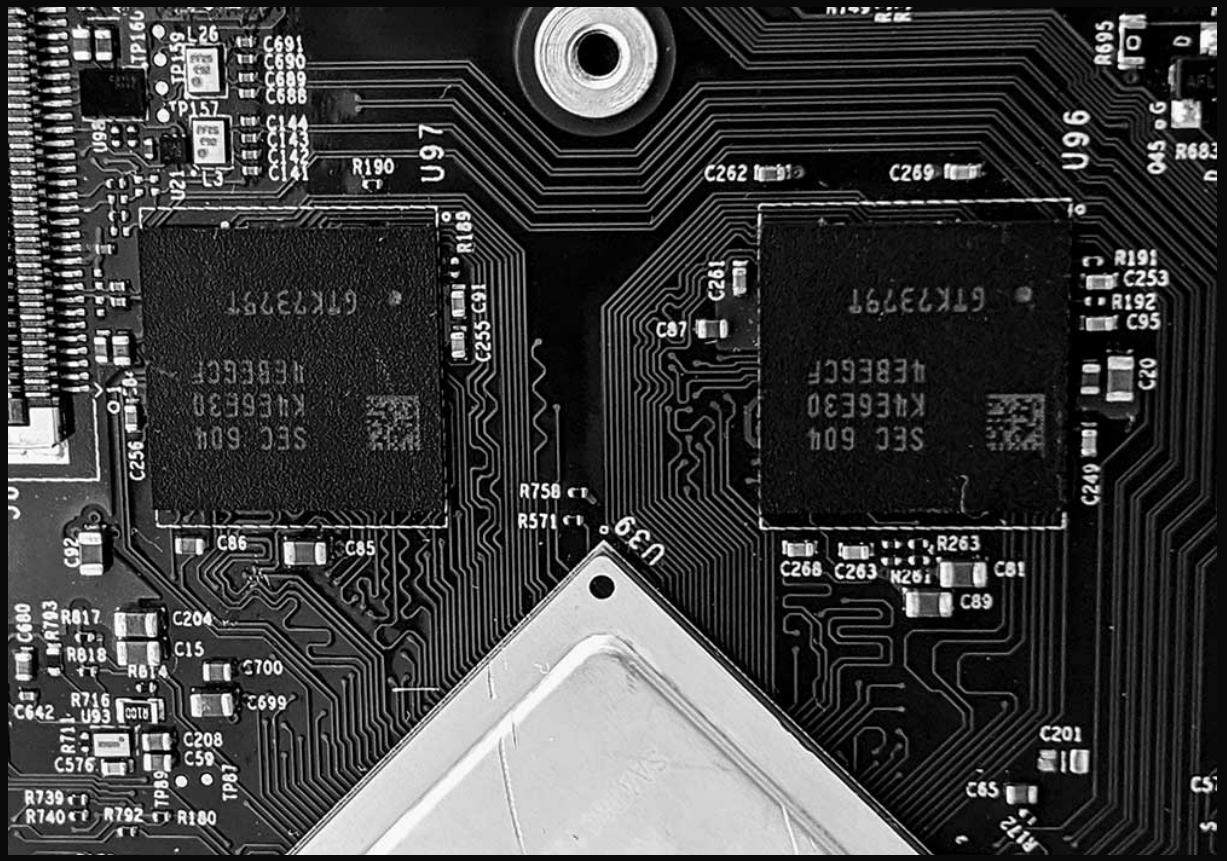
The routing on the substrates becomes part of the length calculation. We add a tailored amount of pin delay to each of the processors and the memory device pins in the constraint editor.

The routing lengths on the board will be unequal but still in tune when we consider the entire time of flight.

Note that there is also a difference in propagation delay depending on if the traces are routed on an outer layer or an inner layer. Capturing all of this information in the electrical constraint manager is no small feat. Get all of that done and it's time for step-2. Are you ready for this?

2. Fan-out the entire memory chip(set) and memory section of the processor. DDR devices are packaged with generous pin pitch. In many cases, you can do yourself some favors with creative fan-out directions rather than uniformly fanning outward from the center of the device. Look to improve the crossovers and lengthy connections during the fan-out stage. All of the members of each group should use the same number of vias. Vias are such a pain for signal integrity that they need to be shared equally.

3. Get the decoupling caps and terminations and everything else placed. Connect the caps to power and ground with utmost concern for short inductive loops from the cap's power and ground paths. Make the most of good design practices for the power distribution. There are not that many power pins so take good care of them.



Section 8. Figure 3. Image 8.3. Credit: Author - The microcontroller was designed to use the outer layer for routing of certain byte lanes. The spacing is tighter than I'd like but it worked.

The termination resistors form the boundary of the chip's area. In addition to the shunt resistor at the end of the line, the technology may require a damping resistor in series somewhere along the lines between the processor and the memory. Resistor packs with four or eight individual resistors are popular for this application.

The resistors are becoming less common as On Die Termination (ODT) has integrated the terminations onto the processor. That's a nice development. It takes a lot of 49.9 ohm resistors for all of those terminations. Why not 50 ohm? When you use 1% tolerance on the resistors, the nearest value is 49.9 ohm. The consistency of the resistor value is good so deal with the price of the 1% tolerances.

4. Route all of the connections keeping the various colors together on the same layer. Leave some extra space for step 5. If you have two primary routing layers, then the aim would be to route even-numbered byte lanes on one layer and the odd-numbered byte lanes on the other. This will tend to distribute the pins and vias that use each layer.

The idea is to increase the launching options that you already optimized during fan-out. Revisiting fan-out during routing is a common occurrence. Sometimes placement also gets a second thought. Setting aside the auto-router demonstrations, a properly routed memory bus takes time. The length-matching tools are helpful but the results will be more compact if you drive the bus yourself.

5. Tune one-byte lane at a time - make the longest member as short as possible. There's guidance on sharp corners but no other reason to stay on 45-degree angles if it helps cut corners. Imagine stretching a rubber band around all of the routing obstacles.

Can anything be done by looking at the inflection points and moving a via or a passive part? If it's no longer the longest, shorten the new longest one. Anything you can do to shorten the longest member of the group is a good thing.

Have you ever looked at a strobe light for any length of time? It can be nauseating. The rest of the signals feel the same way about being routed near the strobes so keep some extra distance between the DQ traces and everything else. They are essentially clock nets.

6. When you can't shorten any more, lengthen the clock diff-pair until it is long enough to meet the spec, (not necessarily as long as the longest line in the group). Phase-match the pair very closely.

The closer the Positive and Negative sides of the diff-pair are to matching, the more that is left as tolerance for tuning the single-ended connections.

It may seem more intuitive to tune the clocks first. The thing is that they will be too long and too short. Where do you stop? Knowing the length of the longest tent-pole allows you to stretch the clocks until they are too long but not too short. You solve the too-long errors by lengthening the too-short data lines. When all of those meet the clock length minus the tolerance, then the clocks will be clear of length matching errors.

7. Meander the short ones to get within spec of the clock/strobe tolerance, again, not necessarily as long as the longest or the clock. The clock/strobe target should be the middle ground. This can be a soothing process or a painful one.

The meanders should be laid out to maximize the use of space. Route one trace along the edges of the previous one, hugging that trace and adding the wrinkles for tuning in ways that preserve the routing area for other functions. It's something else to learn.

8. Highlight the lane that meets timing to a special color and don't touch it unless you have to. Sometimes, it works out and sometimes the other lane needs a little more space. Bite the bullet and make the change and make sure that group is still tuned. Move on to the next one.

9. All done with tuning? Route VREF and other non-length-matched lines. VREF requires a thick line and a wide air-gap. It is a partner of VTT which would use copper planes. It is typical to bury the high-speed connections on the inside layers of the board. That gives you the outer layers for the voltage distribution and miscellaneous signals. Every pin on a memory chip can be considered critical in some way.

10. Submit to SI/PI before continuing with the layout but leave that area alone. Will a memory device work at all without doing all of this length and impedance matching? Maybe but it will bog down when the cyclic redundancy check counts up all of the bits sent and compares that number to the bits received.

If those numbers don't match, it means that a packet of data was dropped and the chip says, "What?" and the whole set of instructions have to be sent again. The result is glitches in the data stream that can cause the video to stutter. If it's bad enough, the program or even the whole system program can crash. Then you have to go back to step one; coffee!

9. Analog Routing

Section nine looks at the different factors of analog routing along with the difference between microstrip and stripline traces.

9.1 Analog Routing

9.2 Estimating impedance for a given stack-up

9.3 Selecting the Correct Dielectrics. It's All in the Materials

9.4 Layer Assignment: Typical Routing Solutions Based on the Number of Layers

9.5 RF Modules - A Solution in a Box

9.6 Graduate School Zone

9.1 Analog Routing

Oh, students of the PCB Design future, this is where it gets real. The digital data is often coupled to the real world where logical ones and zeros are not enough. In between those two, there are shades of grey which are filled in with analog circuits. That data stream is converted into digital approximations and all of a sudden, 'Bam' room for lots of little transmitters and receivers!

"Summing it all up: When it comes to analog layout, impedance is the name of the game. The transmission lines become components unto themselves. Placement is built around the traces rather than the other way around. You don't really place and route so much as build a route incorporating the components along the way."

Dependence on keeping the impedance constant is a strong incentive to use surface-mount components and make the connections on the same layer using microstrip or coplanar waveguides. These transmission lines are noisy by nature so our instinct is to route them on an inner layer where they can be sandwiched between a pair of ground planes as a stripline trace. That plan is preferable to having a long trace on an outer layer and sometimes, you just can't help transmission lines that cross over one another.

The best case is when you can create a placement where all of the RF connections are between pins that are as close as the assembly and test rules allow.

When a choice has to be made to lengthen one trace or another, defer to the receive (RX) chain allowing the transmission (TX) line to have the longer trace. Receivers have big ears and small mouths while transmitters are loud mouths that are relatively hard of hearing. Obviously, the two do not make good neighbors.

9.2 Estimating impedance for a given stack-up

A few dimensions and one constant are what determine the characteristic impedance of a transmission line. The dielectric constant (Dk) is a key factor that determines propagation delay as well as the loss tangent. All insulators have a dielectric constant. For example, air has a Dk of one. Standard FR4 material has a Dk of around 4.5 and this varies over frequency, temperature and, of course, the exact type of material since FR4 covers a lot of ground.

On an outer layer, there is one dielectric thickness of note and that is the distance to the copper plane below the transmission line. The electrical and magnetic fields live primarily in this space. An elegant design will keep them there while a dodgy design will cause them to spread out. In order to achieve the typical 50 ohm impedance, the simple formula is to make the microstrip trace width equal to the dielectric thickness.

That can be an issue if the dielectric material is very thin as it will be when you use a micro-via. Bear in mind that you might use Layer-3 as the reference plane and keep Layer-2 free of copper. Anything on Layer-2 will disrupt the impedance and also get massive coupling between the transmission line and anything in that dielectric space. Take the trace width of the microstrip and multiply it by three to get a good safe distance for any metal on the cleared layer(s).

9.3 Selecting the Correct Dielectrics. It's All in the Materials

The other key factors for controlled impedance are the thickness of the dielectric material and the trace width. The actual z-height of the copper, coating materials and fiber weave considerations are secondary effects that come into play as the rise and fall times increase. There are other metrics such as loss tangent and the coefficient of thermal expansion (tg) of the material to evaluate. "Everywhere you go, there's an equivalent material to cut in."

At that point, you may have to consider exotic materials with trade names like GETEK, Rogers or Megtron. Doing so will require a different approach to rough impedance calculations.

In all cases, you MUST get your PCB fabrication vendor(s) into the conversation sooner than later. There are a lot of choices and many of them are regional. Everywhere you go, there's an equivalent material to cut in.

The analog lines are purposefully wide in Figure 1. This was done by selecting a 0.7 mm dielectric thick. Given the dk, the 50 ohm lines calculated out at a width that is pretty close to the width of the typical pads. This old-school through-hole board also opens up all of the solder mask over the vias to provide a probepoint or potential rework path. Analog technicians are no stranger to scraping away a little bit of solder mask.

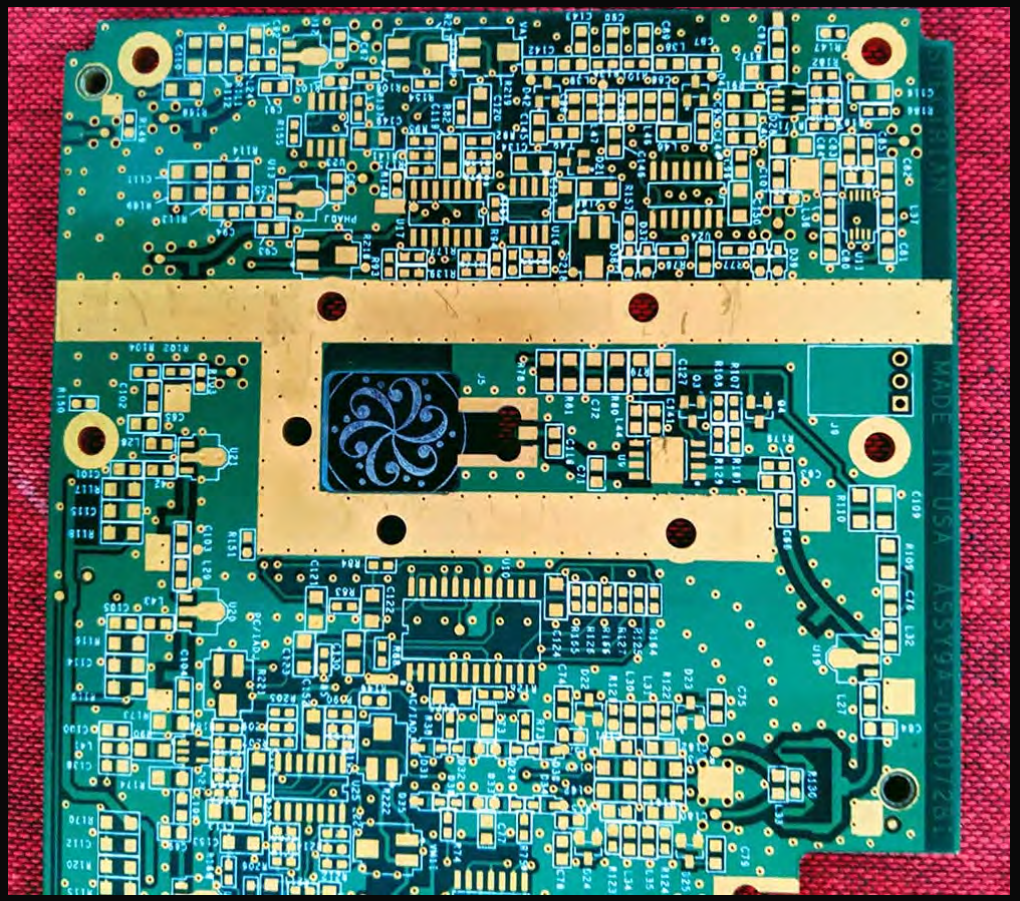
Also not the absurd measures taken every time one of the micro-strip traces takes a dive into the stripline zone.

Not one ground via but four looking almost like a little SMA connector. We're making a coax in the z-axis to shield the via. As the trace approaches the ground vias, the trace narrows and the ground pour converges creating a coplanar waveguide for the segment connecting to the signal via. The trace is meant to be coplanar as it travels over the gap between the ground planes and the signal via. So let's explore the inner workings.

Moving on to stripline traces, the deal is that there are two plane layers, one above and one below the trace. Use the same one-to-one formula but look at the stack-up and make sure you are considering the thinnest dielectric for the rule. Some routing layers in the stack up make this simple because they are symmetric. (1)

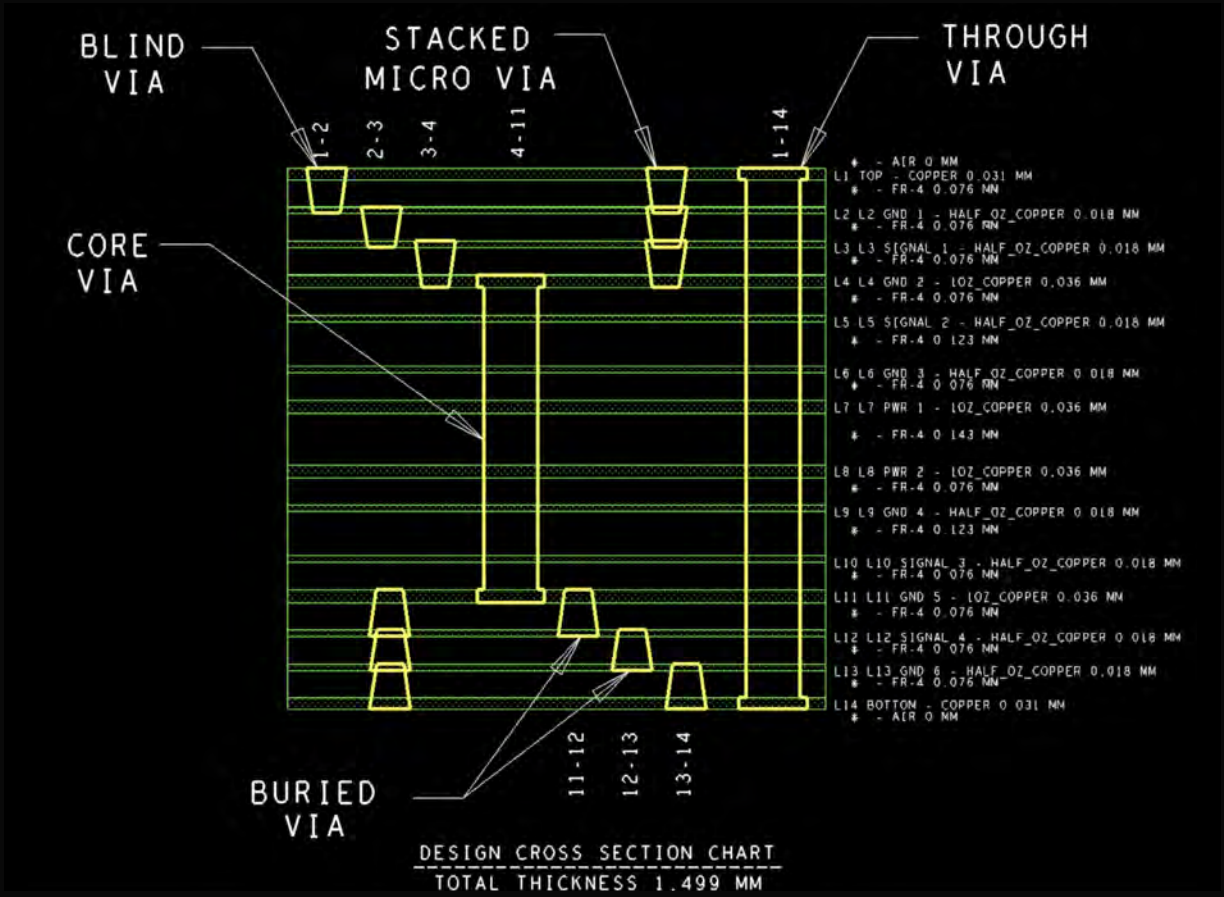
When routing a differential pair, the same rule of thumb applies to the two traces but only if they are separated by a minimum of three times the trace width. Those are considered uncoupled differential pairs. In this case, the impedance value doubles so the one-to-one ratio that gave you 50 ohms now is approximately 100 ohms.

As the two traces get closer and become loosely coupled (between 3x and 1x trace width) or tightly coupled (1x trace width and below) the impedance goes down so the two lines have to be thinner than the dielectric to get back to the common 100 ohm impedance. Again, this drives the PCB into line-width manufacturability issues. For this reason, I've seen loosely coupled and uncoupled differential pairs on the majority of board designs.



Section 9. Figure 1. Image 9.1. Credit: Author - Analog traces like to be sequestered.

9.4 Layer Assignment: Typical Routing Solutions Based on the Number of Layers



Section 9. Figure 2. Image 9.2. Credit: Author - A so-called 3 N 3 stack up where N is equal to the span of the core via.

This 14-Layer stack-up provides four internal routing layers. A 12-Layer version of this one would remove layers 6 and 9 which are ground planes between the power planes and the inner 2 routing channels. Layers 5 and 10 would be the choice for low-speed traces while Layers 3 and 12 would be safer for high speed and RF routing. That's because you would continue to have ground planes above and below the transmission lines rather than a ground plane and a power plane. Best case in that regard is a voltage match between the logic and the power plane.

Additionally, cutting out layers 4 and 5 yields a 10-Layer board. This price point eliminates the ground plane between the inner layer traces. This is called a Ground-Signal-Signal-Ground stack.

The notion is to use the two routing layers orthogonally to each other with the aim of not running the two traces broadside, one over another. Some of that may be inevitable as you fan-out a dense device or connector.

Ten layers is a sweet spot. An 8-Layer board will take you back to Ground-Signal-Ground but you will only have two inner layers for routing. At six layers and below, we're not really distinguishing a layer for a certain purpose.

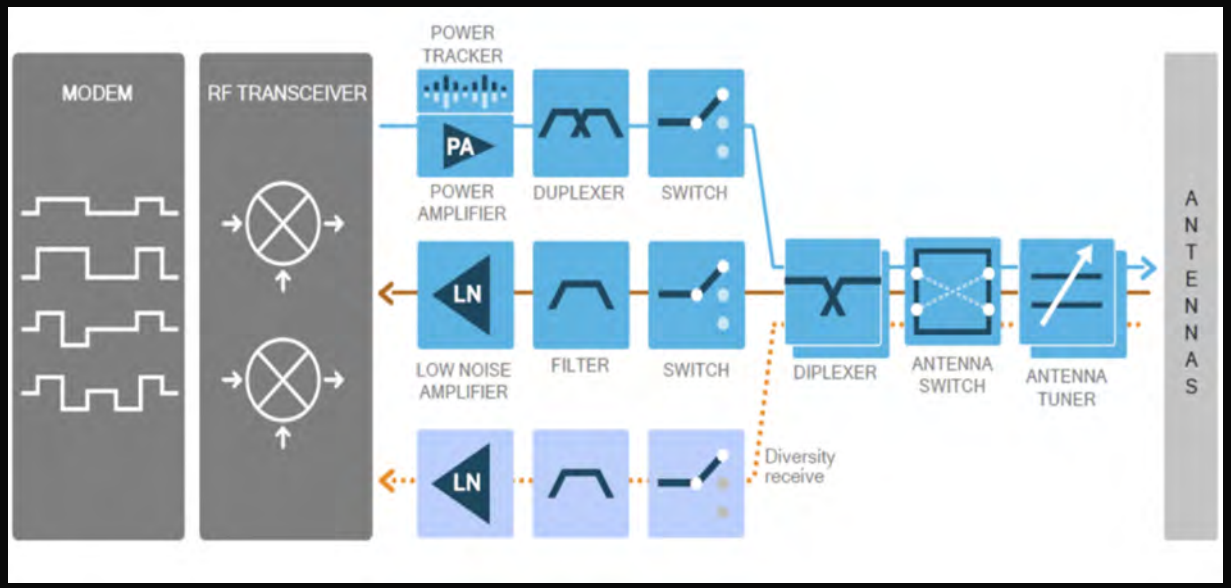
All the layers are ground layers. Then they're component layers, power planes, routing layers and the usual decorations. Those low layer-count boards can be tricky. You always have to bear in mind what is above and below the layer you're working on.

When it comes to analog layout, impedance is the name of the game.

9.5 RF Modules - A Solution in a Box

A popular solution is to use RF modules that include a radio chip with the required bias circuits and matching networks in a rectangular package with pins around the edges. RF modules are similar to QFN packages but may have SMD pins that are larger and located a little more in-board. Depending on the frequency, a module, the substrate can be regular FR4 or a high-speed/low loss version on up to ceramics and other exotic materials.

Front End Modules (FEMs) combine all of the TX and RX circuits in a convenient package. Starting at the RF input pin, they may include a low noise amplifier, filters, a duplexer, switches, a diplexer, the antenna matching network, the power conditioning, and the final stage amplifier leading to the TX out pin. It all fits under a nice RFI shield to help get through FCC regulations.



Section 9. Figure 3. Image 9.3. Credit: EE Online - You can cobble the whole thing together with various components or get an off-the-shelf solution that takes most of the risk out of the equation.

Summing it all up: When it comes to analog layout, impedance is the name of the game. The transmission lines become components unto themselves.

Placement is built around the traces rather than the other way around. You don't really place and route so much as build a route incorporating the components along the way.

9.6 Graduate School Zone

Maintaining impedance through the whole analog chain is easier if the width of the trace and the SMD pads are equal or very close in width. That will call for relatively small pads and wide traces. A narrow trace entering a big pad creates a discontinuity. Meeting the impedance geometry will mean having a thicker dielectric between the transmission line and the reference plane.

As I mentioned in the part about impedance estimation, that plan is in direct conflict with typical HDI design rules that call for thin dielectrics to meet the aspect ratio design guideline of the microvia structure. The working solution in that case is to cut a channel in the ground plane on layer-two right below all of the transmission lines and use layer-three or even further into the board.

The width of the channel in the ground plane should match the metal pull-back on the trace layer, normally 3x the trace width. That way, a line of vias can be dropped along the length of the trace and tie the actual ground plane to the component layer. The saving grace is that analog circuits are typically not the dense part of the board so you should have layers to spare. That is assuming that the other parts of the board are the reason HDI technology is in use to begin with.

(2) A stack-up is symmetric when it is a mirror image from the center line outward. In addition, the three layers that make up a Ground-Signal-Ground routing channel would also have equal dielectric thicknesses surrounding the trace in the z-axis.

Boards are constructed out of core materials and prepregs so just as often, the thickness will vary. The main point of symmetry is to avoid PCB warpage or other similar mechanical variances.

10. Flex Circuits

Section ten looks at what's unique about flex circuits – their plating processes and use cases along with examples of stackups.

10.1 Flex Circuits

10.2 Flex Circuit Materials - A World of Their Own

10.3 Beyond Polyimide, Copper, Adhesive, Coverlay, Stiffeners, and Solder Mask

10.4 Button Plating vs. Panel Plating

10.5 Bend Regions - The Main Point of Being Flexible

10.6 Flex To Assemble

10.7 Dynamic Flexing

10.8 Rigid-Flex Designs

10.1 Flex Circuits

This is a whole different thing. A lot of flexible printed circuits (FPCs) are nothing more than a substitute for a ribbon cable with bespoke architecture to get a bus from one board to another. There would be a connector at either end and a number of traces and perhaps a power line between the connectors. Even this simple form has a lot of aspects that are particular to FPCs.

10.2 Flex Circuit Materials - A World of Their Own

Easily, the biggest difference between rigid boards and flexes is in the material call-outs. Polyimide is the dielectric standard but there will be other dielectric materials in use on the same flex. The stack-up diagrams go into detail on these elements.

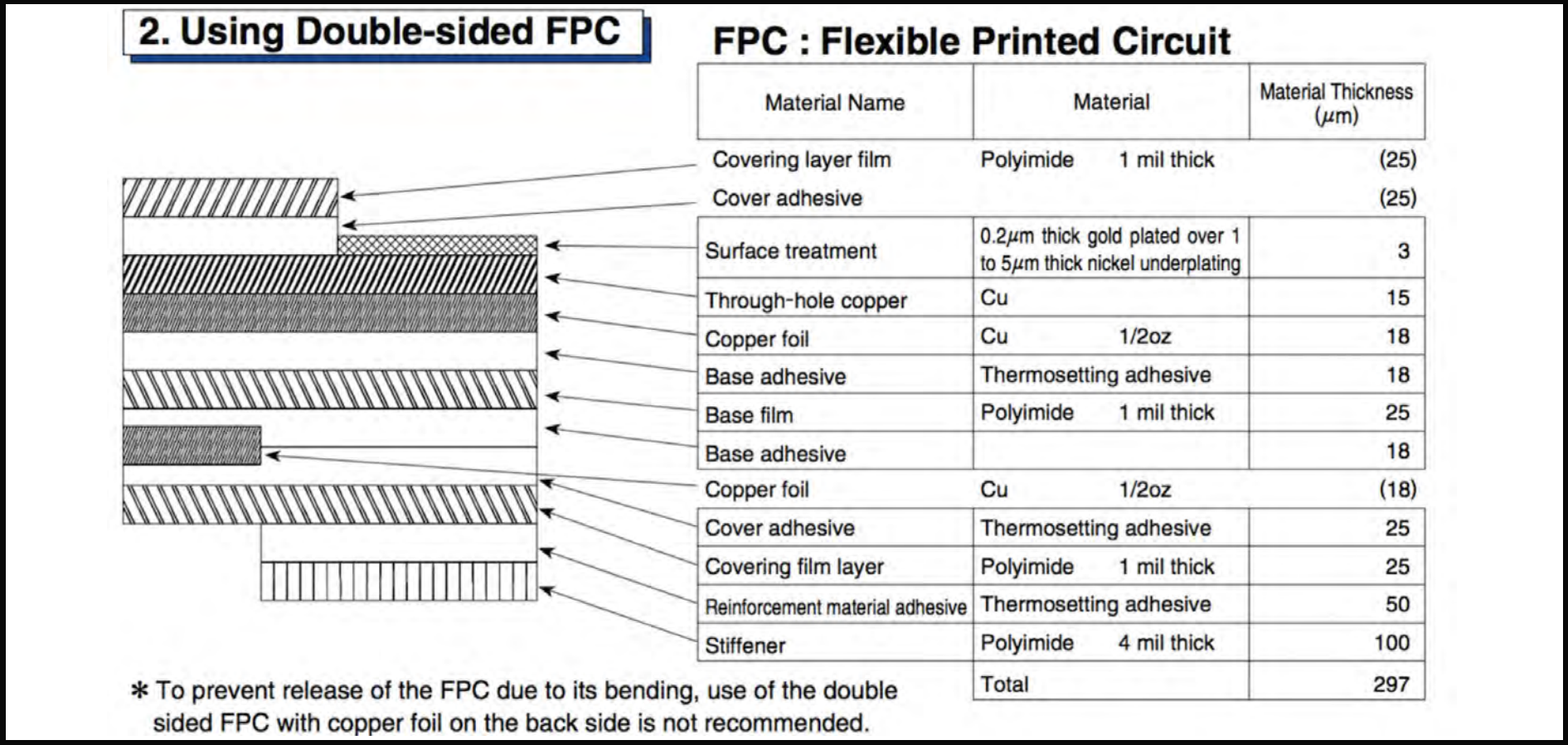
Adhesives are used to bond the various laminated materials. Pressure-sensitive adhesive (PSA) is commonly used to attach the coverlay and polyimide or FR4 stiffeners. When stainless steel or aluminum stiffeners are used, a thermoset adhesive (TSA) is usually called out. The adhesive in this case would also be electrically conductive as it forms part of the thermal path. Metal stiffeners have a pullback from the outline edges while polyimide or FR4 stiffeners will be the same size as the rest of the material. The advantage of metal is that it is stiffer per unit of thickness but it adds cost.

Coverlay is the term for the polyimide material that protects the traces from the elements. It helps keep the traces from lifting or getting damaged by abrasion. Black is a popular color for coverlay and solder mask. The solder mask will only be used in the stiffened parts of the flex circuits where parts are soldered down. The area that transitions from coverlay to stiffener and solder mask is a complex region with some carefully controlled overlap of the materials.

A zero-insertion-force (ZIF) connector is a popular option for connecting a flex to a rigid board. It is a little different from a usual edge connector. The gold fingers are not simple rectangles set side-by-side. It is typical to have two rows that are staggered. The mating connector has a hinged locking mechanism to keep the ZIF connector in place.

The 25 micron (one mil) thick polyimide base film is very common. Two mils is also available. The location where the FPC changes from stiffened area to flexible area is called the transition zone. Pulling the via back 0.4 mm from either side of the transition zone should keep you from getting a call/question from the vendor.

Also, note that the coverlay tucks under the rigid section and comes to an end in the transition zone. The overlap goes on for 0.4 mm. The solder mask overlaps the coverlay by 0.2 mm when a connector or other component is soldered down. Of course, these numbers and material types will vary with different manufacturers. It's even more important to get out in front of the stack-up when it comes to flexes.



Section 10. Figure 1. Image 10.1. Credit: Hirose - When is a Two-Layer Not Really Two Layers? When it's a Flex.

10.3 Beyond Polyimide, Copper, Adhesive, Coverlay, Stiffeners and Solder Mask

Another glue-down covering can be added over the top of the coverlay for EMI suppression. High speed/high frequency traces will radiate through the ground mesh. When you use the EMI suppression film, the coverlay has to be notched to reveal areas of where the ground mesh is exposed. Putting a patch of solid copper with vias is useful to create a good contact between the EMI shield layer and the ground. There should be at least one exposed area near each connector. A longer stretch of flex would have some openings enroute but not in the bend regions.

Yet another technique uses double-sided tape rather than hardware to keep the flex in place. Adhesive foam is another material that will fill gaps.

Both of these materials come with a peelable film over the outward-facing adhesive so that it can be removed when it's time to stick the flex into its final location.

" ...a bend table that shows the desired radius and number of times the area will be bent back and forth will be part of the fabrication drawing."

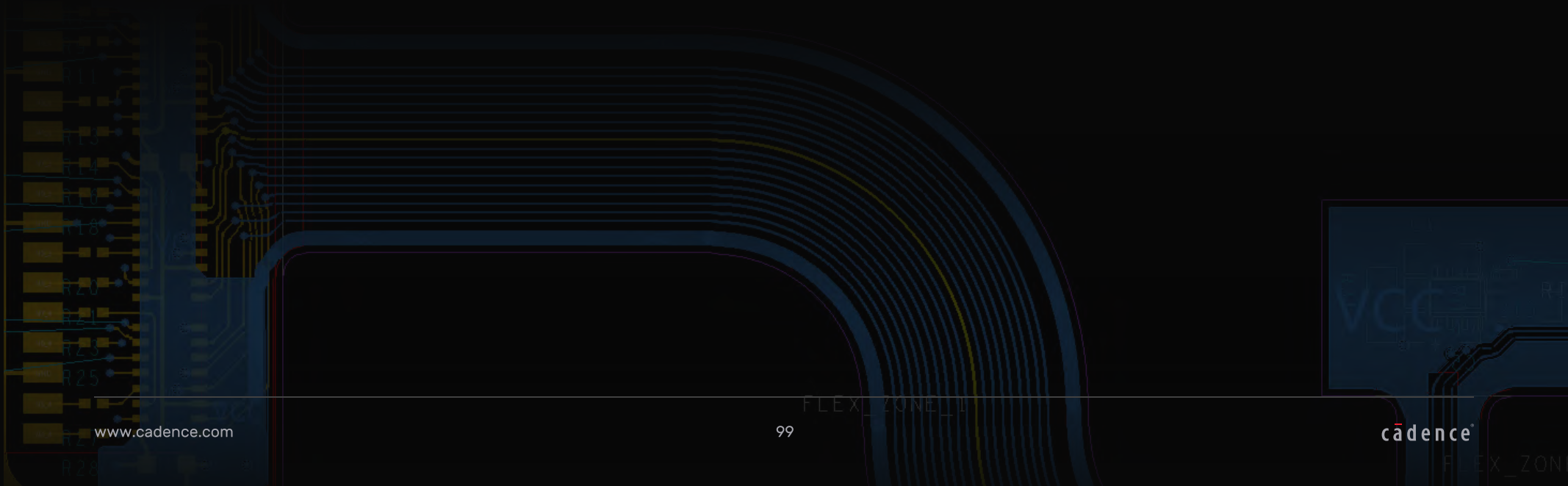
When it comes to conductors, rigid boards can get by with electro-deposited (ED) copper. The grain structure is vertical so the surface is rough and the plating is too brittle for reliable flexes. Rolled annealed (RA) copper has a smoother horizontal grain which may be good for approximately 10,000 bend cycles. The automotive sector sees a lot of potential vibration which compels them to call out "super flexible" HA copper in the material stack-up section.

In any case, a bend table that shows the desired radius and number of times the area will be bent back and forth will be part of the fabrication drawing.

10.4 Button Plating vs. Panel Plating

Unlike rigid stack-ups, the flex stack-ups call out the plate-up copper as a separate item. Plate-up is always done as a means of plating the via barrels whether it is a rigid or flex design. This means electro-deposited copper which we may not want on the traces in the flexible areas whether then bend or not.

To get copper into the vias without getting it everywhere, a special fabrication step is included where they mask off everything except the vias. That's the button plating while doing a plate-up of all of the metal prior to etching is known as panel plating.



10.5 Bend Regions - The Main Point of Being Flexible

Some FPCs remain flat in their installation and usage. Most have to be bent to meet different connector locations. Some have to be flexed during usage. Depending on whether it is a one-time flex to install or a dynamic flex, there are design considerations to follow. The hinge on any laptop will be a very complex part of the solution since the power and data have to get from the graphics processor (GPU) to the display screen.

Not only that, most laptops include a camera module at the top of the screen, maybe a flash or other sensors including a touch sensor for those laptops that convert to a tablet. The various bit streams have to flow through a number of flexes from one side of the case to the other. The repeated bending causes mechanical stress every time the lid is opened or closed.

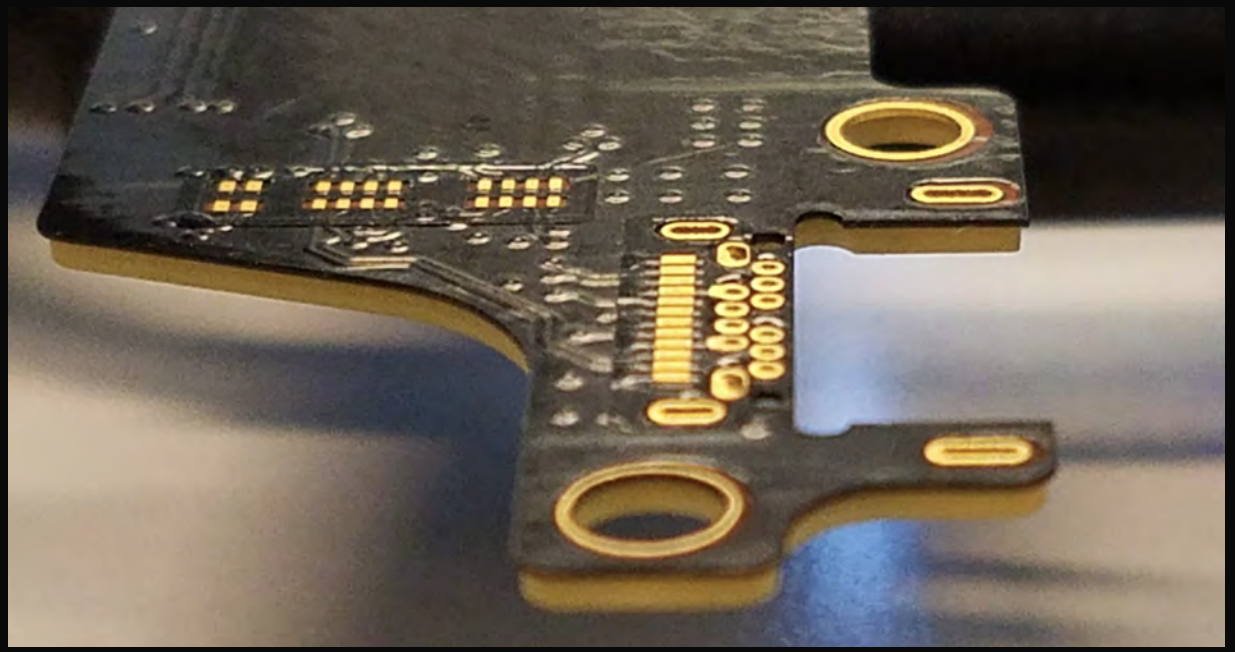
The 360 degree bends are the most stressful. The specific area that has to be actuated over and over is called a dynamic bend region. Special rules are in play.

When there are four or more layers in a flex, consider a loose leaf approach to the stack-up in the bend regions.

10.6 Flex To Assemble

The guidelines are somewhat relaxed depending on the minimum radius of the bend. A one-time bend that is over a small 180 degree arc is typical so that the flex circuit can double over on itself to save space. That type of bend will take more effort than a mild bend over a large radius.

The fabricator can pre-bend the flex so that it has a natural tendency to bend in the required direction. In any case, there is a ratio of flex thickness to minimum inside bend radius. A safe number is ten-to-one and the most radical bend would be twenty-to-one. This will depend on the specific stack-up. A ballpark estimate for the thickness of a three-layer FPC including top and bottom coverlay will be 200 microns or 0.2 mm. The safe bend radius will be 2 mm measured from the inside of the curve.



Section 10. Figure 2. Image 10.2. Credit: Author - A USB Type C connector at the end of a flex.

The first rule of the bend region is to have no vias in the region or even the immediate vicinity. Vias are like little I-beams through the z-axis that act to stiffen the flex.

Flexing them against their will causes them to crack and fail. When an FPC is bent, there is compression of the inner side of the bend and elongation on the outer side.

Something has to give. The most reliable place for routing is right in the middle of the stack-up.

A really long bend region might require a few "redundant" ground vias for the sake of EMI suppression. Special coatings can be added in lieu of the vias. The thing is that the more material you have, the stiffer the flex becomes. The best compromise isn't always clear.

The second thing is to consider the direction of the bend with respect to the direction of the traces. The traces should run straight through the bend region. A wide trace should be necked down or divided into multiple thin traces that join back together after the bend. Solid shapes are generally discouraged in flexes but if used, should convert to a mesh through the bend regions.

For the sake of impedance control, it is common to use a mesh with a certain ratio of metal-to-open area. The cross-hatch usually resembles a chain-link fence. Depending on the outline, the direction of travel may put the traces in alignment with the mesh lines. So, if a flex has a branch that launches on a 45-degree angle, it would be wise to consider rotating the flex to a 22-degree angle or whatever is appropriate for the outline geometry.

The same is true for multiple trace layers in a flex. Traces should not run right above other traces. Stagger one layer so that it runs in the gaps of the other layer even if there is a mesh in between the two. High speed or any other controlled impedance on a flex circuit usually involves three layers. The outer layers are ground mesh and the inner layer is for signals.

A failure in the ground mesh is less likely to cause the entire flex to fail. The outer layers are the stressed layers while the center one isn't stretched or compressed as much.

To get higher signal density using a narrow strip of flex material, the traces can be run on the outer layers with a mesh between them. Ultimately, the mesh can be deleted and you would have just the two layers or even one side for traces and nothing on the other side. If the traces are going to be exposed like that, the enclosure of the unit has to account for radiation into and out of the box.

10.7 Dynamic Flexing

More measures are required for flexes that are in motion. Obviously, you want the largest bend radius that can be managed. Picture the area inside the laptop hinge to know what we're up against. Go with a wide flex rather than a thick multilayer flex. Adhesiveless coverlay is slightly thinner - and a winner!

The changes in linewidth have a gradual taper and come before and after the flex region, not right at the border. Just say no to vias in a dynamic flex zone. It may help to cut slots in the FPC to reduce the mass where the flex sees a lot of bending.

The edge of the flex material should have a copper line that serves as a tear stop to prevent any cracks from propagating to the signal traces. This is especially critical on any inside bends of the outline.

It is common to have a flexible area just after the connector where the outline gets wider to accommodate the connector. That creates an inside bend at exactly the wrong spot. All of the traces are being funnelled into the flex zone.

The border between the stiff area for the components and the flex area is a high stress point. This is true whether there is a bend-region in that location or not. Treat those boundaries as if they were dynamic bend regions. A bead of epoxy along the edge of the stiffener will act as a stress relief.

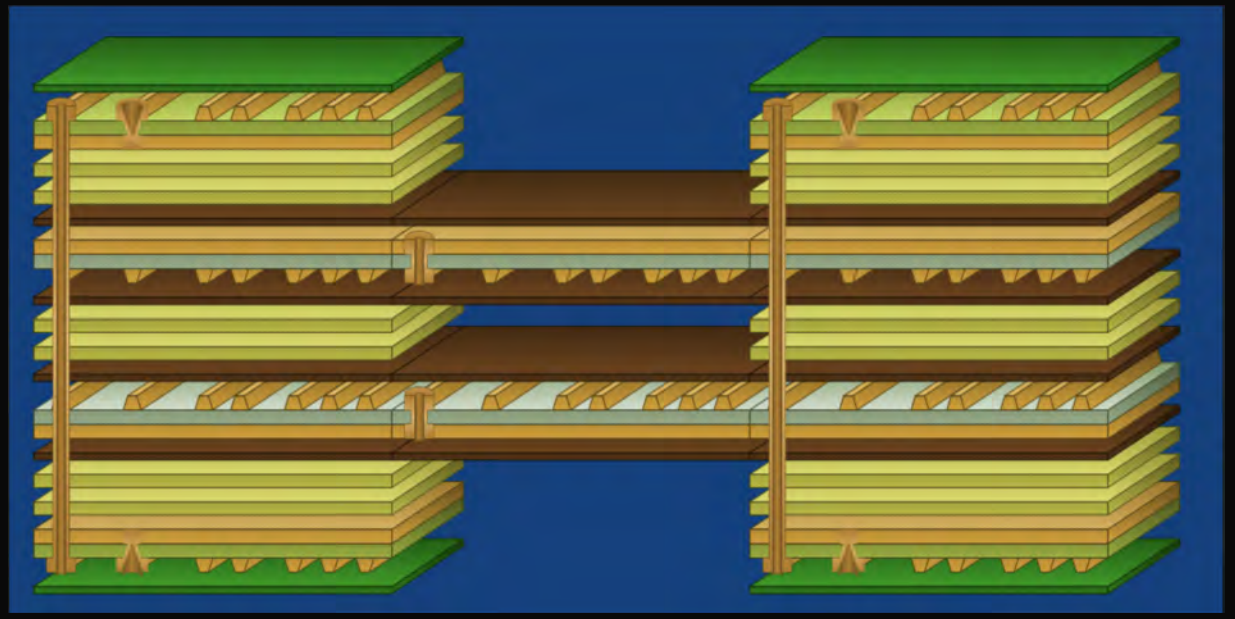
10.8 Rigid Flex Designs

The above design guidelines still apply, Fewer layers is better. Less copper is better. One way to mimic fewer layers is to not glue all of the layers together. When there are four or more layers in a flex, consider a loose leaf approach to the stack-up in the bend regions. A workaround for a high layer-count is to create flex cores on multiple levels so that two or three pieces of flex are embedded between the rigid materials. Using three stacked flexes with two or three layers each is better than a six-layer flex region when it comes to pliability.

I would be careful about how close the vias are located with respect to the rigid to flex transition area. Similar to a regular flex, pulling the via back 0.4 mm from either side of the transition zone should keep you from getting a call/question from the vendor. Also note that the coverlay tucks under the rigid section as it did the stiffener.

When you have a trace running over/under/between a mesh layer in the flex region, the mesh should continue into the rigid section.

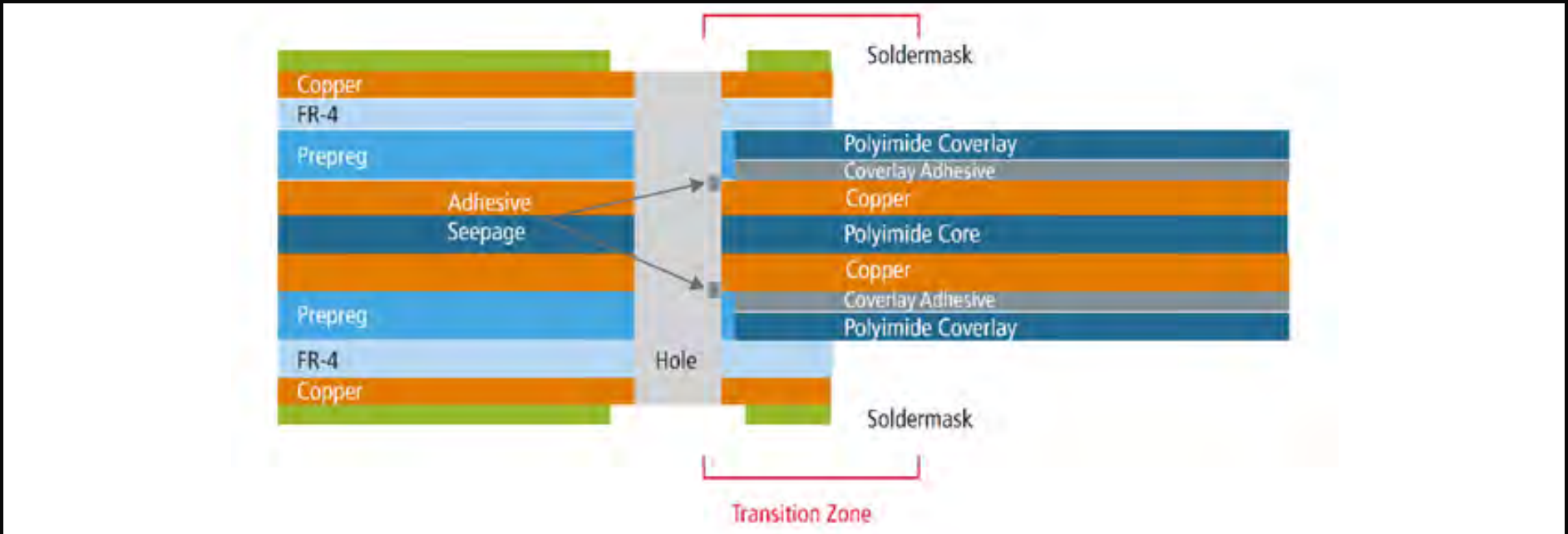
Not to make the rigid section more flexible but to keep the same impedance geometry along the path of the traces.



Section 10. Figure 3. Image 10.3. Credit: Polar Instruments - A rigid/flex with two stacked flexible subsections.

You could make the mesh global on those layers but it's really only needed over and under the controlled impedance traces that run across the flex.

If you know how to do flex and rigid boards, then rigid-flex is just about combining the two in a stack-up and managing the space where it turns from one to the other.



Section 10. Figure 4. Image 10.4. Credit: OrCAD - Subtle distinctions in the transition zone.

11. Multi-Board Systems

Section eleven uses a smartwatch as an example to illustrate rigid, flex, rigid-flex, and a molded board to complete the circuit.

11.1 Multi-Board Systems

11.2 Design Review on Day One?

11.3 Mother Card to Daughter Card: A Flex Usually Spans the Gap

11.4 Massive Data and Voice Transport Systems

11.5 Computers and Peripherals - A Tightly Connected World

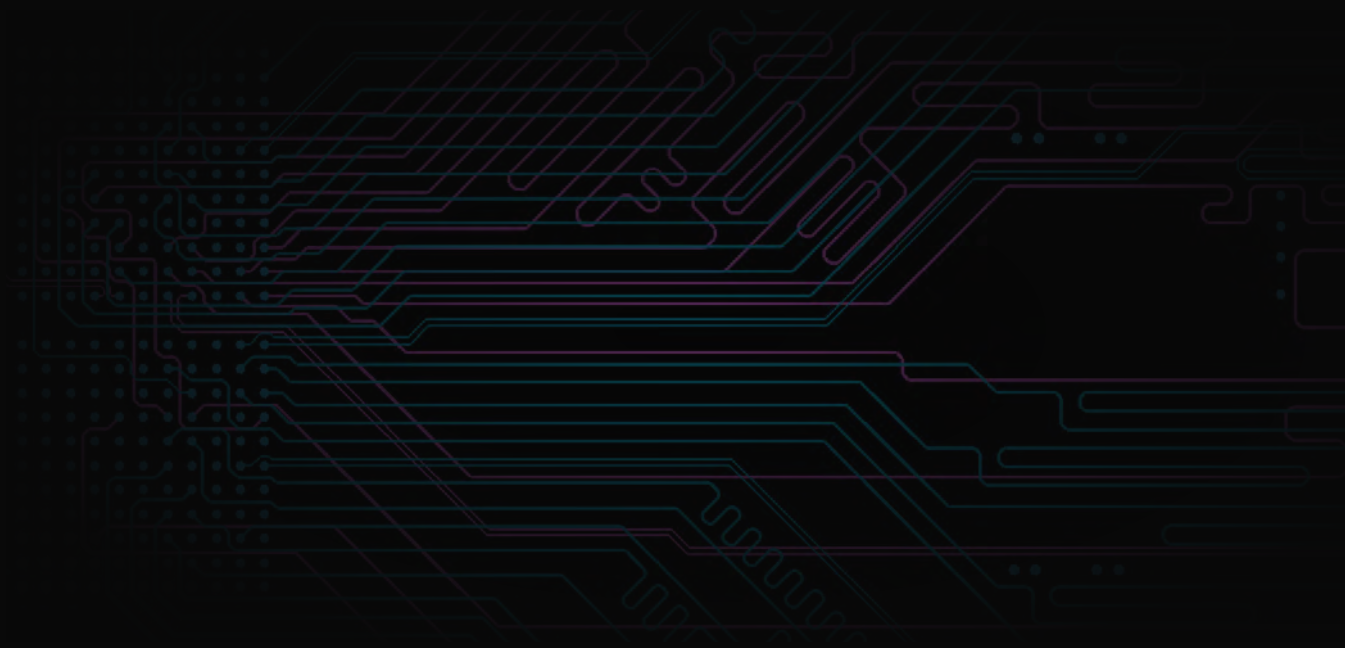
11.1 Multi-Board Systems

Thanks for sticking with me after that brutal lesson on flex circuits. Those same FPC's form the ABC's of multi-board systems. The rigid-flex designs are a culmination of years of putting things together in the most efficient way. The efficiency comes in the use of less overall volume (without the connectors).

While those interconnects are typically small in number, there are exceptions when the flex zone ends up with hundreds of connections. All of the various rigid areas are driven off of the same schematic. The flexes in between go along for the ride. Going for the rigid/flex investment benefits by bringing all of those connections under one roof.

Part of that investment is the RFPC design time. Lately, that design time is at home.

Whether you are doing the rigid-flex or separate rigid and flex boards, it's going to require a crisp hand-off from mechanical outline to PCB design rules. Saved by the internet! We can screen-share anywhere.



11.2 Design Review on Day One?

Right after importing the outline geometry is a good time to ask for a screen share with the Physical Designer. Getting a view of the big picture where a flex circuit is shown folded into its final location is next. Finding out where she can give you a little more or not is helpful. Can the PD explain how it is assembled?

Can you follow pin one by net name through the system interconnects from end to end? There is board A connected to flex B on to board C. All the connections on the inter-board flexes are meant to be straight shots with all of the traces running side-by-side. A high number of flexes are built to translate between two incompatible connectors. By getting those two PCBs to join, there's some kind of new function to be had.

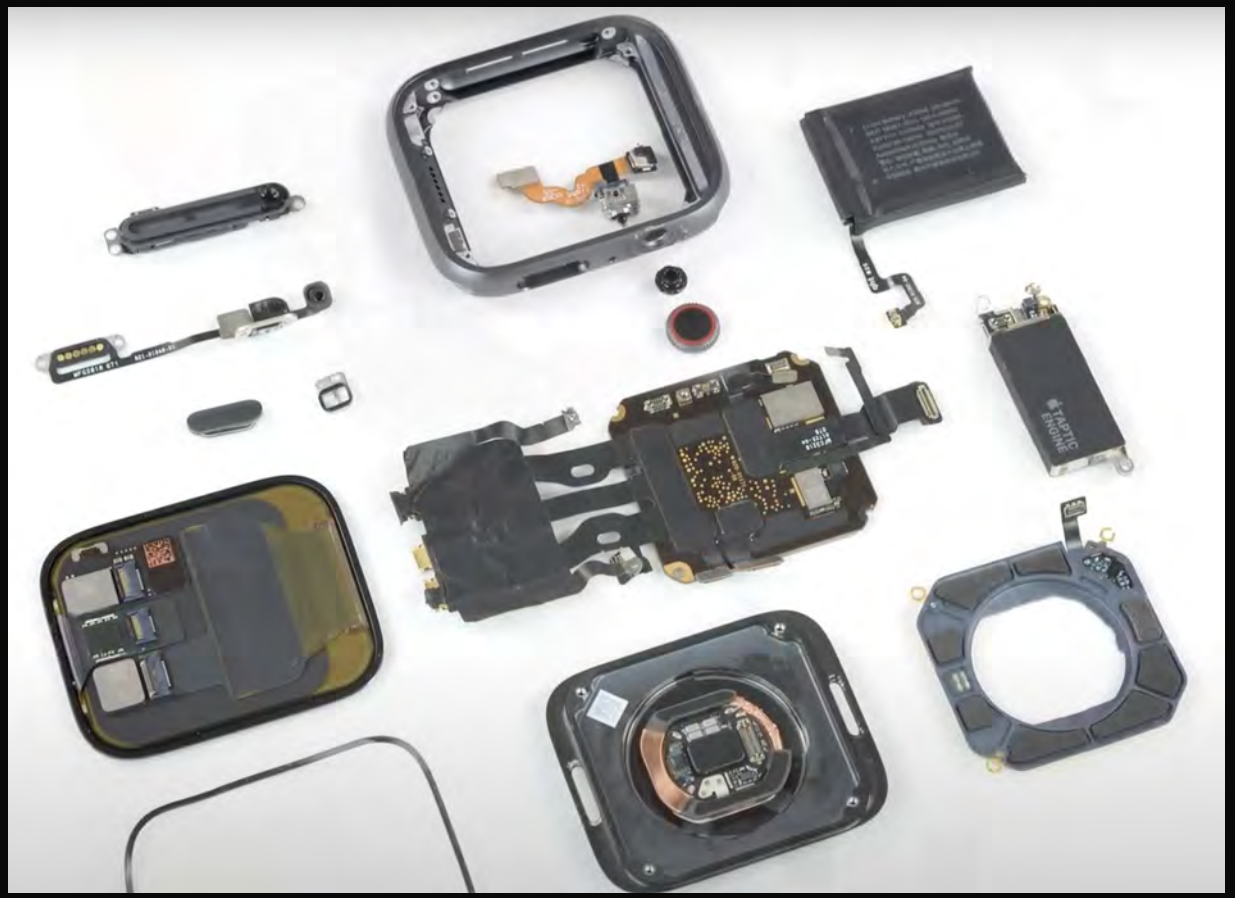
Solving those incompatibility problems with something that isn't a "science project" keeps the FPC designers busy. This path is usually shorter than the one where the two original PCBs are combined as one - even if you own the IP for both.

Note that connectors and custom cabling can also save your project if the inter-board routing is somehow compromised. There is a lead time and a tooling charge that will be highly dependent on where you go to ask for a one-off or limited run of connectors. It might be easier than searching for all of the possible connector configurations related to your parameters. There are a staggering number of off-the-shelf connectors. Cabling them up one way or another is just a thing. Call it out correctly and you're done.

11.3 Mother Card to Daughter Card: A Flex Usually Spans the Gap

I've found the most synergy is gained in the wireless sector or wherever we're sensing things. Sensors and antennas are especially nice to isolate. Ever wonder where they put the Wi-Fi antenna on your phone or laptop? When we're talking about the phone coexistence of all of the different radios is a real chore. The little antennas are scattered as much as can be. You have to use the Wi-Fi to get the stream and the Bluetooth to hand it off to the speaker. Solving that conundrum on a watch takes some flexibility and usually lots of time.

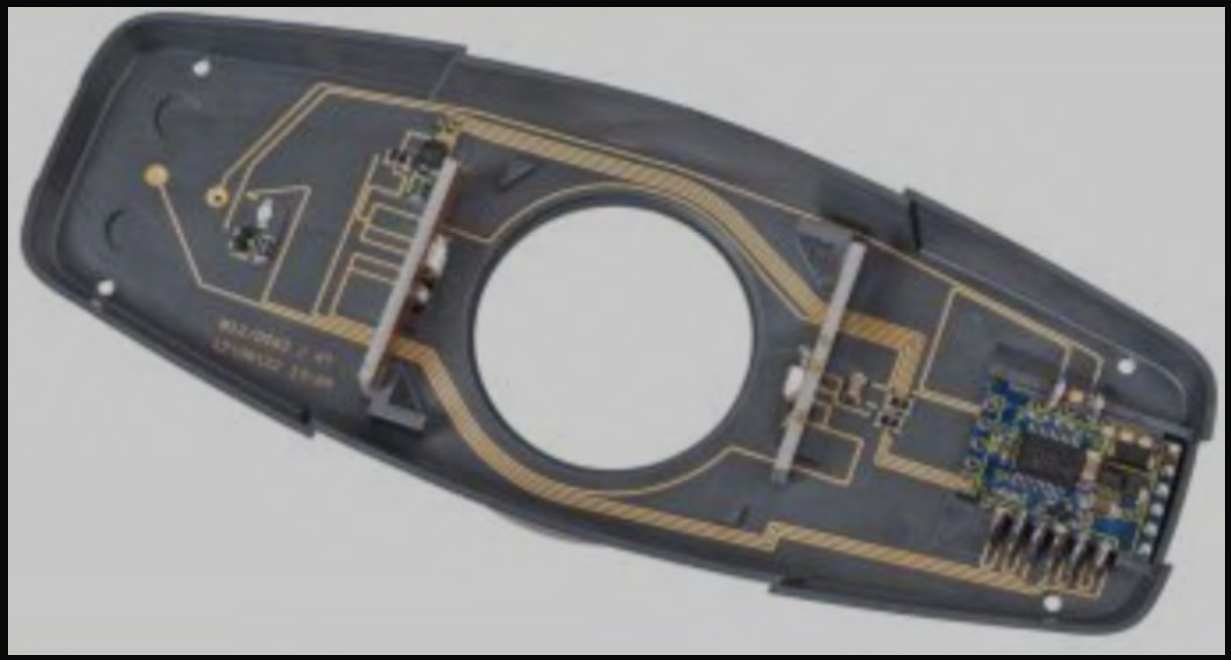
Whether it is capacitive charging or counting steps, the features run on some kind of hardware. It wouldn't surprise me if the watch has a barometer. The integration of different circuit boards is astonishing.



Section 11. Figure 1. Image 11.1. Credit: IFIXIT - A watch teardown shows several flex connectors and even a molded board on the left.

For a long time, all I wanted was for the housing of the device to be responsible for RF shielding. Just mold that shield into the cover!

They did that and more. Molded boards such as the one above and below are part of the external material. Several devices are mounted and routed. Of course, one-layer boards are the easiest. Note below a small PCB on the right and the housing both have a few components mounted.



Section 11. Figure 2. Image 11.2. Credit: Lee Teschler, Design World - Antennas and simple circuits can live on a 3 dimensionsional space.

11.4 Massive Data and Voice Transport Systems

Two and three board systems are only the beginning. Before I got into PCB Design I was building seven-foot high equipment racks for AT&T. There would be six large card shelves stacked up between heat shields and a few two rack-unit enclosures. (1) Each of the shelves held two groups of eight boards plus a 48 VDC power supply.

That's 54 big 10.5 x 14" boards and whatever was inside the smaller enclosures. Our little factory in Santa Clara stood up 350 of those racks in a quarter. An empty rack with just the back-panels (bristling with 640 pin connectors) sold for the price of a mid-size Mercedes.

Fill it up with analog to digital converters, D to A converters, modulators, demodulators, a very interesting analog T1 interface board and a couple of others.

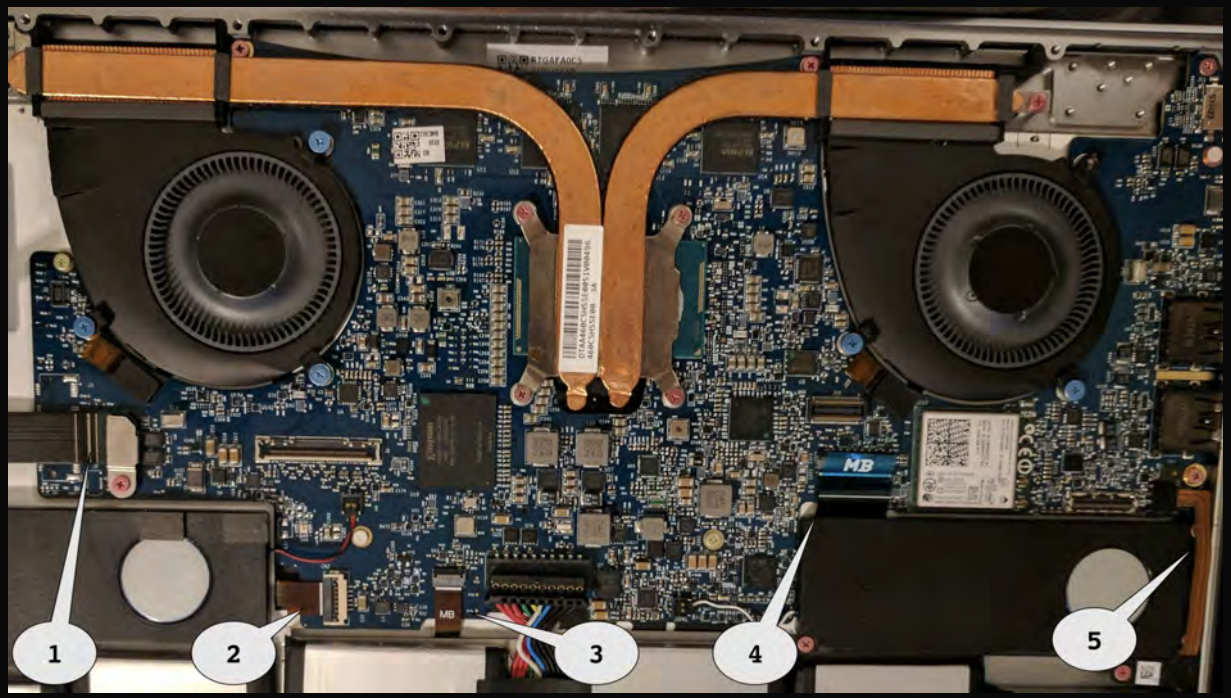
Now you're talking about a mid-size Bay Area house. Times 350; all to become part of a massive multi-board system. The internet backbone is one wing of this system.

Ever wonder where they put the Wi-Fi antenna on your phone or laptop?

11.5 Computers and Peripherals - A Tightly Connected World

Once the internet was up and running, the world needed to shrink the computer and go mobile. So, after years of telecom, I got into consumer electronics. Below is a gut-shot of one of the laptops I got to do when I was with Google. We wanted the longest battery life, the highest resolution, the most accurate touch screen, etc, etc, etc.

The Pixel 2 laptop was a showcase of 2015 technology for developers to have all of the possibilities we could think of. My wife is not a developer but she uses hers every day. It still runs long, quietly, and cool with those two exhaust fans. Step 3 of the teardown instructions is optional but I highly recommend it. “Step 3: Admiring the Compact Chaos”.



Section 11. Figure 3. Image 11.3. Credit: IFIXIT - A handful of flex circuits tie various functions together.

That first flex is special to me. One of the design goals was a rugged laptop. I don't encourage you to try it but the laptop has been tested with a three-foot drop onto concrete. Drop it on it's edge, on the corner, let it slam flat on the bottom or top first onto the concrete pad. Make note of whatever breaks and redesign it to be stronger. This laptop has a USB-type C connector on either side of the chassis; an industry first.

The one that is connected by the USB-flex failed when it became detached from the connector on the main logic card. As soon as I heard about the failure, I started re-routing all of the traces that passed by the board-to-board connector. Fortunately, a day later, the mechanical team came up with a one-screw solution that cantilevered out over the connector to keep it from disconnecting.

The thing we didn't see coming was the stiffness of the flex. There are four superspeed lanes, two traditional USB 2 lanes, and two pairs of power pins along with ground. While only half of the connections are in use at any time, all of them have to be provisioned so that you can plug it in without wondering if it is upside down. A six-layer flex made sense but a wider four-layer FPC might have been better in hindsight.

So, when I'm looking at my wife's laptop, I can picture the guts and plug into the USB port that is soldered directly to the main logic board. If you need to review section 10 on flexes, you'll note that the USB flex image is that exact one. The Pixel doesn't fall far from this PCB designer.

(1) Rack units are 1 ¾ inches high and 19 inches wide. The bolt pattern of the "rack ears" is common to many racked products like telecom but also pro-audio gear. Our shelves were six rack units and the boards were installed vertically like books but into card guides that aligned the board connectors with the back-plane connectors.

12. How to Design for Test

Section twelve outlines test methods and provides some advice on how to implement ICT using telecom equipment as an example.

12.1 How to Design For Test

12.2 Placing Test Points Prior to Routing

12.3 Single-Sided Test Access is Preferred

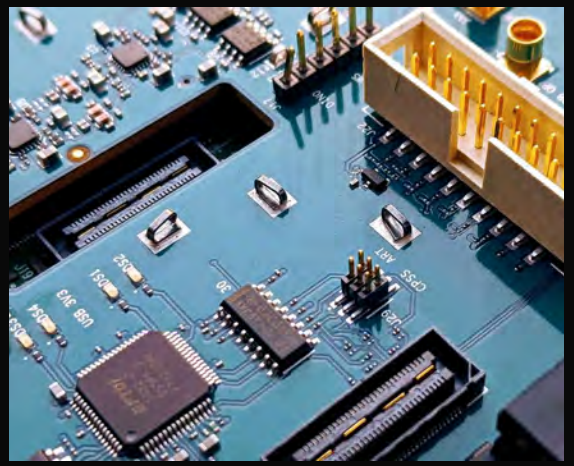
12.4 Graduate School Zone

12.1 How to Design For Test

We design for a lot of different factors. One of the key factors is to be able to isolate and replace defective components. Beyond non-functioning parts, defects can result from polarized components that were placed in the wrong orientation, incorrect part values, and bad solder joints. Solder joints can go bad in numerous ways creating opens or shorts, the worst of which are latent defects. They look good until they don't.

Test access isn't cheap. It takes up valuable PCB space. More than just the size of the probe point, the points have to clear other components, as well as maintain a spacing from one test point to another. Tall components require extra spacing.

Oftentimes, the goal is to look at a few key nets for testing by hand. These are touch-points so a larger reference designator and perhaps a functional description taken from the net name would make the board more user-friendly. Those cases may include terminals where a meter can be hooked to a physical pin. In that case, they are included in the schematic and placed along with the other components. The test hooks are usually surface mount but also come in through-hole versions.



Section 12. Figure 1. Image 12.1. Credit: Author - Basket handle loops can be used to clip the probe hook to the ground or other net.

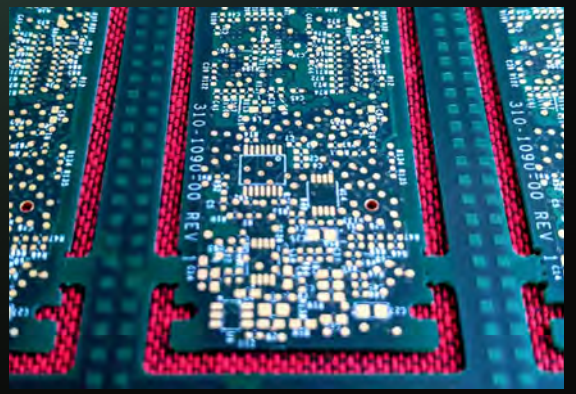
12.2 Placing Test Points Prior to Routing

In-circuit test points should be distributed across the board without being concentrated in any one area. Even placing them in a nice line of test points with more than the required spacing can be a problem. The bed-of-nails test fixture puts pressure on the board so a test point pattern that meets the written design criteria can be rejected by the test engineering team.

If "they" had their way, I'd put all of the test points in a special pattern over and over and then place and route the board around that. To be honest, that's over-stating it but I bet you could find people who would get behind a one-size fits all array of test points. A good ICT plan cannot be an afterthought. Placement and fan-out studies are more tedious when incorporating test points.

The pay-off is that it is much more tedious to graft them onto an existing layout where everything is already buttoned up.

The goal is to have a test point on every net. But wait, there's more! Distribute a handful for each power domain and salt the whole board with ground test points. It got to the point where I don't like resistors below a certain value because the TE wants two test points on either side of those resistors; something about the resistance of the probes themselves. Ok, we need to plan ahead for these things.



Section 12. Figure 2. Image 12.2. Credit: Author - The secondary side of the board is usually a better place for the ICT points if you're only using one side.

12.3 Single-Sided Test Access is Preferred

It would be nice if you could get all of the test points on one side of the PCB. A regular bed-of-nails is expensive. A clamshell fixture is really expensive. Another cost driver has to do with the size of the test points. If you could set your grid to 2.54 mm and then use every other point and stagger each row, they might like the test plan. Welcome to the real world.

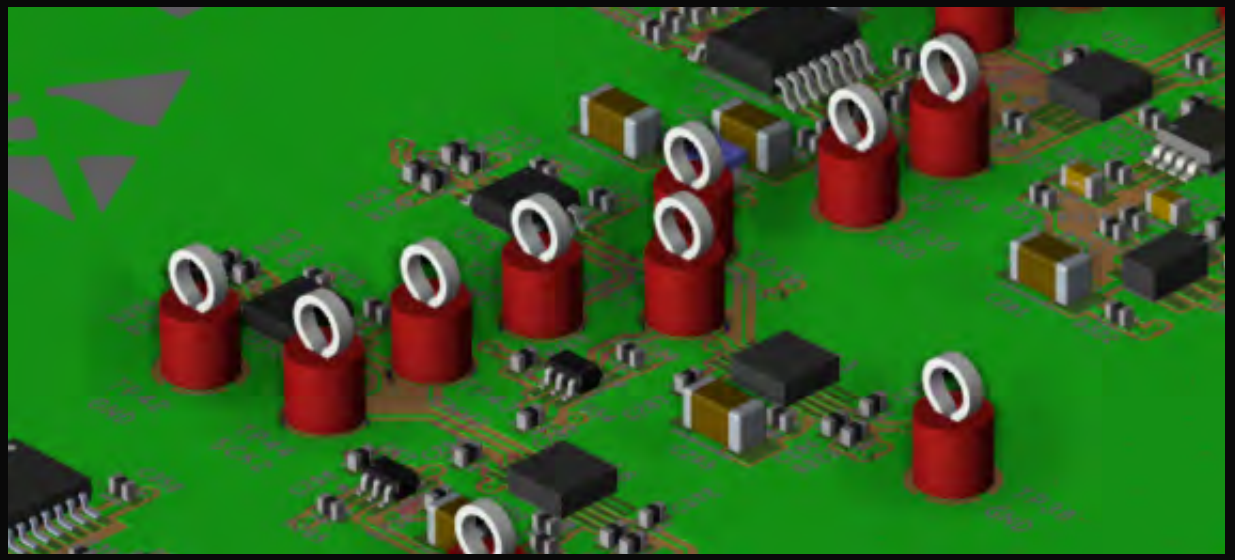
Life is a little better when the plan is to use a flying probe head. For one thing, you can do a revision of the board and they can reprogram the machine to find the new test point locations. The probe points can be a little smaller and it isn't such a big deal to distribute them in an irregular way. The trade-off is that the flying probe machine takes longer to complete each board as it goes around with two robotic probes and checks each connection one-by-one.

Beyond non-functioning parts, defects can result from polarized components that were placed in the wrong orientation, incorrect part values, and bad solder joints.

12.4 Graduate School Zone

Getting a little more real is giving up on 100% coverage and only testing power and ground nets. The thought process is that there are too many connections for the amount of space between the components anyway. Add in the percentage of traces where a test point would be too much of an impedance discontinuity and all that's left is the power and ground planes plus a few points of interest.

As density becomes too much for even that, the fall-back plan is more about functional testing. Boundary scan via a JTAG connector. Boards that fail are set aside. If the tech can't troubleshoot and repair the board in a few minutes, it's not worth the effort in some cases.



Section 12. Figure 3. Image 12.3. Credit: Author - Alternative test points can take several forms. This type allows the tester to hook the probe to the test point to have a free hand.

The factory wants the product to flow with a minimum of human hands on it. It seems wasteful but hand-operations aren't a money maker.

So called second-op tasks that happen after the PCB comes out of the reflow oven have to be eliminated for mass production. It's a cruel world.

13. Solder Mask

Section thirteen looks at quality control and what happens in a receiving inspection lab when PCBs come into the factory.

13.1 Solder Mask

13.2 Solder Mask Slivers - A Leading Cause of PCB Defects

13.3 Containing the Solder During Reflow

13.4 Vias and Solder Mask

13.5 Graduate School Zone - Solder Mask Defined Pads

13.1 Solder Mask

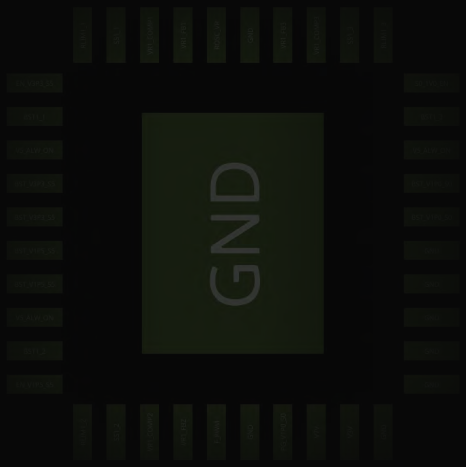
Solder mask is a vital piece of the PCB puzzle. When looking at a bare printed circuit board, you see mostly the mask and the ink. Oddly enough, they are often left for last as we focus first on the actual connectivity. Let's start with the solder mask and we will get to the marking in the next section. There are two basic types of solder masks, liquid photo-imageable (LPI) and dry film with LPI being the most common.

Dry film photo imageable solder mask is more like a sticker or two stickers held together with the solder mask. One side is peeled off and the remaining material goes over the entire board mask-side down. The areas where the metal is to remain exposed are masked off before curing it with ultraviolet light. Peel off the remaining side and clean the board. That's the short version. Liquid mask follows a similar process except there's no sticker.

Section 13. Figure 1. Image 13.1. Credit: Author - The padstack editor is where the mask is defined for the footprints.

It used to be green 100% of the time but then we got blue followed by one color after another. Black is the new blue as far as popularity. It provides the highest contrast with the white or yellow ink of the silkscreen and it looks cool too. You can buy a solder mask in a little bottle with a brush applicator or a pen to do touch-up work. For production, it comes in gallon-size jugs on up to big barrels.

The low-cost method of applying the LPI mask is by silkscreening it onto the board. The more accurate and costly method involves lasers to define the features. It is essentially an ink that cures to a high melting point and doesn't readily burn. It will affect the impedance of outer-layer traces so there is a good reason to use a specific kind and control it to a thin coating.



13.2 Solder Mask Slivers - A Leading Cause of PCB Defects

Solder mask adheres well to the surface but a narrow segment may still flake off. The so-called solder mask slivers are a reject. The designer needs to account for the minimum width of 100 microns (four mils) for a solder dam between any two mask openings. The IPC has developed a test method where adhesive tape of a specific stickiness is stuck on and ripped off of the board. (1) Then, we can check the tape to see if anything, conductive or non-conductive came off.

Once upon a time, I was the receiving inspector for the wireless group in a telecom outfit. When PCBs came in, the tape test was one of my jobs. We had the mylar 1:1 artwork to compare the circuit pattern on a light table.

We had a machine that measured the thickness of the gold on the fingers and another that determined the average thickness of the copper barrel in the via. I could check for cracks in the barrel under a microscope.

We didn't mask over the vias back then. I think most fab shops would still prefer that you didn't. The holes don't always fill completely with the mask. That potentially leaves a little pocket of air that might cause a blow-out when the board goes through the soldering process. It's the shrinking electronics that drive the situation.

Anyway, there were pin gauges to check hole sizes, feeler gauges and a big block of granite to evaluate bow and twist.

My lab even had an optical comparator to show the silhouette at 10X actual size to measure details of the board outline. It was an interesting job but I left before a toxic manager took over. I gave a two-week notice right there in the meeting with the out-going manager when the announcement was made. Then I lined up a cush job running a stock-room that lasted until I went off to ECAD school.

So, I digress. Out of all of the possible defects, more boards failed the tape test than the others. That knowledge stays with me as I design my footprints and boards. Most of the other issues were cosmetic having to do with contamination and a few were not flat enough to pass the spec. Working in assembly and inspection was a good background for the work I'm doing now. Moving on.

13.3 Containing the Solder During Reflow

Solder mask protects the bare copper while keeping the solder from spreading beyond its intended location. Plated through-hole and surface mount components both require a mask to ensure proper solderability. Through-hole pins can be drag soldered where the board is lowered into a bath of molten solder. Another method is wave soldering where the solder is made to well up under the stationary board.

Finally, there is pin-in-paste for mixed technology boards. The idea here is to add a relatively large area of paste on the side where the pins protrude through the board. The size of the paste stencil opening will be much bigger than the solder joint. The idea is to deposit enough paste in the area to fill the space between the pin and the hole with extra for a solder fillet.

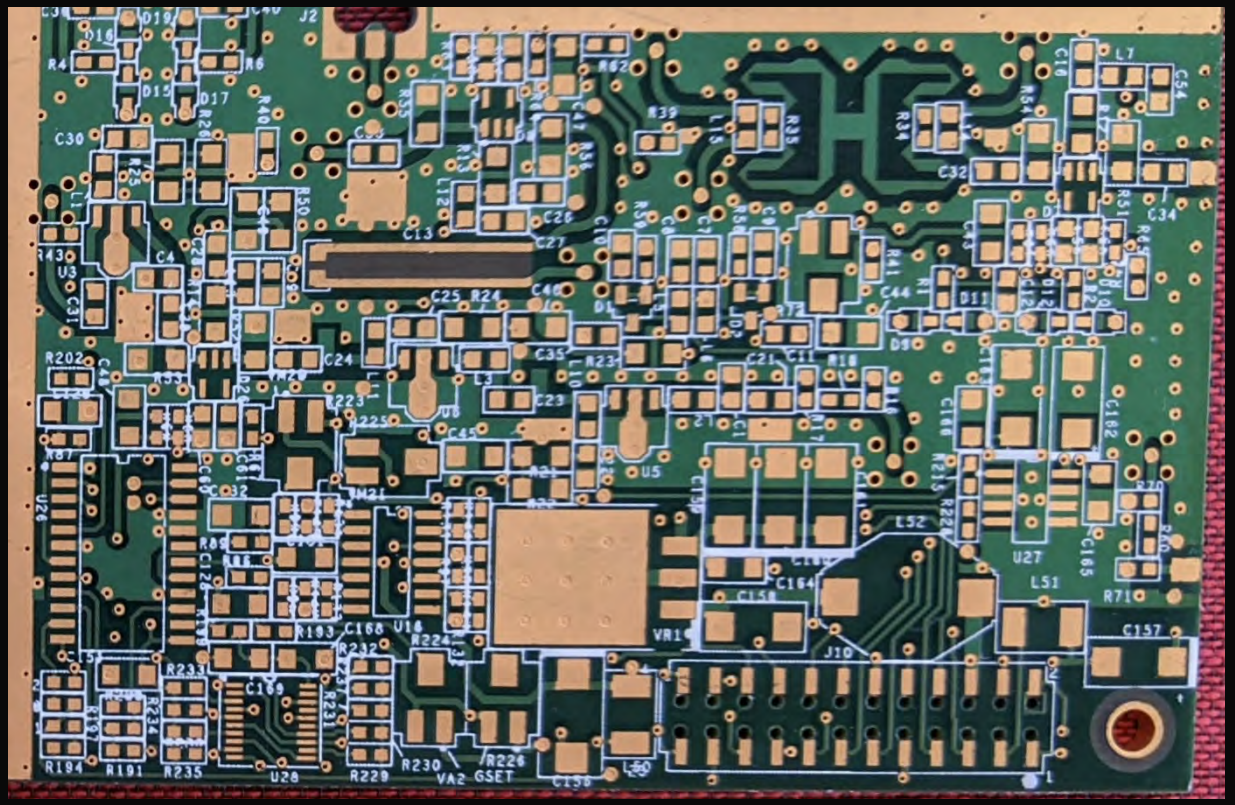
Besides the hole diameter and pin size, the other factors are the thickness of the board and the thickness of the stencil. The paste is only added on the secondary side; it will be acceptable to have no solder fillet on the through-hole component side. It also means that the area around the bottom of the through-hole component has to be clear of other components or exposed metal.

We're counting on a wicking action to draw in the solder paste while the SMD components are also reflowed. While this process can be tricky to dial in, the benefit is that the through-hole and secondary side components are all soldered at the same time. I've also seen over-pasting like that for BGAs where the aperture is a square that surrounds the circular BGA pad.

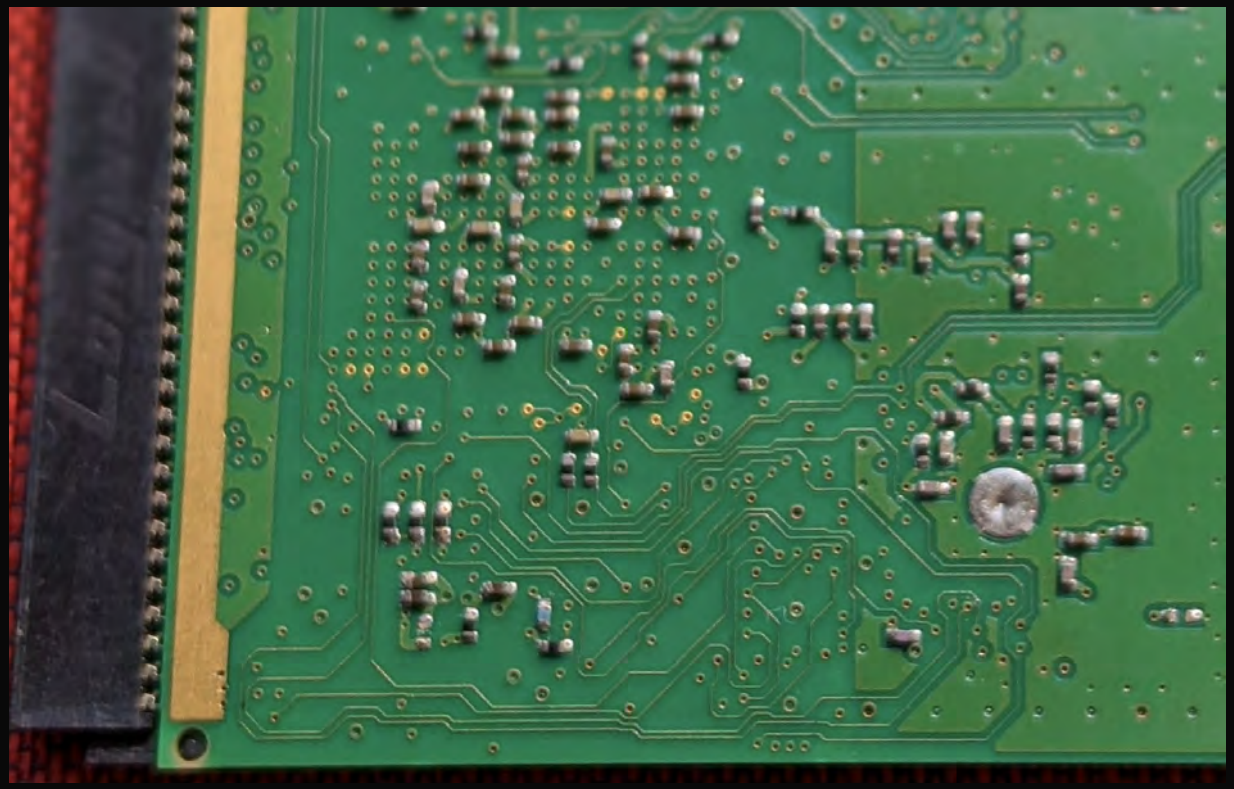
13.4 Vias and Solder Mask

Most PCBs have the vias covered with solde rmask. When the vias are designed with the solder mask opening, the fan-out vias have to account for the minimum dam plus expansion of the mask over both the soldered land and the opened via. Without the dam, the solder is prone to migrating into the via and leaving the connection with insufficient solder.

Good design practices require short power and ground loops so the vias are best held closely to the SMD pad. When the vias are open metal, you have to strike a balance between the two requirements. The easy way out is to cover the vias with a tent of solder mask on both sides.



Section 13. Figure 2. Image 13.2. Credit: Author - Open mask on all of the vias can lead to some compromised fan-outs, especially on a densely populated board. The via-in-pad locations are filled while the vias in the random solder mask openings are left open.



Section 13. Figure 3. Image 13.3. Credit: Author - Underside of the board in a BGA area with selected vias open to access the pin on the device.

While it is common with BGA packages to tent the component side and relieve the solder side, the vendors typically squawk at that idea because it's hard to know where the solder mask is going to flow in that situation. The last thing they want to do is eat (or try to rework) those boards after getting all the way to the solder mask step.

Doing bring-up and trouble-shooting on analog systems can be a high-touch affair. A via that is tented with a mask doesn't allow access with a circuit probe. While the technician can scrape off a little solder mask for rework, it shouldn't be necessary just to take a voltage or current reading. It's not an automated test solution because the vias are too small.

Opening up the mask on all of the vias makes arranging the silkscreen more of a hassle. A workaround that satisfies most cases is to mask over the ground-vias while keeping the power and signal vias exposed. Finally, as above, a specific list of nets that require probing for bring-up are cleared of solder mask.

Board designers should be fluent in everything from angstrom units to inches.

13.5 Graduate School Zone - Solder Mask Defined Pads

Fine pitch components may be required to have solder mask overlapping the pad. The threshold is the 100 micron minimum width mentioned above. Take, for instance, a 0.4 mm pitch BGA. It will have a ball diameter around 250 microns. A rule-of-thumb is that the pad size can be a minimum of 80% of the ball size. The only way to fan-out the BGA of that pitch is with micro-via-in-pad.

Not many fab shops, especially in the United States, will sign up for a 200 micron pad for the micro-via. Going overseas, I will comfortably design with 250 micron pads for the via with 50 micron overall expansion for the mask. On a 400 micron pitch, that leaves the IPC compliant 100 microns left over for the solder dam. I may get down to 224 microns for the capture pad if the mask expansion has to be 76 microns. It's about silkscreening the mask vs. laser defined.

Bringing the fabrication for a 0.4 mm pitch BGA to the homeland, I would expect the vendor to require a 320 micron via pad which would then become the SMD pad size. That leaves an 80 micron air-gap. There's no option to expand the mask and keep the 100 micron minimum dam. Contracting the mask by the 76 micron value creates a mask-defined pad-size of 244 microns. That leaves us a very acceptable 156 micron solder mask width.

You probably have noticed that I don't usually go that deep into the numbers. When I do, it's usually the metric system because that is the unit of measure for most of our components these days. Even when the pin-pitch is 100 mils, I'm likely to call it 2.54 mm. Board designers should be fluent in everything from angstrom units to inches.

Avoiding the numbers makes you go and find out from your vendor. They all have a technology roadmap and no two are exactly the same.

14. Silkscreen Marking

Section fourteen goes over typical silkscreen requirements such as vendor’s required markings, UL certification and date codes.

14.1 Silkscreen Marking

14.2 Marking the Secondary Side - With or Without a Silkscreen

14.3 Proper Marking of the PCB

14.1 Silkscreen Marking

A good silkscreen can really make or break the final product. On the assembly side, the rotation and location are added to the reference designator to inform the placement. From a traceability angle, the part number, revision are a start. The location for other identity labels like serial numbers and the bar codes are defined in the silkscreen with an outline. The labels are subsequently referenced on the assembly drawing and/or bill of materials.

All of this usually goes on the primary side of the PCB along with your company logo and a description of the board function. Connectors and test points may have a secondary call-out of the net names in addition to the reference designator.

Larger components may have a system that shows the assembly and test group the pin designations beyond just pin-one. A BGA may have numbers and letters around the outside to indicate rows and columns. A quad-flat-pack or connector may have a little tick mark for every tenth row.

Safety information warns of high voltage or ESD sensitivity. The Reduction of Hazardous Substances (RoHS) compliance indicator looks like a little garbage can. If necessary the Underwriters Laboratories UL seal of approval will make the main side of the board complete.

Section 14. Figure 1. Image 14.1. Credit: Author - The vendor is responsible for certain information including the flammability rating and traceability.

14.2 Marking the Secondary Side - With or Without a Silkscreen

The secondary side is where we can expect to put the bare board part number and revision. If there are no components down there, the part number can be etched into a metal shape as a negative image or added as text if it is not a PCB with copper pour on the outer layers. The vendor adds their logo, date code and country of origin. This is usually applied in some way on the secondary side but can be inked onto the primary side.

The Quality Assurance people who inspect and/or test the board have a stamp they use when they "buy off" the board. They will put it anywhere unless you give them a special box for their stamps. Whatever is agreed to should be covered in a note on the fabrication drawing.

I try to do an adequate job on the silkscreen but I have to hand the crown over to one of my followers who added polygons around the functional circuits on the board with a label for each one. It was amazingly clean. I wish I could remember his name or at least have a screen-shot of the board.

There are not a lot of boards that I can recall where there was that much extra space but it does happen. The roomy boards like the one I'm using as illustrations in this chapter are typically internal-use test fixtures. There is usually an abundance of extra marking on boards used to bring-up and test integrated circuits in post silicon validation.

Section 14. Figure 2. Image 14.2. Credit: Author - A motherboard/daughter-card pairing with the typical stick-on labels for tracking purposes.

14.3 Proper Marking of the PCB

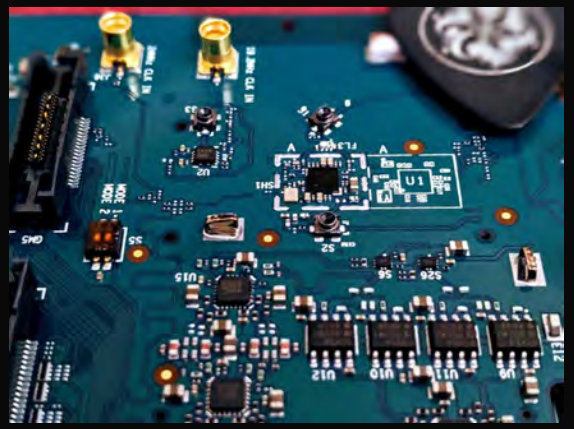
When you don't have enough room for all of the above, there is a hierarchy of things that can be left off. Before you make the decision to start removing any marking, you should get buy-in from the stakeholders in assembly and test. It is generally agreed that the first thing you can remove is the component outline and all of that extra text and tick marks that describe the pin numbers beyond pin-one.

"When the rush-schedule is over, the shortcuts taken remain."

As an aside, using a line between the pads of a passive component smaller than an 0603 package is not a great idea. The line will have at least some height which could act as a fulcrum to start the tombstoning process during reflow.

Going down to 0201 size and smaller components, it would most likely work best without an attempt to draw an outline around the individual parts. You're using those components because board real estate is at a premium.

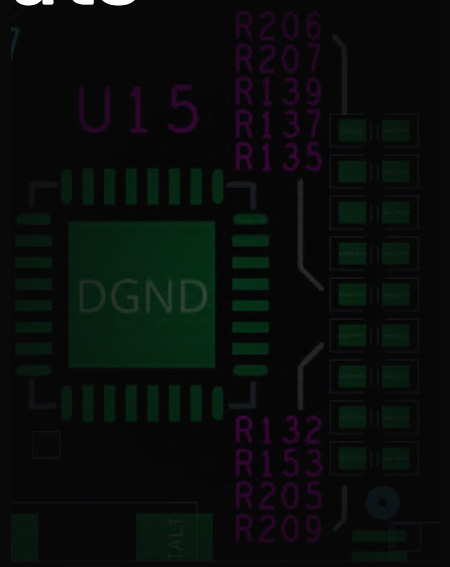
After removing the component outlines, the next thing to sacrifice is the reference designators on those same components. The exception is for test points and connectors which should always have a reference designator on the board. If you can find a remote location where the marking for a dense field of parts can be offset, that can be a fall-back plan. If I was taking that approach, it would be done late in the design cycle. Otherwise, I would advocate for arranging the text earlier to help with placement review.



Section 14. Figure 3. Image 14.3. Credit: Author - Quick and dirty marking scheme.

In *Figure 3*, the important items within the shield have remote labels. The touch-points are labeled with reference designators and functional descriptions. Many components are not marked at all. When the rush-schedule is over, the shortcuts taken remain.

“When the rush-schedule is over, the shortcuts taken remain.”

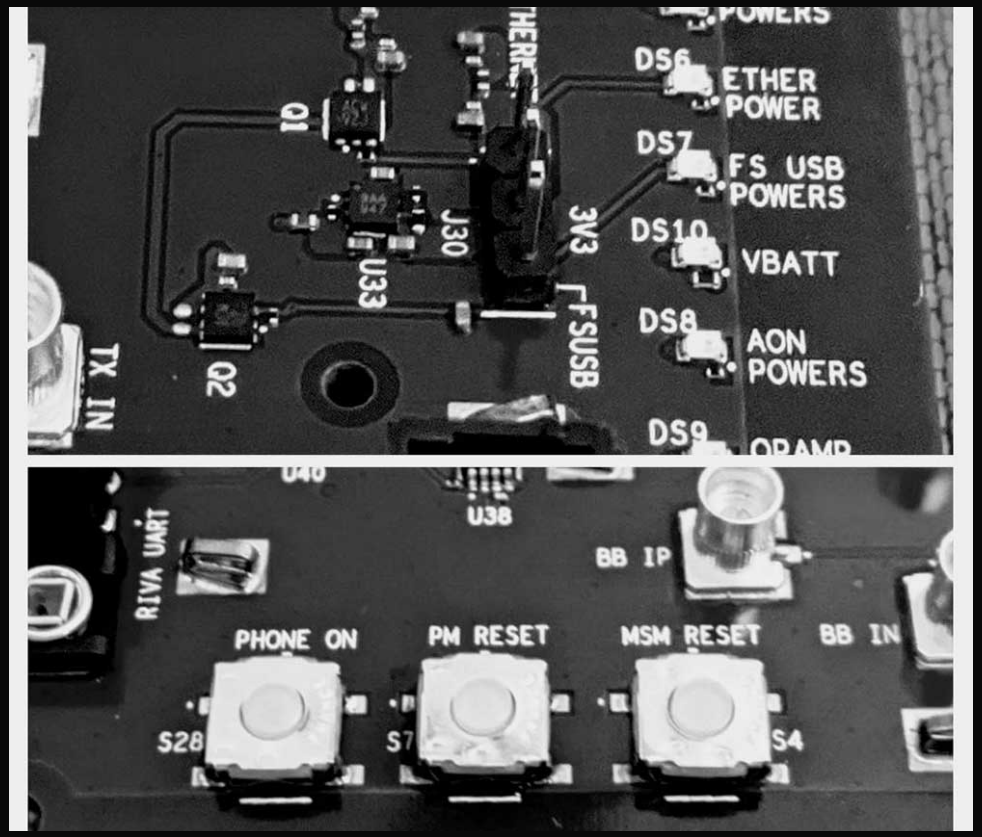


Count on the last minute of the schedule to be hijacked by something out of the blue. Strive to do your part a full day before the last minute.

Polarity marks are a safety and sanity thing so they should stay no matter what. The IPC suggests a dash that protrudes from the component outline rather than a dot. The dash takes up less space but is part of a line which helps it adhere to the board. The same kind of tape test used to check solder mask adhesion is used on the silkscreen.

If you helpfully added the text “3.3V”, I’d put my inspector hat on and get out my adhesive tape then rub it into that period really well so I could rip it off and reject your board. Call it “3V3” and we can all go home happy. It will help the text stay put if it has a stroke width of at least 127 microns. Even better would be 178 micron stroke width.

That thick manufacturable line width could make the tiniest text look blurry. Things like the top of the *R* fill in but the overall shape of the character is defined well enough. The assembly, test, inspection and field service people can usually make it out under good lighting and magnification. And again, just like solder mask, we can sharpen the linework using laser jet technology. Putting decent marking on the PCB is your way of letting all of those people know that you care about them. You ought to since they're depending on you.



Section 14. Figure 4. Image 14.4. Credit: Author - Buttons deserve a description as well as connectors, LED's and, of course, test points.

15. Creating a Document Package

Section fifteen discusses the items that should be included in a typical document package to ensure completeness of a tapeout.

15.1 Creating a Document Package

15.2 Standard Fabrication Notes

15.3 Assembly Drawings

15.4 Projecting Views

15.5 Standard Assembly Notes

15.6 Graduate School Zone - Formatting Data

15.1 Creating a Document Package

The CAD system can do a lot for you, then, you're on your own. Eyeballing the layers two or three at a time will help you find the hidden traps. A camera-ready board is just the beginning. Documenting the PCB requirements with dimensions and other details including hole chart(s), stack-up diagram(s), a list of intentional shorts, etc. puts the necessary guardrails around the fabrication and assembly.

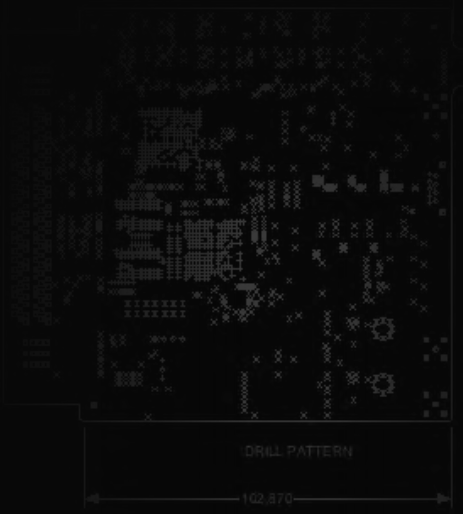
Fabrication notes will change with the technology used for the PCB. The general thrust of this list of example notes is for a multi-layer board targeted for consumer electronics with components on both sides and controlled impedance on inner as well as outer layers.

Section 15. Figure 1. Image 15.1. Credit: Author - A gleaming high-tech sensor factory.

Flexes or high reliability boards would have significantly different call-outs. In any case, there needs to be traceability for all of the attributes of the drawing. If you're calling out a specific process like cleaning with "ultra pure water", then it should be accompanied by a requirement for a Certificate of Conformance. Otherwise, who can tell if the process was followed?

Note 1A covers a lot of ground by itself. The notes that follow are best used to describe options within the spec and exceptions to the spec. For instance, you may call for the class 2 (as shown) but not want to allow the 90 degree break-out of the drilled hole from the pad. In that case, a note calling for no break-out, tangency permitted would be necessary and sufficient for that exception to the general class 2 fabrication rules.

If you ask for that option, you then have to make sure that it is possible by increasing the pad size so that the tolerance stack makes sense.



"If you want to have a good time, hook up with an outfit that will talk 2581 with you."

15.2 Standard Fabrication Notes

Notes: unless otherwise specified.

1. Standards:
Fabricate PCB in accordance with the current revision of IPC-6012, class 2. Interpret dimensions and tolerances in accordance with the current revision of asme y14.5m. Do not scale drawing.

2. Material:
Laminate and prepreg material shall be woven e-glass / epoxy in accordance with ipc-4101/126 or equivalent. Equivalent material shall be rohs compliant, halogen free with a minimum tg of 170 degrees c and approved by company. Thickness of individual copper clad sheets shall be as defined in stack-up diagram.

3. Flatness:
Bow and twist of assembly sub-panel or singulated pwb shall not exceed .025 cm per cm. Test in accordance with the current revision of ipc-tm-650 2.4.22.

4. Etch geometry:
Measure width from the base of the metallization. Minimum line width: 0.nn mm outer, 0.nn mm inner layers. Finished line width and terminal area shall not deviate from the 1-to-1 master pattern image by more than +/- 0.025 mm or 20%, whichever is less.

5. Surface finish: (select appropriate finish):
Enepig plating in accordance with the current revision of ipc4556. Exposed metal shall have 118–236 micro inches electroless nickel, 2–6 micro inches electroless palladium, and 1.2 micro inches gold.

Enig plating per current revision of IPC-4552. Exposed metal shall have 118–236 micro inches electroless nickel and 2–5 micro inches gold.

6. Destructive testing:
Micro section sample and report shall be provided to company design engineering. Solder sample processed through lead-free soldering shall be included with each shipment. X-out panels may be used for solder sample.

7. Holes:
Plating in holes shall be continuous electrolytic copper with 0.025 mm minimum barrel thickness. Minimum finished hole size: 0.nn mm. Hole size measured after plating.
See drill chart for finished hole size and tolerance. All holes shall be located within 0.08 mm of true position as supplied in cad data.

8. Solder mask:
Solder mask over bare copper (smobc) on primary and secondary sides using supplied artwork in accordance with current revision of ipc-sm-840 type b. Color: matte green. Liquid photo-imageable (lpi) 0.001 mm to 0.002 mm thickness, halogen free. No bleed-out allowed over exposed smd pads. No exposed traces.

9. Silkscreen:
Silkscreen primary and secondary side with white epoxy, non-conductive, non-nutrient ink. Any unspecified stroke width shall be 0.13 mm. Clip silkscreen away from any exposed metal. Vendor date code, logo, ul and any additional marking to be located on the secondary side. Bag and tag acceptable for pwbs that are too small for marking.

10. Remove all burrs and break sharp edges r0.01 min.

11. Non-destructive evaluation:
All pcbs shall pass 100% electrical test using supplied ipc-356 netlist in accordance with current revision of ipc-9252, class 2. Certificate of conformance shall be supplied with each shipment.

12. X-outs:
X-out boards that do not meet all specifications using permanent marking on both sides of the affected pcb. Panels that do not have any x-outs shall be packaged together. Panels that have n or fewer x-outs shall be packaged separate from non-x-out panels. Panels with more than n x-outs shall be rejected.

13. Packaging requirements:
Pwbs shall be packaged in vacuum sealed inner containers. Outer containers shall be sufficient to prevent damage during shipping and handling.

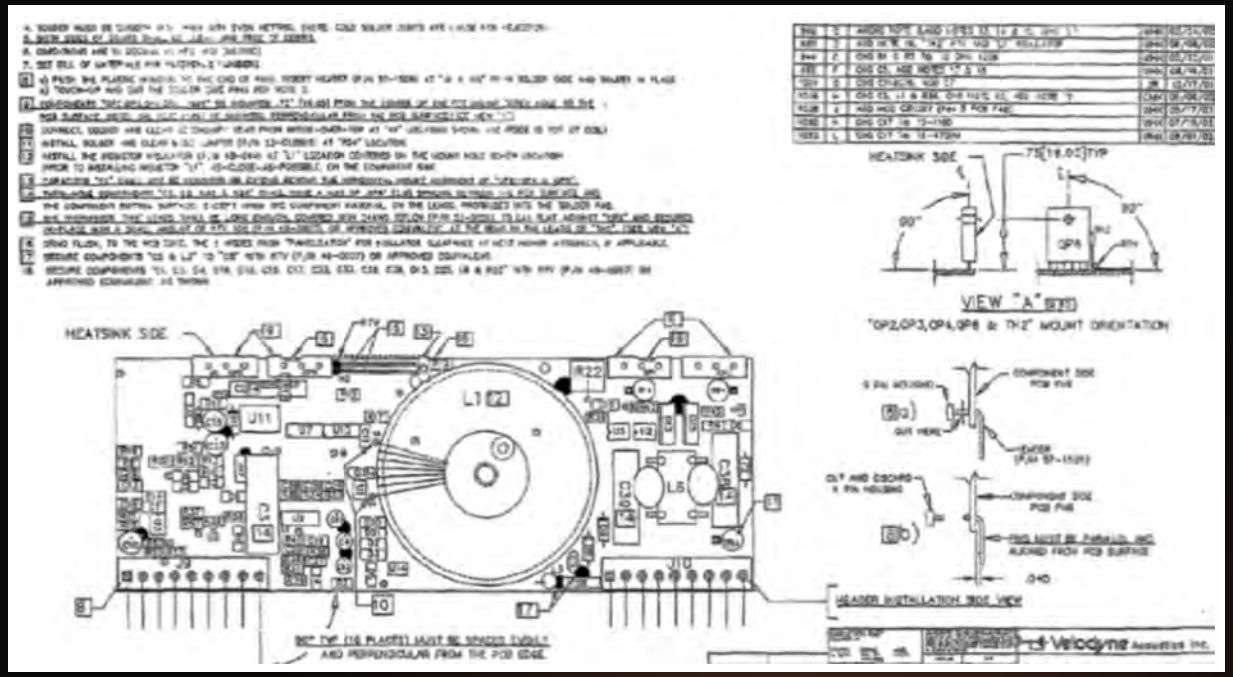
14. Impedance (all tolerances +/- 10%)
All 0.nn mm wide traces on outer layers shall be 50 ohms. All 0.nn mm wide/0.nn mm space pairs on outer layers shall be 90 ohms. All 0.nn mm wide/0.nn mm space pairs on inner layers shall be 90 ohms. Vendor may adjust design geometries up to +/-20% to achieve target impedance.

Adjustments beyond 20% of line width, spacing or dielectric thickness shall require approval from company engineering.

15.3 Assembly Drawings

Another inspection document is the assembly drawing. The primary goal is to show what the finished assembly looks like. If you read my blogs, you probably know that I insist that this is a what-is, not a how-to document.

How-to documents are subject to frequent revision as the process evolves through continuous improvement. I urge you not to get yourself wrapped around that axle.



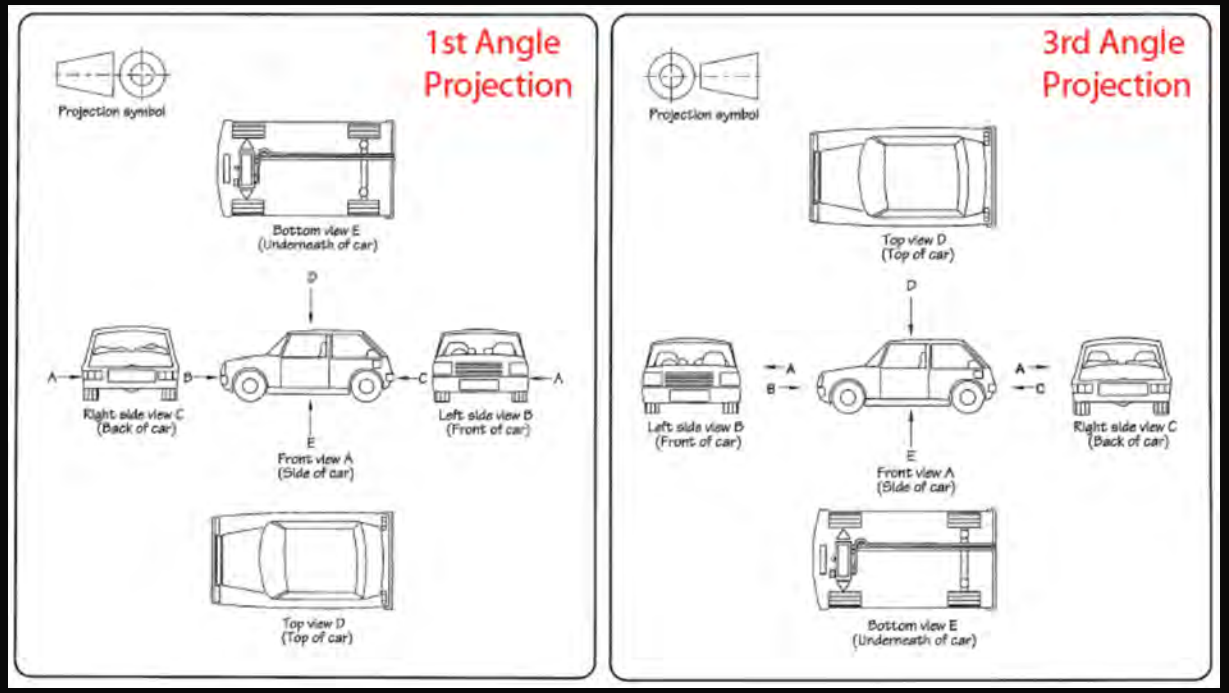
Section 15. Figure 2. Image 15.2. Credit: Velodyne - Assembly drawings come in two main types, SMT assembly and 2nd-operation drawings like this one. Note that the details use third angle projection.

15.4 Projecting views

There will always be a top side plan view of the PCB outline. It may also be augmented with a side view or a cutaway view to show details such as the beveled edge for gold fingers. The United States along with Australia and possibly some other locations use what is called *third angle projection* while *first angle projection* is the norm in Europe and elsewhere.

For third angle projection, imagine the board at the bottom of a bowl. The top and side views are created by sliding and rotating the view up the sides of the bowl so that the side you see in the edge view faces towards the plan view.

On the other hand 1st Angle Projection would invert the imaginary bowl and allow the additional views to slide off the bowl.



Section 15. Figure 3. Image 15.3. Credit: Practical Machinist - The graphics in the upper left corner of these views are included in the title block to establish the projection of various views.

15.5 Standard Assembly Notes

Notes unless otherwise specified.

1. Interpret drawing in accordance with asme y14.5m 1994.
2. Workmanship shall be in accordance with the current revision of j-std-100 and ipc-hdbk-001.
3. Inspection shall be in accordance with the current revision of ipc-a-610, class 2.
4. The controlling document for component placement shall be xy-nnn-nnnn-nn.
5. Mark dash and revision level in permanent black ink as shown.
6. Affix barcode, date code and serial number labels on secondary side as shown. Bag and tag acceptable for assemblies that are too small for labels.
7. Company reference documents as follows:
   - Bill of material nnn-nnnn-nn.
   - Placement file nnn-nnnn-nn.
   - Paste stencil nnn-nnnn-nn.
   - Schematic nnn-nnnn-nn.
   - FPGA programming nnn-nnnn-nn.
   - Test procedure nnn-nnnn-nn.
   - Whatever else is relevant and not documented elsewhere.

15.6 Graduate School Zone - Formatting Data

We have been using Gerber data for about as long as we have been using computers to aid us in our design efforts. Gerber data has evolved to allow unlimited apertures. Originally, this was a mechanical process that involved aperture wheels with 24 different openings that were used to generate the photo tools.

The aperture wheels evolved to an electronic equivalent. Every PCB design was taped out with a separate file for the apertures. The aperture list is now embedded with each artwork layer to make it more resilient. The information conveyed in this standard is just the raw geometry. File names supplemented with a readme.doc help the fabricator determine how to use the data.

Eventually, a de facto standard came along from Valor. It is native to the software used in the front end of the fabrication process. The format is called ODB++ and it has some intelligence along with the geometry data. Fabrication and assembly data can be combined in a single archive. Most PCB factories prefer this type of data. This was ok as long as Valor was a stand-alone enterprise.

The ECAD industry has come together around a new specification that adds more intelligence and is not encumbered by commercial interests. This new thing is the IPC-2581 spec. If you want to have a good time, hook up with an outfit that will talk 2581 with you. The artwork still has to be camera-ready; it's not doing your drawings. What it will do is a thorough extraction and allow the designer to define what goes to the manufacturing partner.

According to my colleague, Hemant Shah, *"IPC-2581 not only makes design data handoff efficient, it also makes manufacturing the board more efficient. IPC_2581 connects to IPC-CFX to help with smart factory automation. with revision C of IPC-2581, manufacturing partners can provide technical queries in 2581 format making it very easy to cross probe, track, resolve, approve, reject TQs."*

You'll still find many fab shops relying on Gerber data since they have a different CAM tool. A majority of the high-capability shops are using Valor and prefer that. It will be up to us, the design community, to help launch the new CAD-neutral IPC format.

16. Vendor Management

Section sixteen brings it all together and talks about the team effort that it takes to bring forth a PCB/PCBA.

16.1 Vendor Management

16.2 Getting Your Impedance Message Across

16.3 Not Defining the Impedance Callouts

16.4 Design Reuse Can Be Useful But Always Recertify the Plan

16.5 Graduate School Zone: PCB Designers as Vendors

16.1 Vendor Management

As the designer, manufacturability, reliability and performance are in your hands. When something gets in the way of those goals, pushing back as the lone voice of reason isn't enough. If your internal stakeholders don't want to hear what you have to say, it's time to loop in the outside vendor(s) for another opinion.

"Handling this correctly in the first place enhances your reputation for getting things done."

Your team might be creating a more difficult process but it's ok as long as everyone is aware and onboard. A collective decision that the product will have a higher production cost is one thing. "I told you so" isn't going to cover your six-o-clock in the aftermath of a lost opportunity. "We told you so" at least spreads the risk.

Section 16. Figure 1. Image 16.1. Credit: Author - The humming assembly floor is where the machines finally come to life.

Vendors end up feeling the brunt of those challenges so they have to be in the loop. Handling this correctly in the first place enhances your reputation for getting things done.

16.2 Getting Your Impedance Message Across

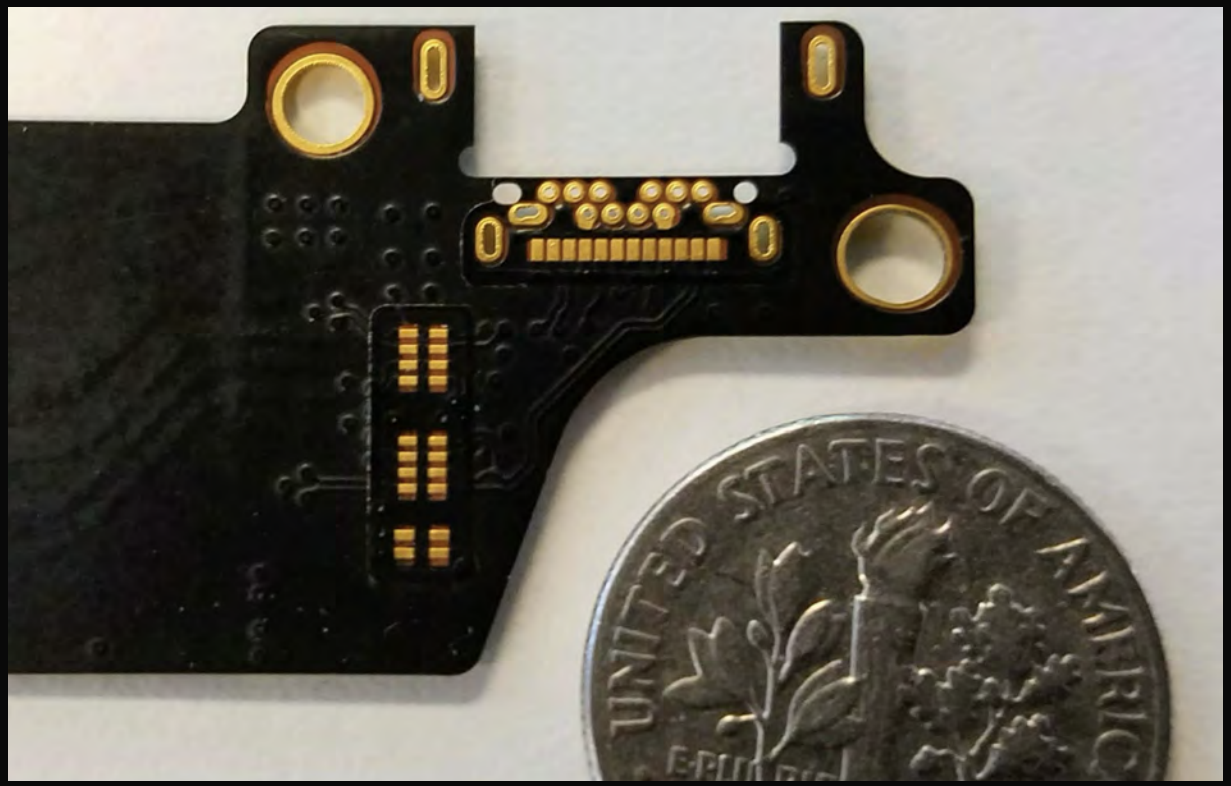
We have to start somewhere. The easy way is to have a "ball-park" stack-up and line width and let the vendor roll with it. They may ask for material substitutions to use equivalent material. It may have equivalent dimensions but different dielectric constants. If you're ok with whatever comes out of the factory, then it's ok. That's your preferred plan but I'm not so sure you will get consistent results from one vendor to another.

“Handling this correctly in the first place enhances your reputation for getting things done.”

16.3 Not Defining the Impedance Callouts

Another way that I'll refer to as the Qualcomm way would be to prescribe exact materials and not allow substitutions for the reference designs. You would not call out impedance requirements in that case. They become a product of the materials used along with the designed trace geometry. If the vendor built the board using anything but the exact materials, then performance was their problem. No deal without getting a good hard look at the proposed construction.

The more accurate way is to get the stack-up and line width(s) from the vendor up front and use those materials and geometries. That's the Microsoft way. The fab shops would give us that information without NRE charges. That's my advice if you can swing it. It pays to have a relationship with the vendor.



Section 16. Figure 2. Image 16.2. Credit: Author - The working end of a USB type-C flex. Flex vendors are very good at some things and not always that good at everything. They said they couldn't follow the tolerances on the fab drawing for this flex. We ended up making them responsible for the turnkey flex with connectors and ESD dordes pre-installed. We're out of the middle.

Optionally, telling the vendor that specific line widths are meant to be a certain impedance in the notes works too. This typically involves the generation of an impedance table that specifies all of the impedance requirements for each routing layer and reference planes as required. For single ended impedance, that is just a line width. For differential routing, a line width and an air-gap is necessary. How you handle neck-down regions is another matter. Of course, you'll need to be prepared to adjust your design based on their feedback.

Asking the fabricator to test and prove the impedance is within the stated tolerance is advisable since there are so many factors in play. Requiring test coupons is normal for quick development when a good impedance is required.

The typical notes would allow the vendor to use alternate materials and adjust line widths by a certain small percentage, say 10%. Deviation more than that would require a consultation with you. The exact amount of latitude would depend on your comfort zone.

16.4 Design Reuse Can Be Useful But Always Recertify the Plan

My most recent board design reused a stack-up. When we sent the vendor's own previously recommended stack-up back to them for approval, you know and I know what happened next. I got to go through the board and revise all of the 50 ohm and 100 ohm differential line-widths. Don't forget to update the design rules, the stack-up geometry and the impedance table!

Aside from the stack-up, technical questions are likely to arise once the vendor has your data in their hands. It is usually a question of plating, especially the barrel thickness of the vias. The amount of plating the vias can get is a function of the minimum air-gap on the outer layer. Our trace/space geometry hems them in. There are a variety of potential questions that may arise.

Hole tolerances are another common *negotiating point*. The goal is to avoid providing missing or conflicting information. The clock is ticking once you get the list of waivers so it's better to be agreeable when there is a hot job on the plate.

Digging in your heels over a feature that is part of a vendor's recommended footprint might be an exception to this advice. If the fabricator says they can't meet the spec, then you're in the middle. This is a risky situation. You have to pull strings on both sides to get a representative of the fab shop and the part supplier to agree on a way forward. One of the things about a high-priority job is that you sure don't have a second chance.

16.5 Graduate School Zone: PCB Designers as Vendors

To get from the first stage to the last. Our plan is to loop in the help we need at key points in the timeline. You know what progress looks like, right? You know what the schedule is. So keeping progress and time in alignment is the big picture here. Hitting the market window is a requirement of a design win. This is why we need access to a second shift when we're done for the day.

Freelance PCB Design is a cottage industry with global reach. Your design might be best served spending the night on another continent. One of the hallmarks of a good service bureau is having offices in different time-zones. The amount of work I do in Silicon Valley? I could do just as much in Fiji. Spring time might be wet but CAD is indoor work so we're good.

Section 16. Figure 3. Image 16.3. Credit: Author - Always be prepared for a minor roadblock. Delays are inevitable.

The point is that you need this outside layout option on tap to execute an aggressive schedule. You have your finger on the pulse of all of your projects and there is always at least one that is on a fast track. Pull them into the hottest job and define the scope of their effort on an on-going basis.

Your design partner's work will be a reflection of the information you give both verbally and in terms of design rules. Put yourself in their shoes for a moment. A remote meeting or call that goes with each hand-off is a really good investment in everyone's time; once a day minimum.

Now that you actually have your job done a day early, don't squander that lead with a slow procurement cycle. The first thing that will snag you is materials if you didn't square that away with the fabrication vendor before routing got started.

Section 16. Figure 4. Image 16.4. Credit: Author - A clean desk is more productive.

Board design will humble you, confuse you, inspire you, madden you, give your confidence, take it away, give it back. Doing this job takes determination and attention to detail. From sat-comm to wireless, to consumer electronics, autonomy, mixed reality, and back to sat-comm, I've had a long ride and it was a labor of love to bring it to you in this series. Thanks for reading.

John enjoys playing bass and racing bikes when he's not writing about or performing PCB layout.

John Burkhert Jr is a career PCB Designer experienced in Military, Telecom, Consumer Hardware and lately, the Automotive industry. Originally, an RF specialist – compelled to flip the bit now and then to fill the need for high-speed digital design. You can find John on LinkedIn.

More from John Burkhert.

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